



Lecture
Notes

2026

Mechanics of Continua

Prof. Barak Kol

77606 · HUJI Physics · Eran Rehani

Contents

I	Elasticity of Solids	6
1	Lecture 01 — Introduction, Continuum Limit, and the Stress Tensor	7
2	Lecture 02 — The Prismatic Bar in Uniaxial Stress	10
2.1	Stress and Strain in 1D	10
2.2	Lateral Contraction and Poisson’s Ratio	11
2.3	Worked Example — Steel Column Under Compression	12
2.4	The Stress Tensor and Its Symmetry	13
2.5	Static Equilibrium Equations	14
2.6	Brief Note on Tensors	14
3	Lecture 03 — Stress, Strain, and Tensor Decomposition	16
4	Lecture 04 — Tensor Algebra, Coordinate Systems, and 1D Constitutive Laws	19
5	Lecture 05 — Hooke’s Law and Linear Isotropic Elasticity	24
6	Lecture 06 — Equations of Equilibrium and the Lamé–Navier Equation	30
7	Lecture 07 — Rotating Cylinder and Elastic Waves in Solids	36
8	Lecture 08 — Torsion and Bending of Beams	44
9	Lecture 09 — Euler–Bernoulli Beam Equation	50
II	Mechanics of Fluids	57
10	Lecture 10 – From Elasticity to Hydrostatics	58
10.1	Final Recap of the Elasticity Chapter	58
10.2	What Is a Fluid?	59
10.3	Hydrostatic Stress: Only Pressure Survives	60
10.4	Units and Scales of Pressure	60
10.5	Hydrostatic Equilibrium Equation	61
10.6	Incompressible Fluid: Pressure in Water	61
10.7	Compressible Hydrostatics and the Equation of State	62
10.8	The Gibbs Potential Per Unit Mass	63
10.9	Example: Isothermal Atmosphere	64
10.10	Summary and Bridge to the Next Lecture	65
11	Lecture 11 – Hydrostatics Applications and Buoyancy	67
11.1	Pascal’s Principle and Hydraulic Machines	67
11.1.1	Uniform Pressure in a Small Hydraulic Device	67
11.1.2	Force Transmission Without Amplification	68
11.1.3	The Hydraulic Press	68
11.1.4	Quasistatic Assumption and Boundary Forces	69
11.2	Archimedes’ Principle	69
11.2.1	Statement of the Buoyant Force	69
11.2.2	Displaced-Fluid Proof	70
11.2.3	Point of Application	70
11.3	Floating Bodies and Ship Stability	70

11.3.1	Force Balance for Floating	71
11.3.2	Small Tilt and the Restoring Moment	71
11.4	Resultant Hydrostatic Force on a Dam	71
11.4.1	Pressure Distribution	72
11.4.2	Total Force	72
11.4.3	Center of Pressure	72
11.5	Lift on a Truncated Cone Filled with Water	73
11.5.1	Setting Up the Integral	73
11.5.2	Total Upward Force	73
11.6	Summary	74
12	Lecture 12 – Dynamics of an Ideal Fluid	75
12.1	Fields and the Ideal-Fluid Approximation	75
12.1.1	Pressure Versus General Stress	75
12.2	Mass Conservation: The Continuity Equation	76
12.2.1	Mass Current	76
12.2.2	Steady Flow Through a Tube	76
12.3	The Material Derivative	77
12.3.1	Definition	77
12.3.2	Derivation from a Particle Trajectory	77
12.4	Momentum Balance and Euler's Equation	78
12.4.1	General Momentum Balance	78
12.4.2	Euler Equation for an Ideal Fluid	79
12.5	Mechanical Closure and Equations of State	79
12.5.1	Barotropic Closure	80
12.5.2	Incompressible Fluids	80
12.5.3	Isentropic Ideal Gases	80
12.5.4	Enthalpy per Unit Mass	81
12.6	The Closed Ideal-Fluid System and Boundary Conditions	81
12.6.1	Counting Unknowns and Equations	82
12.6.2	Impermeable Boundary	82
12.7	How These Field Equations Are Solved	82
12.7.1	Analytic Versus Numerical Solutions	82
12.8	Bernoulli's Equation as Energy Conservation	83
12.8.1	Mechanical Analogy	83
12.8.2	Derivation for an Ideal Isentropic Flow	84
12.8.3	Incompressible Form	84
12.9	Summary	85
13	Lecture 13 – Bernoulli's Equation and Applications	86
13.1	Complete Derivation of Bernoulli's Equation	86
13.1.1	Setup: Steady Flow with a Conservative Body Force	86
13.1.2	Dot with $\mathbf{v}$	86
13.1.3	Important Special Cases	87
13.2	Application 1: Draining Tank (Torricelli's Theorem)	88
13.2.1	Part (a): Exit Speed — Torricelli's Theorem	88
13.2.2	Part (b): Draining Time	89
13.2.3	Part (c): Stream Profile Below the Tap	89
13.3	Application 2: The Venturi Effect	90
13.3.1	Analysis	91
13.4	Application 3: Bernoulli and Airfoil Lift	91
13.4.1	Standard Argument	91

13.5	Preview: Conservation Laws and Lagrangian Description	92
13.5.1	The Two Descriptions of Fluid Motion	92
13.5.2	Key Points for Next Lecture	92
14	Lecture 14 – Circulation, Vorticity, and Two Descriptions of Fluid Motion	94
14.1	Brief Review: Venturi and Airfoil Lift	94
14.2	Kelvin’s Circulation Theorem	95
14.2.1	Definition: Circulation	95
14.2.2	Material Curves	95
14.2.3	Key Kinematic Lemma	95
14.2.4	Proof of Kelvin’s Theorem	95
14.2.5	Physical Discussion	96
14.3	Helmholtz Decomposition	96
14.3.1	Fourier-Space Picture	97
14.3.2	Connection to Circulation	97
14.4	Vorticity	97
14.4.1	Stokes’ Theorem: Connecting Integral and Differential	98
14.4.2	Differential Forms of Kelvin’s Theorem (stated without proof)	98
14.5	Two Descriptions of Fluid Motion	98
14.5.1	Eulerian (Spatial) Description	98
14.5.2	Lagrangian (Material) Description	99
14.5.3	Equivalence and Conversion	99
14.5.4	Advantages	99
14.6	Streamlines and Pathlines	100
14.7	Conservation Laws: Energy and Momentum	100
14.7.1	Energy Conservation	100
14.7.2	Momentum Conservation	101
14.8	Action Principle for Ideal Fluids (Preview)	101
14.8.1	Compressible Ideal Fluid	101
14.8.2	Incompressible Ideal Fluid	102
15	Lecture 15 – Potential Flow	103
15.1	Irrotational Flow: Motivation and Definition	103
15.1.1	Velocity Potential	104
15.1.2	Dimensions of ϕ	104
15.1.3	Equipotential Surfaces	104
15.2	Bernoulli-Like Equation	104
15.2.1	Vector Identity for the Convective Term	104
15.2.2	Euler’s Equation in Potential Form	105
15.2.3	Summary of Equations for Potential Flow	105
15.3	Incompressible Potential Flow	106
15.3.1	Laplace’s Equation	106
15.3.2	Decoupling and the Role of Pressure	106
15.3.3	Boundary Conditions	106
15.3.4	Solution Methods	107
15.4	Example 1: Point Source in 3D	107
15.4.1	Symmetry and the Form of $\mathbf{v}$	107
15.4.2	Velocity Potential	108
15.4.3	Pressure Field	108
15.4.4	Minimum Source Size	108
15.5	Example 2: Point Source Near a Rigid Wall	108
15.5.1	Image Method	109

15.5.2	Stagnation Point	109
15.5.3	Equipotential Surfaces – Qualitative Picture	109
15.5.4	Pressure Maximum from Bernoulli	109
15.6	Local Flow Near a Stagnation Point	109
15.6.1	Taylor Expansion of ϕ	110
15.6.2	Constraints on the Quadratic Form	110
15.7	Example 3: Time-Dependent Syringe	110
15.7.1	Continuity: v Depends Only on Time	110
15.7.2	Velocity Potential	110
15.7.3	Bernoulli-Like at Top and Bottom	111
15.7.4	Equation of Motion for $h(t)$	111
15.7.5	Effective-Potential Interpretation	111
16	Lecture 16 – Potential Flow Around a Sphere and D’Alembert’s Paradox	113
16.1	Recap: The Potential-Flow Program	113
16.2	Physical Setup and Two Reference Frames	114
16.3	The Boundary-Value Problem in the Sphere Frame	116
16.3.1	Electrostatic Analogy	116
16.3.2	Multipole Expansion and Selection of the Dipole	116
16.3.3	Matching the Neumann Condition	117
16.3.4	The Solution in the Sphere Frame	117
16.4	Transforming to the Fluid Frame	118
16.5	Streamlines, Equipotentials, and Stagnation Points	118
16.5.1	Stagnation Points	118
16.5.2	Equipotential Surfaces	119
16.6	Generalization to Arbitrary Motion	119
16.6.1	Vectorial Rewriting	119
16.6.2	The Accelerating Sphere	119
16.7	The Force on the Sphere	120
16.7.1	Assembling the Pressure	120
16.8	Added Mass (Effective Mass)	121
16.8.1	A Tensor Integral over the Sphere	121
16.8.2	Evaluating the Force	121
16.8.3	The Effective Mass	122
16.8.4	Generalization: The Effective-Mass Tensor	122
16.9	Resolution of D’Alembert’s Paradox	123
17	Lecture 17 – Planar Potential Flow and the Complex Potential	125
17.1	What is Planar Flow, and Why Study It Separately?	125
17.2	Mathematical Interlude: Two Operations on Planar Vectors	126
17.3	The Two Potentials: ϕ and the Stream Function ψ	127
17.4	Equations of Motion Are the Cauchy–Riemann Equations	128
17.5	Boundary Conditions	129
17.6	The Complex Velocity Potential	129
17.6.1	Recovering the Velocity Field	129
17.7	Example: Uniform Flow	130
17.8	Example: Source and Sink	130
17.9	Example: The Point Vortex	131
17.10	Physical Meaning of the Stream Function	132
17.11	Example: Flow Around a Corner	132
17.12	Example: Flow Around a Disk	134

III Homework Solutions	136
18 Chapter 1 – Fundamental Equations §7	137
19 Fluid Mechanics — Draining Vessels (from Lecture 13)	146

Part I

Elasticity of Solids

1 Lecture 01 — Introduction, Continuum Limit, and the Stress Tensor

Overview

This lecture introduces the field of continuum mechanics, describing the historical and conceptual shift from discrete point particles (Newtonian mechanics) to continuous media. The framework covers both solid mechanics (elasticity) and fluid dynamics (liquids and gases). The principles discussed—such as continuous fields, field equations, and tensors—are foundational for engineering applications as well as advanced physics, including electromagnetism and quantum field theory. The lecture defines the continuum limit, introduces the concept of the stress tensor, and demonstrates how surface contact forces are converted into a volume force density via the divergence theorem.

Definitions

Physical Concept 1.1: Continuum Limit

The regime where a discrete collection of atoms or molecules (with typical interatomic spacing $a \sim 10^{-10}$ m) can be well-approximated as a continuous material. This macroscopic description is valid when the characteristic length scale L of the system is much larger than the atomic scale ($L \gg a$). In this limit, discrete masses map to a continuous mass density $\rho = \lim_{\Delta L \rightarrow 0} \frac{\Delta m}{\Delta L}$.

Physical Concept 1.2: Area Element $d\mathbf{S}$

An infinitesimal vector representing a surface patch. Its magnitude corresponds to the differential area dS , and its direction is perpendicular to the surface, pointing strictly along the *outward normal*.

Physical Concept 1.3: Stress Tensor σ_{ij}

A rank-2 tensor (often represented as a 3×3 matrix) that relates the area element $d\mathbf{S}$ of a given plane to the internal contact force $d\mathbf{F}$ exerted across that plane by adjacent volume elements.

Physical Concept 1.4: Levi-Civita Symbol ϵ_{ijk}

A completely antisymmetric rank-3 tensor used for expressing vector cross products and moments in index notation. $\epsilon_{ijk} = +1$ for even permutations of $(1, 2, 3)$, -1 for odd permutations, and 0 if any index is repeated.

Key Results

Physical Law 1.1 Contact Force on a Surface

The i -th component of the contact force $d\mathbf{F}$ acting on a surface element $d\mathbf{S}$ is strictly proportional to the area vector via the stress tensor:

$$dF_i = \sigma_{ij} dS_j \quad (1)$$

where the repeated index j implies summation over the spatial coordinates (Einstein summation convention).

Physical Law 1.2 Force Density via Divergence

The total contact force per unit volume (force density) acting on a continuum element is given by the spatial divergence of the stress tensor:

$$\boxed{f_i = \partial_j \sigma_{ij}} \quad (2)$$

If the stress tensor is uniform ($\partial_j \sigma_{ij} = 0$), the net contact force on the macroscopic volume element vanishes.

Derivations**Total Contact Force and Force Density**

To find the total contact force F_i^{total} acting on a macroscopic volume element V bounded by a closed surface S , we integrate the surface contact forces over the entire boundary:

$$\begin{aligned} F_i^{\text{total}} &= \oint_S dF_i \\ &= \oint_S \sigma_{ij} dS_j. \end{aligned} \quad (3)$$

This expression represents a flux integral for each vector component i . By applying the divergence theorem (Gauss's theorem), we can convert this closed surface integral into a volume integral over V :

$$\oint_S \sigma_{ij} dS_j = \int_V \partial_j \sigma_{ij} dV. \quad (4)$$

Thus, the total force evaluates to:

$$F_i^{\text{total}} = \int_V (\partial_j \sigma_{ij}) dV. \quad (5)$$

For an infinitesimal volume element dV , the integrand $f_i = \partial_j \sigma_{ij}$ is precisely the net force per unit volume. For the element to remain in static equilibrium, this differential force must balance external body forces (such as gravity).

Examples**Worked Example 1.1 (Forces on the Faces of a Cube)**

Consider an infinitesimal cubic volume element aligned with the Cartesian axes (x, y, z) .

- **Top Face:** The outward normal points in the positive z -direction. The area vector is $d\mathbf{S} = (0, 0, dS)$, meaning $dS_3 = dS$. The force on this face is $dF_i = \sigma_{i3} dS$.
- **Bottom Face:** The outward normal points in the negative z -direction. The area vector is $d\mathbf{S} = (0, 0, -dS)$. The force on this face is $dF_i = -\sigma_{i3} dS$.

If σ_{i3} is uniform (constant) across the vertical height of the cube, the forces on these two opposing faces are equal and opposite, leading to exact cancellation.

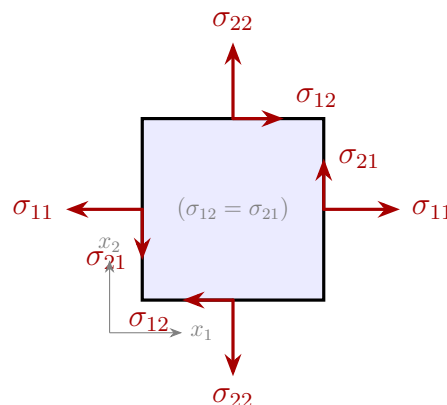
Connections

- **Classical Mechanics:** Continuum mechanics builds upon the discrete point-particle mechanics developed by Newton, Euler, and Lagrange, generalizing it to bodies with infinite degrees of freedom.

- **Electromagnetism:** The mathematical description of continuous material fields conceptually preceded and inspired the formulation of the electromagnetic fields by Maxwell and Faraday.
- **Quantum Field Theory (QFT):** The classical field theories (and field equations) established in continuum mechanics form the classical basis that is eventually quantized in fundamental particle physics.
- **Engineering & Physics Applications:** The principles taught directly underly civil/mechanical engineering (steel, bridges, building structures), fluid engineering (hydraulics, aerodynamics), and modern physics domains (stellar structure, nuclear models, quark-gluon plasmas).

To Remember

- The continuum approximation replaces discrete molecular structures with continuous fields, valid for macroscopic length scales $L \gg 10^{-10}$ m.
- The local state of stress is encoded in a 3×3 tensor σ_{ij} .
- The placement of tensor indices (upper vs. lower) carries no physical distinction in this specific course, in contrast to General Relativity.
- The Einstein summation convention dictates that any index repeated twice in a multiplicative term is implicitly summed over the three spatial coordinates.
- A net volume force arises strictly from the non-uniformity (spatial derivatives / divergence) of the stress tensor.



Exam / HW Hints

- **Grading:** The course grade is based 100% on a 2.5-hour final exam.
- **Homework:** Due to budget cuts, there are no official TAs and no graded homework assignments.
- **Practice:** The lecturer will use one hour per week for active, guided problem-solving. Students are highly encouraged to participate actively in these sessions.
- **Exam Preparation:** The best way to prepare for the exam is to solve the final exams from the previous two years.
- **Textbooks:** Recommended reading includes Landau & Lifshitz (Vol. 7 *Theory of Elasticity* and Vol. 6 *Fluid Mechanics*), as well as Timoshenko & Gere (*Mechanics of Materials*) for the engineering perspective.

2 Lecture 02 — The Prismatic Bar in Uniaxial Stress

Overview and Motivation

This lecture revisits stress from the ground up by studying a single, concrete object: a *prismatic bar* (a rod of constant cross-section) under *uniaxial loading*. The bar is our **leading example** for the entire theory of elasticity — simple enough to solve exactly, yet rich enough to introduce every key concept: stress, strain, Young’s modulus, Poisson’s ratio, the stress tensor, and the static equilibrium equations.

We consider a bar of:

- length L ,
- uniform cross-sectional area A (square, circular, or any shape — the analysis is identical),
- isotropic, linearly elastic material (e.g. steel, concrete, bone).

An **axial load** F is applied inward at both ends (compression) or outward at both ends (tension). Static equilibrium means the two forces are equal and opposite; the bar as a whole does not accelerate.

2.1 Stress and Strain in 1D

Internal force and the concept of stress

Imagine cutting the bar at an arbitrary cross-section and isolating the right half. That half is in equilibrium: the external pull F to the right must be balanced by an equal and opposite *internal contact force* F to the left, exerted by the left half through the cut surface. Because the same argument applies at *any* cross-section, the internal force is equal to F everywhere along the bar.

The key insight is to normalise by area. We define the **stress**

$$\sigma = \frac{F}{A}, \quad (6)$$

which is the force per unit area acting on the cross-section. Stress has the same SI units as pressure — *pascals* (Pa, named after Blaise Pascal, 1647):

$$[\sigma] = \frac{\text{N}}{\text{m}^2} = \text{Pa}.$$

Typical engineering values are in the range of megapascals (MPa) to gigapascals (GPa).

Physical intuition: Two bars made of the same material but with different cross-sections, loaded by the same force, deform by the same *strain* only if acted on by the same *stress*. It is stress, not force, that drives deformation.

Deformation of a volume element and strain

Consider an infinitesimal volume element of length dz and area dA . It is pulled by forces $dF = \sigma dA$ on both faces. By the same proportionality argument for any element,

- Each element elongates by some amount $d(\Delta z)$ proportional to its length dz .
- Summing over all elements, the total elongation ΔL is proportional to the original length L .

We define the **longitudinal strain** (also called *elongation*) as the dimensionless relative deformation,

$$\varepsilon_{zz} \equiv \frac{\Delta L}{L}. \quad (7)$$

Note on notation: Landau & Lifshitz use a different (older) convention; this course adopts the modern standard in which strain is denoted ε_{ij} .

Hooke's Law and Young's modulus

Linear elastic response means stress is proportional to strain:

$$\boxed{\sigma = E \varepsilon_{zz}}, \quad (8)$$

where E is **Young's modulus** (or the *elastic modulus*), first proposed by Euler (1727) and systematised by Thomas Young (1807). Re-substituting the definitions of σ and ε_{zz} gives the fundamental displacement formula:

$$\boxed{\Delta L = \frac{F L}{E A}}. \quad (9)$$

The analogy with resistors: The electrical resistance of a wire is $R = \rho_e L/A$, where ρ_e is the resistivity. Under a voltage V , the current is $I = V/R$. Equation (9) has the identical structure with $F \leftrightarrow V$, $E \leftrightarrow 1/\rho_e$, and $\Delta L \leftrightarrow I$. This is not a coincidence — both phenomena are governed by linear constitutive relations in geometrically similar media.

Dimensions of E : Since ε_{zz} is dimensionless, E has the same dimensions as stress:

$$[E] = \text{Pa} = \frac{\text{N}}{\text{m}^2} = \frac{\text{J}}{\text{m}^3}.$$

The last equality shows that E can also be read as energy stored per unit volume — a useful perspective in thermodynamics of solids.

2.2 Lateral Contraction and Poisson's Ratio

When a bar is stretched longitudinally, it contracts in the lateral (transverse) directions — and vice versa. By symmetry, both transverse directions x and y contract equally. The **Poisson's ratio** (named after Siméon Denis Poisson, 1827) quantifies this coupling:

$$\nu \equiv -\frac{\varepsilon_{xx}}{\varepsilon_{zz}} = -\frac{\varepsilon_{yy}}{\varepsilon_{zz}}, \quad (10)$$

so that $\varepsilon_{xx} = \varepsilon_{yy} = -\nu \varepsilon_{zz}$. The negative sign ensures $\nu > 0$ for any ordinary material (which grows thinner when stretched). As we shall show later from thermodynamic stability arguments, ν is constrained to the interval

$$-1 < \nu \leq \frac{1}{2}.$$

For virtually all engineering materials $\nu > 0$, i.e. the cross-section shrinks under tension.

Compression: Reversing $F \rightarrow -F$ flips the sign of all strains simultaneously; the formulas remain valid with the understanding that $\sigma < 0$ denotes compression.

Typical values of elastic constants

Material	E (GPa)	ν	Notes
Steel (structural)	200	0.28	Fe alloy, most common structural metal
Diamond	1 000	0.20	Hardest known material
Concrete	20–30	0.20	Weak in tension; used in compression
Aluminium	70	0.33	Light structural metal
Gold	78	0.44	Dense, ductile
Bone (cortical)	15–25	0.30	Varies along grain direction
Wood (along grain)	10–15	0.35	Strongly anisotropic
Rubber	~ 0.01	≈ 0.50	Nearly incompressible
Cork	~ 0.05	≈ 0.00	Used as bottle stoppers

Key observations:

- E spans four orders of magnitude across common materials.
- ν is restricted to a much narrower range; most materials cluster between 0.25 and 0.45.
- Cork's near-zero ν is precisely what makes it an ideal bottle stopper: it does not expand laterally under axial compression, so it can be pushed into a bottle without cracking the glass.
- Rubber's $\nu \approx \frac{1}{2}$ means it is nearly *incompressible*: volume is essentially conserved under deformation.

2.3 Worked Example — Steel Column Under Compression

Worked Example 2.1

Problem. A structural steel column has length $L = 3$ m, a square cross-section of side 10 cm, and is subjected to a compressive load $F = 200$ kN.

- For steel: $E = 200$ GPa, $\nu = 0.28 \approx 0.3$.
- Find: the axial shortening $|\Delta L|$ and the lateral expansion $|\Delta x|$.

Step 1 — Convert units carefully.

$$A = (0.10 \text{ m})^2 = 0.01 \text{ m}^2, \quad F = 200 \times 10^3 \text{ N}.$$

Step 2 — Compute the stress.

$$\sigma = \frac{F}{A} = \frac{2 \times 10^5 \text{ N}}{10^{-2} \text{ m}^2} = 2 \times 10^7 \text{ Pa} = 20 \text{ MPa}.$$

(Note: the load is compressive, so strictly $\sigma = -20$ MPa.)

Step 3 — Compute the axial strain.

$$\varepsilon_{zz} = \frac{\sigma}{E} = \frac{2 \times 10^7 \text{ Pa}}{2 \times 10^{11} \text{ Pa}} = 10^{-4}.$$

Step 4 — Find ΔL .

$$|\Delta L| = |\varepsilon_{zz}| L = 10^{-4} \times 3 \text{ m} = 3 \times 10^{-4} \text{ m} = \boxed{0.3 \text{ mm}}.$$

Step 5 — Find the lateral expansion.

$$|\varepsilon_{xx}| = \nu |\varepsilon_{zz}| = 0.3 \times 10^{-4} = 3 \times 10^{-5}.$$

$$|\Delta x| = |\varepsilon_{xx}| x_0 = 3 \times 10^{-5} \times 0.10 \text{ m} = 3 \times 10^{-6} \text{ m} = \boxed{3 \text{ } \mu\text{m}}.$$

Sanity check. The column shortens by 0.3 mm (1 part in 10^4 of its length) and its cross-section widens by only $3 \text{ } \mu\text{m}$. These tiny deformations justify the linear elastic (small-strain) approximation: the geometry is essentially unchanged during loading. The model breaks down when $|\varepsilon| \gtrsim 10^{-3}$, where nonlinear or plastic effects become significant.

Common student error: Confusing $1 \text{ GPa} = 10^9 \text{ Pa}$ with $1 \text{ MPa} = 10^6 \text{ Pa}$ leads to an error of a factor of 10^3 in strain — just as was discussed in class.



2.4 The Stress Tensor and Its Symmetry

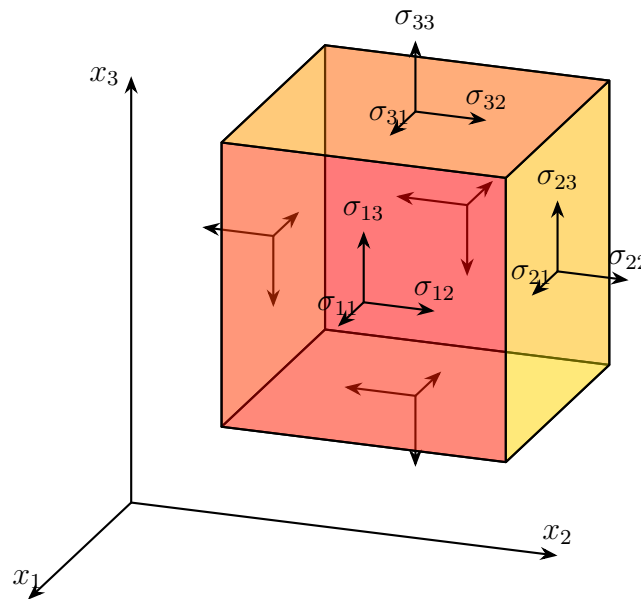
From 1D to 3D: review of σ_{ij}

The 1D example generalises: the force $d\mathbf{F}$ on any oriented surface element $d\mathbf{S} = \hat{n} dA$ inside a solid is given by the **stress tensor** $\boldsymbol{\sigma}$,

$$dF_i = \sigma_{ij} dS_j, \quad (11)$$

where summation over the repeated index j is implied (Einstein convention). The tensor $\boldsymbol{\sigma}$ has $3 \times 3 = 9$ components. The diagonal components $\sigma_{11}, \sigma_{22}, \sigma_{33}$ are *normal stresses*; the off-diagonal components σ_{ij} ($i \neq j$) are *shear stresses*.

Stress vs. pressure: Pressure in a fluid is isotropic — $\sigma_{ij} = -P\delta_{ij}$ — and acts equally in all directions. In a solid, the stress tensor need not be isotropic: different faces of the same volume element may carry different forces. Stress is the more general concept.



Symmetry of σ_{ij} : torque argument

We now show that $\sigma_{ij} = \sigma_{ji}$ must hold for any volume element in static equilibrium.

Consider a small rectangular element with sides Δx , Δy , Δz . Focus on the torque about the z -axis. The shear stress σ_{12} (the y -component of the force on the $+x$ -face) exerts a clockwise torque:

$$\tau_z^{(1)} = \sigma_{12} (\Delta y \Delta z) \cdot \frac{\Delta x}{2} + \sigma_{12} (\Delta y \Delta z) \cdot \frac{\Delta x}{2} = \sigma_{12} \Delta x \Delta y \Delta z.$$

The corresponding shear σ_{21} (the x -component of the force on the $+y$ -face) exerts a counter-clockwise torque of equal magnitude $\sigma_{21} \Delta x \Delta y \Delta z$. Rotational equilibrium ($\tau_z = 0$) requires:

$$\sigma_{12} = \sigma_{21}. \quad (12)$$

Repeating for the other two planes gives $\sigma_{13} = \sigma_{31}$ and $\sigma_{23} = \sigma_{32}$. Hence:

$$\boxed{\sigma_{ij} = \sigma_{ji}}, \quad (13)$$

i.e. $\boldsymbol{\sigma}$ is a **symmetric** tensor with only **6 independent components**.

Symmetry of σ_{ij} : Landau–Lifshitz argument

An equivalent, more algebraic proof uses the total torque $\mathbf{N}$ on a volume element:

$$N_i = \oint_{\partial V} \varepsilon_{ijk} x_j \sigma_{kl} dS_l = \int_V \varepsilon_{ijk} \partial_l (x_j \sigma_{kl}) dV.$$

Applying the product rule, $\partial_l (x_j \sigma_{kl}) = \delta_{jl} \sigma_{kl} + x_j \partial_l \sigma_{kl}$. The second term is the moment of the body-force density $f_k = \partial_l \sigma_{kl}$, which represents the torque due to net forces on the element and is already accounted for in the force-balance equation. The first term must vanish independently:

$$\varepsilon_{ijk} \delta_{jl} \sigma_{kl} = \varepsilon_{ijk} \sigma_{kj} = 0.$$

For $i = 3$: $\varepsilon_{312} \sigma_{21} + \varepsilon_{321} \sigma_{12} = \sigma_{21} - \sigma_{12} = 0$, i.e. $\sigma_{12} = \sigma_{21}$. The same follows for all pairs, confirming (13).

2.5 Static Equilibrium Equations

The net force on a volume element dV comes from two sources:

1. **Contact (internal) forces** from surrounding material, giving a force per unit volume $\partial_j \sigma_{ij}$ (cf. the divergence theorem applied to Eq. (11)).
2. **Body forces** — for gravity this is ρg_i per unit volume, where ρ is mass density.

Newton's second law for static equilibrium ($\mathbf{a} = 0$) yields:

$$\boxed{\partial_j \sigma_{ij} + \rho g_i = 0}. \quad (14)$$

This is the Cauchy equilibrium equation. It is the continuum analogue of $\sum F = 0$ for a particle. The gradient of stress (divergence of $\boldsymbol{\sigma}$) must balance the body-force density.

Boundary conditions: At a free surface with outward normal $\hat{n}$, the traction must equal the applied surface load t_i :

$$\sigma_{ij} n_j \Big|_{\text{surface}} = t_i.$$

For a truly free surface (no applied load), $t_i = 0$.

Remark: The precise sign convention for Eq. (14) and the boundary conditions will be refined at the start of Lecture 03.

2.6 Brief Note on Tensors

A **tensor of rank n** is a multilinear map that takes n vectors and returns a scalar, represented in a given basis by an array of 3^n components. Familiar examples:

- Rank-0 (scalar): temperature T .
- Rank-1 (vector): displacement u_i , body force g_i .
- Rank-2: stress σ_{ij} , strain ε_{ij} .
- Rank-4: the elastic stiffness tensor C_{ijkl} (to be introduced shortly) relates stress and strain in 3D.

Key Takeaways

- **Stress** $\sigma = F/A$ is force per unit area (units: Pa). It is the quantity that drives deformation, not the bare force.
- **Longitudinal strain** $\varepsilon_{zz} = \Delta L/L$ is the dimensionless relative elongation.
- **Hooke's law** in 1D: $\sigma = E\varepsilon_{zz}$, giving $\Delta L = FL/(EA)$. Linear elastic response holds for $|\varepsilon| \lesssim 10^{-3}$.
- **Young's modulus** E (units: Pa) characterises the stiffness of a material. It spans from ~ 0.01 GPa (rubber) to ~ 1000 GPa (diamond).
- **Poisson's ratio** ν (dimensionless) relates lateral contraction to longitudinal extension: $\varepsilon_{xx} = -\nu\varepsilon_{zz}$. Physically $0 < \nu \leq \frac{1}{2}$ for most materials; thermodynamics requires $-1 < \nu \leq \frac{1}{2}$.
- The **stress tensor** σ_{ij} generalises 1D stress to 3D. It is **symmetric** ($\sigma_{ij} = \sigma_{ji}$) because rotational equilibrium must hold for every volume element, leaving only **6 independent components**.
- **Equilibrium equation**: $\partial_j \sigma_{ij} + \rho g_i = 0$, the continuum version of $\sum F_i = 0$.

3 Lecture 03 — Stress, Strain, and Tensor Decomposition

Overview

This lecture transitions from basic one-dimensional mechanics to the generalized three-dimensional formulation of continuum mechanics. We review uniaxial stress, introduce the concepts of shear stress and shear strain, and formulate the Cauchy stress tensor (σ) and the infinitesimal strain tensor (ϵ). We conclude with the mathematical decomposition of these symmetric tensors into their hydrostatic (volumetric) and deviatoric (shape-changing) components.

Derivations

1. Introduction to Shear Stress and Shear Strain

For a uniaxial rod, the normal stress is $\sigma = F/A$ and the stretch is characterized by Young's modulus E , where $\sigma = E\epsilon$. In contrast, **Shear Stress** (τ) arises when the applied force is parallel to the cross-sectional area.

Consider a block of height H with its base fixed. A horizontal force F applied parallel to its top face (area A) generates an average shear stress:

$$\tau = \frac{F}{A} \quad (15)$$

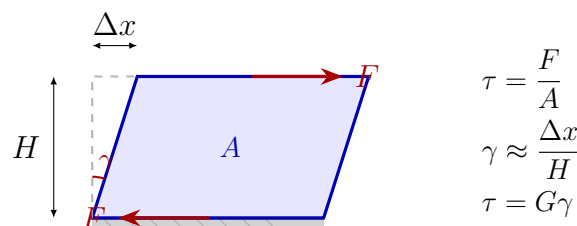
Under this force, the top face displaces by a horizontal distance Δx . The block tilts by a small **engineering shear angle** γ :

$$\gamma \approx \tan \gamma = \frac{\Delta x}{H} \quad (16)$$

For small deformations, the linear elastic relationship is governed by the shear modulus G :

$$\tau = G\gamma \quad (17)$$

Physical Context: The shear modulus quantifies the material's structural rigidity and resistance to shape-distortion under parallel sliding forces.



2. The Cauchy Stress Tensor (σ)

In a 3D continuum, stress is not a single number or vector, but a rank-2 tensor σ (components σ_{ij}) that maps any arbitrary surface normal $\mathbf{n}$ to the traction vector $\mathbf{T}$ (force per unit area) acting upon that surface.

Cauchy's Stress Principle: Consider a thin boundary layer at the surface of the body. To maintain equilibrium at the surface, the internal stresses acting on the area ΔS with outward normal $\mathbf{n}$ must balance the externally applied traction $\mathbf{T}^{\text{ext}}$ applied outside:

$$\begin{aligned} \mathbf{T}^{\text{ext}} \Delta S + \sigma(-\mathbf{n}) \Delta S &= \mathbf{0} \\ \mathbf{T}^{\text{ext}} &= \sigma \mathbf{n} \end{aligned} \quad (18)$$

In indicial notation, this is written as $T_i = \sigma_{ji}n_j$. However, moment equilibrium on a differential element requires that the Cauchy stress tensor is symmetric:

$$\sigma_{ij} = \sigma_{ji} \quad (19)$$

Thus, we commonly write $T_i = \sigma_{ij}n_j$. This symmetry reduces the tensor from 9 to 6 independent components.

Static Equilibrium: Let us write the force per unit volume resulting from a varying stress field. Summing surface forces on a material volume via the divergence theorem and adding the body forces yields:

$$\begin{aligned} \oint_{\partial V} \sigma \mathbf{n} dS + \int_V \mathbf{f} dV &= \mathbf{0} \\ \int_V (\nabla \cdot \sigma + \mathbf{f}) dV &= \mathbf{0} \end{aligned} \quad (20)$$

Since the volume V is arbitrary:

$$\boxed{\nabla \cdot \sigma + \mathbf{f} = \mathbf{0} \quad \text{or} \quad \partial_j \sigma_{ji} + f_i = 0} \quad (21)$$

where $\mathbf{f}$ is the external body force per unit volume (e.g., gravity).

3. The Infinitesimal Strain Tensor (ϵ)

Let a material point initially reside at the Lagrangian (material) coordinate $\mathbf{X}$. After deformation, its Eulerian (spatial) coordinate is $\mathbf{x}$. The displacement field is:

$$\mathbf{u}(\mathbf{X}) = \mathbf{x} - \mathbf{X} \quad (22)$$

To measure strain objectively, we evaluate the change in the square of the distance between two infinitesimally close material points, originally separated by $d\mathbf{X}$:

$$\begin{aligned} d\mathbf{x} &= d\mathbf{X} + d\mathbf{u} = d\mathbf{X} + (\nabla_{\mathbf{X}} \mathbf{u}) d\mathbf{X} \\ dx_i &= dX_i + \frac{\partial u_i}{\partial X_j} dX_j \end{aligned} \quad (23)$$

The new squared distance is calculated by taking the dot product $dx_i dx_i$:

$$\begin{aligned} dx_i dx_i &= (dX_i + \partial_j u_i dX_j) (dX_i + \partial_k u_i dX_k) \\ &= dX_i dX_i + \partial_j u_i dX_j dX_i + \partial_k u_i dX_i dX_k + \partial_j u_i \partial_k u_i dX_j dX_k \\ &= dX_i dX_i + (\partial_j u_i + \partial_i u_j + \partial_k u_i \partial_k u_j) dX_i dX_j \end{aligned} \quad (24)$$

We assume a **small deformation limit** ($|\nabla \mathbf{u}| \ll 1$) and neglect the nonlinear quadratic term $\partial_k u_i \partial_k u_j$. The change in squared length is then characterized by the infinitesimal strain tensor ϵ :

$$\boxed{\epsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial X_j} + \frac{\partial u_j}{\partial X_i} \right)} \quad (25)$$

Physical Context: Similar to the stress tensor, ϵ is a symmetric rank-2 tensor. The diagonal elements (ϵ_{11} , etc.) are the relative normal elongations. The off-diagonal elements represent *half* of the engineering shear angle: $\epsilon_{ij} = \frac{1}{2} \gamma_{ij}$ ($i \neq j$).

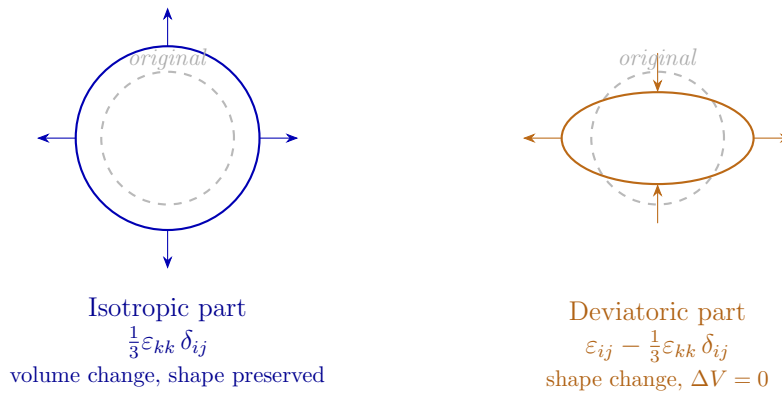
4. Isotropic / Deviatoric Tensor Decomposition

Every rank-2 symmetric tensor can be decomposed into an isotropic (volumetric) component and a deviatoric (shape-distorting) component. We demonstrate this for the strain tensor ϵ , but a parallel breakdown applies to the stress tensor σ .

$$\epsilon_{ij} = \underbrace{\frac{1}{3}\epsilon_{kk}\delta_{ij}}_{\text{Isotropic (Hydrostatic)}} + \underbrace{\left(\epsilon_{ij} - \frac{1}{3}\epsilon_{kk}\delta_{ij}\right)}_{\text{Deviatoric}} \quad (26)$$

Physical Context:

- **Isotropic Part:** Proportional to the trace $\epsilon_{kk} = \nabla \cdot \mathbf{u}$, it describes a homogeneous volumetric expansion or compression across all directions without altering the geometric proportions of the element.
- **Deviatoric Part:** A traceless tensor containing all shear and differential stretching effects, capturing pure, isochoric (constant-volume) geometric distortion.



Connections

- **Constitutive Laws / Mechanics:** The connection between the stress tensor σ and the strain tensor ϵ is established via a constitutive relationship (e.g., generalized Hooke's Law for linear elastic solids), allowing us to close the system of variables conceptually.
- **Linear Algebra:** Being real symmetric matrices, both stress and strain tensors are guaranteed to have real eigenvalues and orthogonal eigenvectors. Their principal eigen-axes define the frames where the state is purely diagonal (free of shear components).

Exam / HW Hints

- **Factor of 2 in Shear Strain:** To avoid errors, rigorously distinguish the engineering shear strain γ and the physical tensor shear strain $\epsilon_{ij} = \frac{1}{2}\gamma$. Check whether given formulas expect γ or ϵ_{ij} .
- **Symmetries in Static Equilibrium:** Ensure symmetry in the continuum stress tensor ($\sigma_{ij} = \sigma_{ji}$). Remember this to quickly relate tangential surface stress conditions (e.g., $\sigma_{xy} = \sigma_{yx}$).
- **Pure Shear Conditions:** In a coordinate system perfectly oriented to the shear force, the block experiences only deviatoric stress. Under these axes, diagonal terms are zero ($\sigma_{11} = \sigma_{22} = \sigma_{33} = 0$).

4 Lecture 04 — Tensor Algebra, Coordinate Systems, and 1D Constitutive Laws

Overview

In this lecture, we expand upon the mathematical foundations of the stress tensor (σ) and the infinitesimal strain tensor (ϵ). We explore tensor diagonalization, isotropic/deviatoric decomposition, and introduce new tensor algebraic operations (dyadic and symmetric tensor products). These mathematical tools are then applied to derive the strain tensor in polar coordinates. Finally, we bridge the gap between abstract tensors and physical reality by introducing the phenomenological 1D stress-strain curve and Hooke's Law, representing our first constitutive relation.

Review: Properties of Symmetric Tensors

Both the stress tensor σ and the strain tensor ϵ are second-rank symmetric tensors ($\sigma_{ij} = \sigma_{ji}$). A fundamental property of symmetric tensors is that they can always be diagonalized by choosing an appropriate orthogonal coordinate system.

- **Principal Axes:** The coordinate axes in which the tensor is diagonal.
- **Principal Values:** The eigenvalues of the tensor (e.g., principal stresses $\sigma_1, \sigma_2, \sigma_3$ or principal strains $\epsilon_1, \epsilon_2, \epsilon_3$).
- **Invariants:** The Trace ($\text{Tr}(\sigma)$) and the Determinant ($\text{Det}(\sigma)$) remain invariant under coordinate transformations.

Note on Tensor Fields: While a tensor field $\sigma(\mathbf{x})$ can be diagonalized at any specific point $\mathbf{x}$, it generally cannot be diagonalized globally across the entire domain using a single Cartesian coordinate system, because the principal directions may vary spatially.

Example 1: Diagonalizing a Pure Shear State

Consider a 2D pure shear stress state. In the standard Cartesian basis, the stress tensor is:

$$\sigma = \begin{pmatrix} 0 & \sigma_0 \\ \sigma_0 & 0 \end{pmatrix} \quad (27)$$

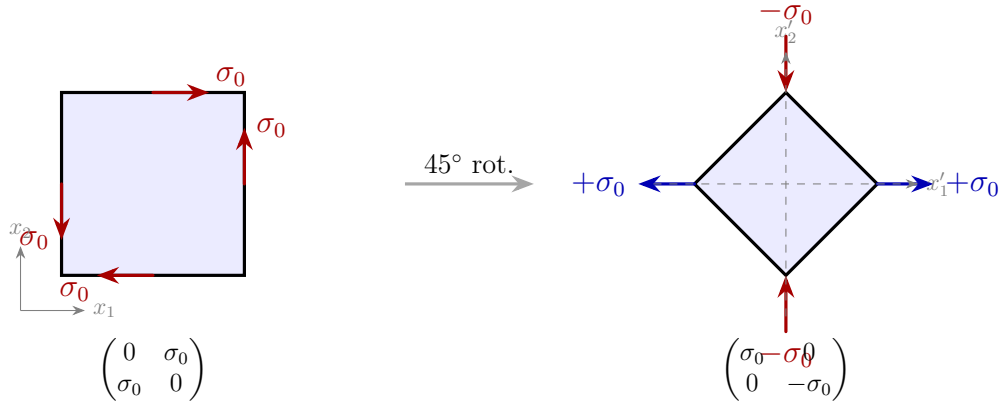
To find the principal stresses, we can calculate the eigenvalues. Several methods exist:

1. **Pauli Matrices Analogy:** Recognizing this as proportional to the Pauli matrix σ_x , the eigenvalues are immediately known to be $\pm\sigma_0$.
2. **Invariants Method:** We know $\text{Tr}(\sigma) = \lambda_1 + \lambda_2 = 0$ and $\text{Det}(\sigma) = \lambda_1\lambda_2 = -\sigma_0^2$. Solving this system yields $\lambda_{1,2} = \pm\sigma_0$.

In the principal basis (rotated by 45°), the tensor becomes:

$$\sigma' = \begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix} \quad (28)$$

Physical Context: A state of pure shear on one set of planes is physically equivalent to a state of pure tension (σ_0) and pure compression ($-\sigma_0$) on planes rotated by 45° .



Example 2: Isotropic and Deviatoric Decomposition

Any second-rank tensor can be decomposed into an isotropic part (a multiple of the identity tensor) and a deviatoric (traceless) part. In the principal axis system:

$$\begin{pmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{pmatrix} = \bar{\sigma} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} \sigma_1 - \bar{\sigma} & 0 & 0 \\ 0 & \sigma_2 - \bar{\sigma} & 0 \\ 0 & 0 & \sigma_3 - \bar{\sigma} \end{pmatrix} \quad (29)$$

where the mean hydrostatic stress is $\bar{\sigma} = \frac{1}{3}\text{Tr}(\sigma) = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3)$. The second term on the RHS is the **deviatoric stress tensor**, σ' . By definition, $\text{Tr}(\sigma') = (\sigma_1 - \bar{\sigma}) + (\sigma_2 - \bar{\sigma}) + (\sigma_3 - \bar{\sigma}) = \sigma_1 + \sigma_2 + \sigma_3 - 3\bar{\sigma} = 0$.

New Tensor Algebra Operations

To generalize our formulation of continuum mechanics, we define new algebraic operations.

1. Tensor Product (Dyadic/Outer Product)

Given two vectors $\mathbf{a}$ and $\mathbf{b}$, the tensor product $\mathbf{a} \otimes \mathbf{b}$ produces a second-rank tensor (matrix).

$$(\mathbf{a} \otimes \mathbf{b})_{ij} = a_i b_j \quad (30)$$

Properties: It is multilinear (linear in both $\mathbf{a}$ and $\mathbf{b}$) but **non-commutative** ($\mathbf{a} \otimes \mathbf{b} \neq \mathbf{b} \otimes \mathbf{a}$).

2. Symmetric Tensor Product

We define the symmetric tensor product of two vectors, denoted by $\odot$, as:

$$\mathbf{a} \odot \mathbf{b} = \frac{1}{2}(\mathbf{a} \otimes \mathbf{b} + \mathbf{b} \otimes \mathbf{a}) \quad (31)$$

In index notation:

$$(\mathbf{a} \odot \mathbf{b})_{ij} = \frac{1}{2}(a_i b_j + b_i a_j) \quad (32)$$

Physical Context: We can now express the infinitesimal strain tensor elegantly without indices. Recalling that $\epsilon_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$, using the Del operator ∇ , we can write:

$$\boldsymbol{\epsilon} = \nabla \odot \mathbf{u} \quad (33)$$

3. Tensor-Vector Contraction (Dot Product)

Given a second-rank tensor σ and a vector $\mathbf{v}$, their inner product (contraction) produces a new vector $\mathbf{u}$:

$$\mathbf{u} = \sigma \cdot \mathbf{v} \implies u_i = \sigma_{ij} v_j \quad (34)$$

For symmetric tensors, left-multiplication and right-multiplication yield the same result.

Derivations: Strain Tensor in Polar Coordinates

We wish to calculate the strain tensor $\boldsymbol{\epsilon} = \nabla \odot \mathbf{u}$ in 2D polar coordinates (r, ϕ) . Let the displacement vector be $\mathbf{u} = u_r \hat{\mathbf{r}} + u_\phi \hat{\boldsymbol{\phi}}$, and the gradient operator be $\nabla = \hat{\mathbf{r}} \frac{\partial}{\partial r} + \hat{\boldsymbol{\phi}} \frac{1}{r} \frac{\partial}{\partial \phi}$.

Physical Context: When operating with ∇ in curvilinear coordinates, the derivatives must act not only on the vector components (u_r, u_ϕ) but also on the basis vectors $(\hat{\mathbf{r}}, \hat{\boldsymbol{\phi}})$, as they change direction in space. The derivatives of the unit vectors are:

$$\frac{\partial \hat{\mathbf{r}}}{\partial \phi} = \hat{\boldsymbol{\phi}}, \quad \frac{\partial \hat{\boldsymbol{\phi}}}{\partial \phi} = -\hat{\mathbf{r}}, \quad \frac{\partial \hat{\mathbf{r}}}{\partial r} = \mathbf{0}, \quad \frac{\partial \hat{\boldsymbol{\phi}}}{\partial r} = \mathbf{0} \quad (35)$$

First, we compute the full tensor gradient $\nabla \otimes \mathbf{u}$:

$$\begin{aligned} \nabla \otimes \mathbf{u} &= \left(\hat{\mathbf{r}} \frac{\partial}{\partial r} + \hat{\boldsymbol{\phi}} \frac{1}{r} \frac{\partial}{\partial \phi} \right) \otimes (u_r \hat{\mathbf{r}} + u_\phi \hat{\boldsymbol{\phi}}) \\ &= \hat{\mathbf{r}} \otimes \frac{\partial \mathbf{u}}{\partial r} + \frac{1}{r} \hat{\boldsymbol{\phi}} \otimes \frac{\partial \mathbf{u}}{\partial \phi} \end{aligned} \quad (36)$$

Evaluating the partial derivatives using the product rule:

$$\frac{\partial \mathbf{u}}{\partial r} = \frac{\partial u_r}{\partial r} \hat{\mathbf{r}} + \frac{\partial u_\phi}{\partial r} \hat{\boldsymbol{\phi}} \quad (37)$$

$$\begin{aligned} \frac{\partial \mathbf{u}}{\partial \phi} &= \frac{\partial u_r}{\partial \phi} \hat{\mathbf{r}} + u_r \underbrace{\frac{\partial \hat{\mathbf{r}}}{\partial \phi}}_{=\hat{\boldsymbol{\phi}}} + \frac{\partial u_\phi}{\partial \phi} \hat{\boldsymbol{\phi}} + u_\phi \underbrace{\frac{\partial \hat{\boldsymbol{\phi}}}{\partial \phi}}_{=-\hat{\mathbf{r}}} \\ &= \left(\frac{\partial u_r}{\partial \phi} - u_\phi \right) \hat{\mathbf{r}} + \left(\frac{\partial u_\phi}{\partial \phi} + u_r \right) \hat{\boldsymbol{\phi}} \end{aligned} \quad (38)$$

Assembling the tensor product:

$$\nabla \otimes \mathbf{u} = \frac{\partial u_r}{\partial r} (\hat{\mathbf{r}} \otimes \hat{\mathbf{r}}) + \frac{\partial u_\phi}{\partial r} (\hat{\mathbf{r}} \otimes \hat{\boldsymbol{\phi}}) + \frac{1}{r} \left(\frac{\partial u_r}{\partial \phi} - u_\phi \right) (\hat{\boldsymbol{\phi}} \otimes \hat{\mathbf{r}}) + \frac{1}{r} \left(\frac{\partial u_\phi}{\partial \phi} + u_r \right) (\hat{\boldsymbol{\phi}} \otimes \hat{\boldsymbol{\phi}}) \quad (39)$$

Symmetrizing to find $\boldsymbol{\epsilon} = \frac{1}{2}(\nabla \otimes \mathbf{u} + (\nabla \otimes \mathbf{u})^T)$, we extract the components:

$$\begin{aligned} \epsilon_{rr} &= \frac{\partial u_r}{\partial r} \\ \epsilon_{\phi\phi} &= \frac{1}{r} \frac{\partial u_\phi}{\partial \phi} + \frac{u_r}{r} \\ \epsilon_{r\phi} = \epsilon_{\phi r} &= \frac{1}{2} \left(\frac{1}{r} \frac{\partial u_r}{\partial \phi} + \frac{\partial u_\phi}{\partial r} - \frac{u_\phi}{r} \right) \end{aligned} \quad (40)$$

Note: The terms $\frac{u_r}{r}$ and $-\frac{u_\phi}{r}$ arise entirely from the rotation of the local basis vectors, a hallmark of curvilinear continuum mechanics.

The 1D Constitutive Law: Stress-Strain Curve

Up to this point, our treatment of stress and strain has been purely geometrical and based on equilibrium. To close the system of equations, we need a relationship that links deformation (ϵ) to the applied internal forces (σ). This relationship depends strictly on the material and is called a **Constitutive Equation**.

For a standard 1D tensile test (e.g., pulling a steel rod), the phenomenological mapping between normal stress σ and normal strain ϵ yields the **Stress-Strain Curve**. Key regions include:

1. **Proportional Limit (A):** The region where σ scales linearly with ϵ .

$$\boxed{\sigma = E\epsilon} \quad (41)$$

This is **Hooke's Law**, where the constant of proportionality E is *Young's Modulus*. For steel, this regime is valid up to ~ 200 MPa, corresponding to a tiny strain of roughly 0.1%.

2. **Elastic Limit:** For metals like steel, this generally coincides with the proportional limit. If unloaded from here, the material returns exactly to its original shape (reversible deformation).
3. **Yield Point (B - *Kniya*):** The point beyond which the material undergoes significant plastic (irreversible) deformation with very little increase in applied stress.
4. **Necking and Fracture (C):** The cross-sectional area decreases rapidly (necking), eventually leading to structural failure (rupture). Note that strains here are 10 – 15 times larger than in the elastic regime.

Engineering Working Stress: In engineering design, components must operate strictly within the linear elastic regime to prevent permanent deformation.

$$\sigma_{\text{working}} = \frac{\sigma_{\text{yield}}}{\text{S.F.}} \quad (42)$$

where S.F. is a *Safety Factor* (e.g., 1.6), accounting for unpredictable environmental loads, material imperfections, or modelling simplifications.

Material Typology:

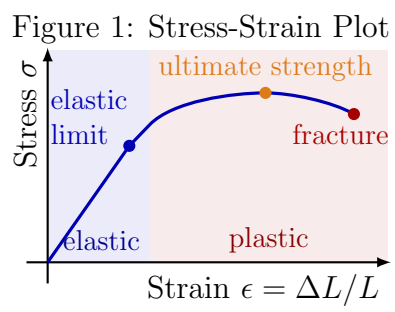
- **Ductile Materials (e.g., Steel, Aluminum):** Exhibit a long plastic deformation region before fracture, providing physical warning before failure.
- **Brittle Materials (e.g., Cast Iron, Ceramics, Glass):** Fail suddenly with almost no plastic deformation.

Connections

- **Linear Algebra:** Finding principal stresses is mathematically identical to finding the eigenvalues of an operator. The trace and determinant properties directly translate from standard matrix invariants.
- **Engineering Mechanics:** Cauchy's stress principle, heavily reliant on the tensor-vector contraction ($\mathbf{t} = \sigma \cdot \hat{\mathbf{n}}$), forms the boundary conditions for practically all structural simulations.

Exam / HW Hints

- When calculating strain in curvilinear coordinates (cylindrical or spherical), **do not** simply take partial derivatives of the components. You must account for the spatial derivatives of the basis vectors ($\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}$).
- Remember the definitions of invariants: if a problem asks you to verify if two stress states are rotations of one another, simply check if their Traces and Determinants are equal.



5 Lecture 05 — Hooke's Law and Linear Isotropic Elasticity

Overview

This lecture has two parts. First, we complete the geometric interpretation of the strain tensor in cylindrical coordinates, illustrating the origin of the hoop-strain term $\varepsilon_{\varphi\varphi} = u_r/r$ and connecting it to differential geometry. Second, we derive Hooke's law for a linear isotropic elastic solid. Using an **effective-field-theory** argument, symmetry alone determines the most general linear relation between σ_{ij} and ε_{ij} , leaving exactly two independent material constants $\tilde{K}$ and $\tilde{G}$. We define the shear modulus $G = \tilde{G}/2$ and bulk modulus $K = \tilde{K}/3$, invert the constitutive relation, and establish their connection to Young's modulus E and Poisson's ratio ν through three worked examples.

Part I — Completion: Strain tensor in cylindrical coordinates

Cylindrically symmetric displacement

Consider a cylindrically symmetric displacement field that depends only on the radial distance r :

$$\vec{u} = u_r(r) \hat{e}_r, \quad u_\varphi = 0.$$

No azimuthal component exists and there is no φ -dependence. The two non-trivial diagonal strain components are:

$$\varepsilon_{rr} = \frac{\partial u_r}{\partial r}, \quad (43)$$

$$\varepsilon_{\varphi\varphi} = \frac{u_r}{r}. \quad (44)$$

The first term is the expected radial (longitudinal) strain. The second term, which contains no derivative, requires geometric explanation.

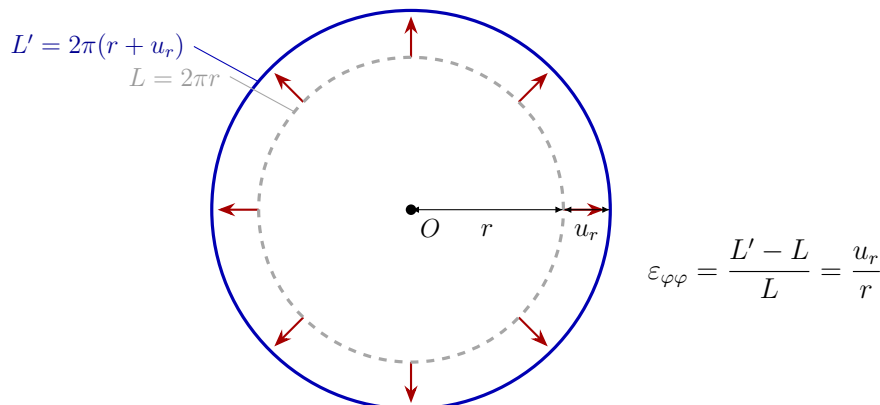
Geometric interpretation of $\varepsilon_{\varphi\varphi} = u_r/r$

- A cylindrical shell at radius r has original circumference $L = 2\pi r$.
- After a radial displacement $u_r(r)$, the new radius is $r + u_r$ and the new circumference is $L' = 2\pi(r + u_r)$.
- The azimuthal (hoop) strain is therefore

$$\varepsilon_{\varphi\varphi} = \frac{L' - L}{L} = \frac{2\pi u_r}{2\pi r} = \frac{u_r}{r},$$

even though u_r is independent of φ .

This is entirely analogous to $\varepsilon_{rr} = \partial_r u_r = \Delta\ell/\ell$ for a radial segment: both express the fractional change in a material line length. The difference is that the hoop length changes because the *radius* of the circle changes, not because u_r varies along the circle.



Remark 1 (Connection to differential geometry). The extra term in (44) arises because the unit vectors $\hat{e}_r$ and $\hat{e}_\varphi$ depend on φ in cylindrical coordinates. When differentiating a vector field in curvilinear coordinates, derivatives of the basis vectors produce additional terms. The coefficients of these corrections are called **Christoffel symbols** (or **connection coefficients**), and they are the foundational objects of **differential geometry** — the mathematics of curved spaces. Although our physical space is flat, the use of curvilinear coordinates introduces the same structure.

Part II — Constitutive relations: Hooke’s law

The problem: connecting σ_{ij} and ε_{ij}

We have established two tensors:

- The **strain tensor** $\varepsilon_{ij} = \frac{1}{2}(\partial_j u_i + \partial_i u_j)$, encoding local deformation.
- The **stress tensor** σ_{ij} , giving surface forces via $dF_i = \sigma_{ij} dS_j$.

The missing link is the material-dependent **constitutive relation** $\sigma_{ij} = \sigma_{ij}(\varepsilon_{kl})$.

Why linear elasticity?

We focus on the linear elastic regime for three reasons:

1. **Universality (Taylor expansion):** For any stress-free reference configuration, $\sigma_{ij}|_{u=0} = 0$. Taylor expansion then forces the leading correction to be linear in the strain.
2. **Physical validity:** Every material is approximately linear for sufficiently small strains; non-linearity and yielding only appear beyond a material-dependent threshold.
3. **Engineering practice:** Structures (bridges, buildings) are designed to remain in the elastic regime. Reaching the yield point constitutes failure.

Effective-field-theory derivation of Hooke’s law

The strategy of **effective field theory** is: *what is not forbidden by symmetry is compulsory*. We enumerate all symmetries that constrain $\sigma_{ij}(\varepsilon_{kl})$ and read off the most general permitted form.

Step 1 — Stress-free reference state

Assumption: the body is stress-free in its rest configuration:

$$\sigma_{ij}|_{u=0} = 0.$$

Taylor-expanding σ_{ij} in powers of u , the constant term vanishes, so σ_{ij} is at least first order in u .

Step 2 — Uniform displacement generates no stress

For a rigid translation $\vec{u} = \vec{c}$ (constant, independent of position), the body is undistorted and $\sigma_{ij} = 0$. Therefore σ_{ij} cannot depend on u_i itself, only on its spatial derivatives:

$$\sigma_{ij} \sim \partial_k u_l.$$

Higher-derivative terms ($\partial^2 u$, etc.) are suppressed when spatial variation is slow and also lead to higher-order equations of motion; we keep only the leading term.

Step 3 — Rigid-body rotation generates no stress

The displacement gradient decomposes into symmetric and antisymmetric parts:

$$\partial_j u_i = \varepsilon_{ij} + \omega_{ij},$$

where $\varepsilon_{ij} = \frac{1}{2}(\partial_j u_i + \partial_i u_j)$ is the strain tensor and $\omega_{ij} = \frac{1}{2}(\partial_j u_i - \partial_i u_j)$ describes infinitesimal rigid rotation. A uniform rigid rotation

$$u_i = \varepsilon_{ijk} \theta_j x_k \implies \varepsilon_{ij} = 0, \omega_{ij} \neq 0$$

produces no deformation and hence no stress. Therefore σ_{ij} depends only on ε_{ij} , not on ω_{ij} .

Remark 2. This conclusion is also forced by the symmetry of the stress tensor: $\sigma_{ij} = \sigma_{ji}$. In a linear theory, a symmetric tensor can only depend linearly on another symmetric tensor. Adding a term proportional to the antisymmetric ω_{ij} would break the symmetry of σ_{ij} .

Step 4 — General linear relation (stiffness tensor)

The most general *linear* map from the symmetric tensor ε_{kl} to the symmetric tensor σ_{ij} is

$$\sigma_{ij} = C_{ijkl} \varepsilon_{kl}, \quad (45)$$

where C_{ijkl} is the **rank-4 stiffness tensor**. With the symmetry $C_{ijkl} = C_{jikl} = C_{ijlk}$ and the minor symmetry $C_{ijkl} = C_{klij}$, it has at most 21 independent components for a general anisotropic crystal.

Step 5 — Isotropy reduces to two constants

For an **isotropic** material (macroscopically identical in all directions), C_{ijkl} must itself be an isotropic rank-4 tensor. The most general such tensor is a linear combination of the three isotropic rank-4 tensors:

$$\delta_{ij}\delta_{kl}, \quad \delta_{ik}\delta_{jl}, \quad \delta_{il}\delta_{jk}.$$

Imposing symmetry $C_{ijkl} = C_{jikl} = C_{ijlk}$ reduces these to two independent combinations, giving:

$$C_{ijkl} = \frac{\tilde{K} - \tilde{G}}{3} \delta_{ij}\delta_{kl} + \tilde{G} (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}).$$

Substituting into (45) and using $\delta_{ik}\delta_{jl} \varepsilon_{kl} = \varepsilon_{ij}$:

$$\sigma_{ij} = \frac{\tilde{K}}{3} \varepsilon_{kk} \delta_{ij} + \tilde{G} \left(\varepsilon_{ij} - \frac{1}{3} \varepsilon_{kk} \delta_{ij} \right) \quad (46)$$

Every isotropic linear elastic material is fully characterised by the two constants $\tilde{K}$ and $\tilde{G}$.

- The first term is proportional to $\varepsilon_{kk} = \text{Tr}(\varepsilon)$, the **volumetric** (isotropic) part of the strain.
- The second term involves the **deviatoric** (traceless) part $\varepsilon_{ij} - \frac{1}{3}\varepsilon_{kk}\delta_{ij}$, which describes shape change at constant volume.

Elastic moduli: shear modulus G and bulk modulus K

Two standard moduli are defined from $\tilde{K}$ and $\tilde{G}$:

$$G \equiv \frac{\tilde{G}}{2} \quad (\text{shear modulus}), \quad (47)$$

$$K \equiv \frac{\tilde{K}}{3} \quad (\text{bulk modulus}). \quad (48)$$

In terms of K and G , Hooke's law (46) reads:

$$\sigma_{ij} = K \varepsilon_{kk} \delta_{ij} + 2G \left(\varepsilon_{ij} - \frac{1}{3} \varepsilon_{kk} \delta_{ij} \right). \quad (49)$$

Both K and G have dimensions of pressure (Pa).

Inverse relation: strain from stress

Taking the trace of (46):

$$\sigma_{kk} = \tilde{K} \varepsilon_{kk} \implies \varepsilon_{kk} = \frac{\sigma_{kk}}{\tilde{K}} = \frac{\sigma_{kk}}{3K}.$$

Substituting back and solving for ε_{ij} yields the inverse Hooke's law:

$$\varepsilon_{ij} = \frac{1}{3\tilde{K}} \sigma_{kk} \delta_{ij} + \frac{1}{\tilde{G}} \left(\sigma_{ij} - \frac{1}{3} \sigma_{kk} \delta_{ij} \right) \quad (50)$$

or equivalently in terms of K and G :

$$\varepsilon_{ij} = \frac{1}{9K} \sigma_{kk} \delta_{ij} + \frac{1}{2G} \left(\sigma_{ij} - \frac{1}{3} \sigma_{kk} \delta_{ij} \right). \quad (51)$$

The motivation for defining $G \equiv \tilde{G}/2$ and $K \equiv \tilde{K}/3$ is precisely that inversion maps $\tilde{K} \mapsto 1/\tilde{K}$ and $\tilde{G} \mapsto 1/\tilde{G}$ cleanly.

Example 2: Uniaxial tension — connecting (K, G) to (E, ν)

Setup

A rod of cross-sectional area A is pulled with force F along the z -axis. The boundary conditions on the free lateral surfaces impose zero traction there, so the stress tensor is diagonal with a single nonzero entry:

$$\sigma_{ij} = \sigma_0 \delta_{iz} \delta_{jz}, \quad \sigma_0 = \frac{F}{A}, \quad \text{Tr}(\sigma) = \sigma_0.$$

Strain from the inverse relation

Using (50) component by component:

$$\varepsilon_{zz} = \frac{\sigma_0}{3\tilde{K}} + \frac{1}{\tilde{G}} \left(\sigma_0 - \frac{\sigma_0}{3} \right) = \frac{\sigma_0}{3\tilde{K}} + \frac{2\sigma_0}{3\tilde{G}}, \quad (52)$$

$$\varepsilon_{xx} = \varepsilon_{yy} = \frac{\sigma_0}{3\tilde{K}} + \frac{1}{\tilde{G}} \left(0 - \frac{\sigma_0}{3} \right) = \frac{\sigma_0}{3\tilde{K}} - \frac{\sigma_0}{3\tilde{G}}. \quad (53)$$

Identification of E and ν

Comparing (52) with the definition $\varepsilon_{zz} = \sigma_0/E$, and (53) with $\varepsilon_{xx} = -\nu \varepsilon_{zz}$:

$$\frac{1}{E} = \frac{1}{3\tilde{K}} + \frac{2}{3\tilde{G}} = \frac{1}{9K} + \frac{1}{3G}, \quad (54)$$

$$\frac{\nu}{E} = \frac{1}{3\tilde{G}} - \frac{1}{3\tilde{K}} = \frac{1}{6G} - \frac{1}{9K}. \quad (55)$$

Inverting (54)–(55) gives the practical formulas:

$$K = \frac{E}{3(1-2\nu)}, \quad G = \frac{E}{2(1+\nu)}. \quad (56)$$

Remark 3 (Stability and the range of ν). Physical stability requires $K > 0$ and $G > 0$ (the elastic energy must be positive definite). Inserting into (56), these conditions translate to

$$-1 < \nu < \frac{1}{2}.$$

At $\nu \rightarrow \frac{1}{2}$, $K \rightarrow \infty$ (incompressible material, e.g. rubber). At $\nu \rightarrow -1$, $G \rightarrow 0$. The poles of K and G as functions of ν make the range $(-1, \frac{1}{2})$ physically inevitable.

Example 3: Simple shear — justifying the name “shear modulus”

Apply a shear force F tangentially to the top face of a block, generating a stress:

$$\sigma_{12} = \sigma_{21} = \sigma_0, \quad \text{all other } \sigma_{ij} = 0.$$

Since $\text{Tr}(\sigma) = 0$, the inverse relation (50) gives:

$$\varepsilon_{12} = \frac{\sigma_0}{\tilde{G}} = \frac{\sigma_0}{2G}.$$

Recalling that the engineering shear strain is $\gamma = 2\varepsilon_{12}$ (twice the off-diagonal entry, corresponding to the total angle of deformation):

$$G = \frac{\sigma_0}{\gamma}.$$

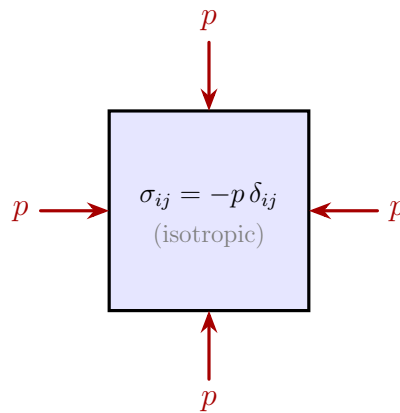
This is precisely the definition of the shear modulus as the ratio of shear stress to shear angle. The factor of 2 between $\tilde{G}$ and G originates in the difference between the physical (tensor) convention and the engineering (Voigt) convention for shear strains.

Example 4: Hydrostatic compression — justifying the name “bulk modulus”

Setup

A solid elastic body is immersed in a fluid at pressure p . Every surface element $d\vec{S}$ experiences a force $d\vec{F} = -p d\vec{S}$ directed inward. The resulting stress tensor is **isotropic**:

$$\sigma_{ij} = -p \delta_{ij}, \quad \text{Tr}(\sigma) = -3p.$$



Strain

The deviatoric part vanishes identically:

$$\sigma_{ij} - \frac{1}{3} \sigma_{kk} \delta_{ij} = -p \delta_{ij} - \frac{-3p}{3} \delta_{ij} = 0.$$

From the inverse relation (50):

$$\varepsilon_{ij} = \frac{1}{3\tilde{K}} \cdot (-3p) \cdot \delta_{ij} = -\frac{p}{\tilde{K}} \delta_{ij} = -\frac{p}{3K} \delta_{ij}. \quad (57)$$

The strain is isotropic (equal compression in every direction), as expected from the symmetry of the loading.

Volume change and definition of K

General result: $\Delta V/V_0 = \varepsilon_{kk}$. Consider an infinitesimal box with edge lengths L_x, L_y, L_z . After deformation, the edges become $L_\alpha(1 + \varepsilon_{\alpha\alpha})$. Expanding the new volume and discarding quadratic and cubic terms (valid for small strains):

$$\frac{\Delta V}{V_0} = (1 + \varepsilon_{xx})(1 + \varepsilon_{yy})(1 + \varepsilon_{zz}) - 1 \approx \varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz} = \varepsilon_{kk}.$$

The trace of the strain tensor is the fractional volume change. This is a general result, not specific to the hydrostatic case.

Application to hydrostatic compression. From (57), $\varepsilon_{kk} = -3p/(3K) = -p/K$, so:

$$\frac{\Delta V}{V_0} = -\frac{p}{K} \implies K = -\frac{p}{\Delta V/V_0}.$$

The bulk modulus K is the ratio of applied pressure to fractional volume decrease. A stiffer material (larger K) undergoes less compression under the same pressure. This is the direct physical meaning of the name *bulk modulus*.

Summary of elastic moduli and their relations

Symbol	Name	Physical meaning
E	Young's modulus	Axial stiffness in uniaxial tension: $\sigma_{zz} = E \varepsilon_{zz}$
ν	Poisson's ratio	Transverse contraction: $\varepsilon_{xx} = -\nu \varepsilon_{zz}$
G	Shear modulus	Resistance to shape change: $\sigma_{12} = G \gamma_{12}$
K	Bulk modulus	Resistance to volume change: $K = -p V_0 / \Delta V$

Any two of $\{E, \nu, G, K\}$ determine all others. The key inter-relations are:

$$K = \frac{E}{3(1-2\nu)}, \quad G = \frac{E}{2(1+\nu)}, \quad E = \frac{9KG}{3K+G}, \quad \nu = \frac{3K-2G}{2(3K+G)}. \quad (58)$$

Stability ($K > 0, G > 0$) constrains $\nu \in (-1, \frac{1}{2})$.

Homework: Constrained uniaxial strain (oedometric loading)

A steel rod of cross-section A is compressed along the z -axis by a force F , but is **prevented from deforming in the transverse directions**:

$$\varepsilon_{xx} = \varepsilon_{yy} = 0.$$

Find the effective **constrained modulus** M defined by $\sigma_{zz} = M \varepsilon_{zz}$, expressed in terms of E and ν . Note that $M \neq E$ because the lateral constraint develops non-zero transverse stresses $\sigma_{xx} = \sigma_{yy} \neq 0$, even though the corresponding strains vanish.

6 Lecture 06 — Equations of Equilibrium and the Lamé–Navier Equation

Overview

This lecture has two parts. First, we consolidate the full set of field equations governing a linear isotropic elastic solid: the constitutive relation (Hooke’s law), its inverse, and the catalogue of elastic moduli including the **constrained modulus** M and the **Lamé parameters** λ and μ . Second, we derive the **Lamé–Navier equation** — the equation of motion (or equilibrium) for the displacement field $\vec{u}$ — by substituting the constitutive relation and the strain–displacement relation into Newton’s law, expressed through the stress divergence. The result is a second-order vector PDE that is the elastic analogue of Laplace’s equation in electrostatics. To build physical intuition we solve in full detail the problem of a **heavy rod hanging under its own weight**: finding the stress profile $\sigma_{zz}(z)$, the strain $\varepsilon_{zz}(z)$, and the displacement field $u_z(z)$, including a qualitative discussion of the lateral (Poisson) contraction.

Part I — Complete Summary of Linear Isotropic Elasticity

The constitutive relation and its inverse

The central result of the previous lecture was Hooke’s law in two equivalent forms. In terms of the bulk modulus K and shear modulus G :

$$\sigma_{ij} = K \varepsilon_{kk} \delta_{ij} + 2G \left(\varepsilon_{ij} - \frac{1}{3} \varepsilon_{kk} \delta_{ij} \right), \quad (59)$$

and its inverse:

$$\varepsilon_{ij} = \frac{1}{9K} \sigma_{kk} \delta_{ij} + \frac{1}{2G} \left(\sigma_{ij} - \frac{1}{3} \sigma_{kk} \delta_{ij} \right). \quad (60)$$

In terms of Young’s modulus E and Poisson’s ratio ν — the two moduli most commonly tabulated — the inverse relation reads:

$$\varepsilon_{ij} = \frac{1}{E} [(1 + \nu) \sigma_{ij} - \nu \sigma_{kk} \delta_{ij}]. \quad (61)$$

These equations are called **local (microscopic) Hooke’s law** to distinguish them from the integral relation $F = k\Delta\ell$ for a spring (macroscopic Hooke’s law). The analogy mirrors that of Ohm’s law: the local form $\vec{J} = \sigma_{\text{el}} \vec{E}$ (with resistivity as the material property) versus the integral form $V = IR$ for a whole resistor. The local relation holds at every point inside the material; integrating it over the geometry of the body recovers the macroscopic relation.

Catalogue of elastic moduli

Any two independent moduli suffice to characterise an isotropic linear elastic material. Several choices are in common use, each convenient for different classes of problems.

Symbol	Name	Key relation
E	Young’s modulus	$\sigma_{zz} = E \varepsilon_{zz}$ (uniaxial tension)
ν	Poisson’s ratio	$\varepsilon_{xx} = -\nu \varepsilon_{zz}$ (uniaxial tension)
$G = \frac{E}{2(1 + \nu)}$	Shear modulus	$\sigma_{12} = G \gamma_{12}$
$K = \frac{E}{3(1 - 2\nu)}$	Bulk modulus	$K = -p V_0 / \Delta V$
$M = \frac{E(1 - \nu)}{(1 + \nu)(1 - 2\nu)}$	Constrained (oedometric) modulus	$\sigma_{zz} = M \varepsilon_{zz}, \varepsilon_{xx} = \varepsilon_{yy} = 0$
$\lambda = \frac{E\nu}{(1 + \nu)(1 - 2\nu)}$	First Lamé parameter	see below
$\mu = G$	Second Lamé parameter (= G)	see below

Constrained modulus M . When a body is compressed along z but is prevented from deforming laterally ($\varepsilon_{xx} = \varepsilon_{yy} = 0$), the constraint requires non-zero lateral stresses $\sigma_{xx} = \sigma_{yy} \neq 0$. The effective stiffness is therefore larger than E :

$$M = \frac{E(1 - \nu)}{(1 + \nu)(1 - 2\nu)} = K + \frac{4G}{3}. \quad (62)$$

Note that $M > E$ for all physically admissible $\nu \in (-1, \frac{1}{2})$. At $\nu = 0$, $M = E$, and $M \rightarrow \infty$ as $\nu \rightarrow \frac{1}{2}$ (incompressible limit).

Lamé parameters λ and μ . A second useful pair of independent constants is (λ, μ) , introduced by Gabriel Lamé in the 19th century. They are defined by:

$$\lambda = K - \frac{2}{3}G = \frac{E\nu}{(1 + \nu)(1 - 2\nu)}, \quad (63)$$

$$\mu = G = \frac{E}{2(1 + \nu)}. \quad (64)$$

In terms of λ and μ , Hooke's law (59) takes the compact form:

$$\boxed{\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij}}. \quad (65)$$

This form is elegant because it separates cleanly into a term proportional to ε_{ij} (with coefficient 2μ) and a term proportional to the identity (with coefficient $\lambda \text{Tr}(\varepsilon)$). The parameter λ is sometimes called the **first Lamé coefficient**; μ is the **second Lamé coefficient** (identical to the shear modulus G). Each parameterisation $\{E, \nu\}$, $\{K, G\}$, $\{\lambda, \mu\}$, $\{M, G\}$ contains exactly two independent constants; the choice depends on which makes a given problem most tractable.

Remark 4 (Poisson's ratio and incompressibility). At $\nu = \frac{1}{2}$, the bulk modulus $K \rightarrow \infty$ (the material resists volume change infinitely strongly) and $\lambda \rightarrow \infty$, while G remains finite. This is the **incompressible limit** (e.g. rubber). At $\nu = 0$ there is no lateral strain in uniaxial tension and $\lambda = 0$. The physical stability conditions $K > 0$, $G > 0$ confine ν to the interval $(-1, \frac{1}{2})$.

Part II — The Lamé–Navier Equation of Equilibrium

The missing equation: equilibrium of forces

So far we have the constitutive relation (65) and the strain–displacement relation:

$$\varepsilon_{ij} = \frac{1}{2}(\partial_j u_i + \partial_i u_j). \quad (66)$$

The unknown field is the **displacement field** $u_i(\vec{x})$. Newton's law (equilibrium condition) for an elastic body with body force g_i per unit mass and mass density ρ reads:

$$\partial_j \sigma_{ij} + \rho g_i = 0, \quad (67)$$

where ρg_i is the body force per unit volume. In most applications, g_i is simply gravitational acceleration: $\rho g_i = \rho g \hat{z}$.

Substituting the constitutive and kinematic relations

The goal is to express (67) entirely in terms of u_i . Starting from (65):

$$\partial_j \sigma_{ij} = \lambda \partial_i (\partial_k u_k) + \mu \partial_j (\partial_j u_i + \partial_i u_j),$$

where the material parameters λ and μ are taken to be spatially uniform (homogeneous material), so they factor out of the derivatives. Using the symmetry $\partial_j \partial_i u_j = \partial_i (\partial_j u_j)$:

$$\partial_j \sigma_{ij} = (\lambda + \mu) \partial_i (\partial_k u_k) + \mu \partial_j \partial_j u_i.$$

In vector notation, $\partial_j \partial_j u_i = \nabla^2 u_i$ is the i -th component of the vector Laplacian $\nabla^2 \vec{u}$, and $\partial_i (\partial_k u_k) = \nabla_i (\nabla \cdot \vec{u})$ is the i -th component of $\vec{\nabla}(\vec{\nabla} \cdot \vec{u})$. Substituting into (67):

$$(\lambda + \mu) \vec{\nabla}(\vec{\nabla} \cdot \vec{u}) + \mu \nabla^2 \vec{u} + \rho \vec{g} = 0. \quad (68)$$

This is the **Lamé–Navier equation**. It is a second-order linear PDE for the vector field $\vec{u}(\vec{x})$ — the elastic analogue of Laplace’s equation $\nabla^2 \phi = -\rho_{\text{el}}/\varepsilon_0$ in electrostatics.

Alternative form via the curl-of-curl identity. Using the vector identity

$$\vec{\nabla}(\vec{\nabla} \cdot \vec{u}) = \nabla^2 \vec{u} + \vec{\nabla} \times (\vec{\nabla} \times \vec{u}), \quad (69)$$

one can rewrite (68) as:

$$(\lambda + 2\mu) \vec{\nabla}(\vec{\nabla} \cdot \vec{u}) - \mu \vec{\nabla} \times (\vec{\nabla} \times \vec{u}) + \rho \vec{g} = 0. \quad (70)$$

This form is sometimes more convenient: when the divergence term $\vec{\nabla} \cdot \vec{u}$ vanishes (incompressible deformation), only the curl term survives; when the curl term vanishes (irrotational deformation), only the divergence term survives. Both forms are equivalent; the choice depends on the symmetry of the problem.

Remark 5 (Physical content). The Lamé–Navier equation says that the net elastic restoring force at each point — a combination of resistance to volume change (coefficient $\lambda + 2\mu = M$) and resistance to shape change (coefficient $\mu = G$) — exactly balances the applied body force. The two types of elastic response correspond physically to the two independent wave speeds in a solid (compressional and shear), which will be revisited when we study elastic waves.

Summary of the complete set of field equations

The full system governing a linear isotropic elastic body in static equilibrium is:

$$\varepsilon_{ij} = \frac{1}{2}(\partial_j u_i + \partial_i u_j) \quad (\text{kinematics}), \quad (71)$$

$$\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij} \quad (\text{constitutive}), \quad (72)$$

$$\partial_j \sigma_{ij} + \rho g_i = 0 \quad (\text{equilibrium}), \quad (73)$$

with unknown u_i and the Lamé–Navier equation (68) as their combination. Boundary conditions prescribe either displacements (Dirichlet) or surface tractions $\sigma_{ij} n_j$ (Neumann) on the boundary of the body.

Worked Problem: Rod Hanging Under Its Own Weight

Problem statement

A uniform elastic rod of length L , cross-sectional area A (circular, radius R , so $A = \pi R^2$), mass density ρ , Young’s modulus E , and Poisson’s ratio ν hangs vertically from the ceiling. Its upper end is fixed to the ceiling at $z = 0$; the free lower end is at $z = L$. The z -axis points *downward*. Atmospheric pressure is neglected. Find:

- (a) The stress profile $\sigma_{zz}(z)$ and applicable boundary conditions.
- (b) The axial strain $\varepsilon_{zz}(z)$.
- (c) The displacement field $u_z(z)$ and the total elongation ΔL .
- (d) A qualitative description of the lateral (radial) deformation.

Physical picture and symmetry argument

Before computing, we identify the key physics:

- The rod is in **tension** throughout: gravity pulls every cross-section downward, and the material above must pull upward to maintain equilibrium. Using the sign convention that tension is positive, $\sigma_{zz} > 0$ everywhere.
- The tension is **largest at the top**: the topmost cross-section must support the weight of the entire rod below it; the bottom free surface supports nothing.
- The axial strain ε_{zz} is proportional to σ_{zz} (via E), so it also **varies linearly** with z , from a maximum at $z = 0$ to zero at $z = L$.
- The displacement $u_z(z)$ is the accumulated stretch from the fixed point $z = 0$ downward; it starts at zero (the ceiling attachment) and increases toward the bottom, with a **parabolic** profile because the strain (its derivative) is linear.

Step (a): Stress profile $\sigma_{zz}(z)$

Symmetry reduction. By the rotational and translational symmetry of the problem (homogeneous rod, gravity in z), all fields depend only on z . The only non-trivial displacement is $u_z(z)$; the lateral displacements are $u_x = u_y = 0$ to leading order (see Step (d)). The only non-trivial stress component needed for equilibrium is $\sigma_{zz}(z)$.

Equilibrium equation. The z -component of (73) reduces to an ordinary differential equation:

$$\frac{d\sigma_{zz}}{dz} + \rho g = 0, \quad (74)$$

where $g > 0$ is the magnitude of gravitational acceleration (the z -axis points downward, so the body force per unit volume is $+\rho g$ in the z -direction).

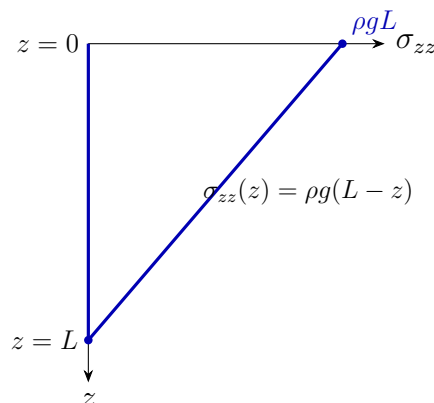
Boundary condition. The lower face at $z = L$ is a **free surface**: no external traction acts on it, so the normal stress must vanish:

$$\sigma_{zz}(L) = 0. \quad (75)$$

Solution. Integrating (74) with condition (75):

$$\sigma_{zz}(z) = \rho g (L - z). \quad (76)$$

The stress is linear in z : it vanishes at the free bottom ($z = L$) and reaches its maximum $\sigma_{zz}^{\max} = \rho g L$ at the fixed top ($z = 0$), confirming the physical picture.



Step (b): Axial strain $\varepsilon_{zz}(z)$

From Hooke's law (uniaxial tension, lateral faces free so $\sigma_{xx} = \sigma_{yy} = 0$):

$$\varepsilon_{zz}(z) = \frac{\sigma_{zz}(z)}{E} = \frac{\rho g}{E}(L - z). \quad (77)$$

The strain is also linear in z , vanishing at the free end and maximal at the fixed end.

Step (c): Displacement field $u_z(z)$ and total elongation

Kinematic relation. The displacement $u_z(z)$ is related to the strain by:

$$\varepsilon_{zz}(z) = \frac{du_z}{dz}. \quad (78)$$

This is a first-order ODE in u_z .

Boundary condition. The rod is attached to the ceiling at $z = 0$, so:

$$u_z(0) = 0. \quad (79)$$

Solution. Integrating (78) with ε_{zz} from (77) and applying (79):

$$u_z(z) = \int_0^z \varepsilon_{zz}(z') dz' = \frac{\rho g}{E} \int_0^z (L - z') dz' = \frac{\rho g}{E} \left(Lz - \frac{z^2}{2} \right). \quad (80)$$

Equivalently:

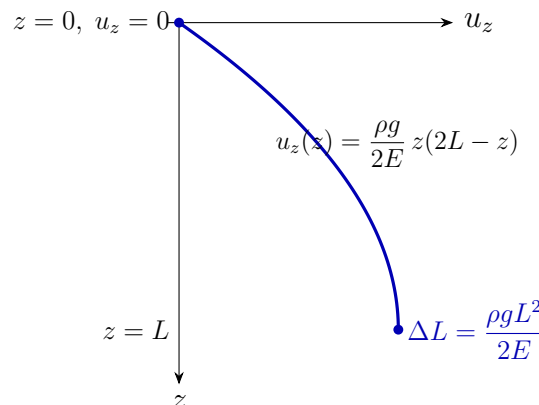
$$u_z(z) = \frac{\rho g}{2E} z(2L - z). \quad (81)$$

The displacement is a **downward-opening parabola** in z (the coefficient of z^2 is negative): it starts at $u_z(0) = 0$ (fixed ceiling), increases monotonically, and reaches its maximum at the free end $z = L$.

Total elongation. The total elongation is the displacement of the free bottom end:

$$\boxed{\Delta L = u_z(L) = \frac{\rho g L^2}{2E}}. \quad (82)$$

Remark 6 (Comparison with uniform tension). If the same rod were pulled by a uniform external force $F = \rho g L \cdot A$ (equal to its own weight) applied at *both* ends, the stress would be uniform $\sigma_{zz} = \rho g L$ throughout, giving elongation $\Delta L_{\text{uniform}} = \rho g L^2 / E$. The hanging rod gives exactly *half* this value. Physically, the stress in the hanging rod varies from $\rho g L$ at the top to 0 at the bottom — its average is $\frac{1}{2}\rho g L$ — so the total elongation is half that of the uniformly stressed rod. This factor of $\frac{1}{2}$ is the area under the triangular stress profile, and is confirmed by the exact integral.



Step (d): Lateral deformation (qualitative)

By Poisson's effect, the axial tension $\varepsilon_{zz} > 0$ induces lateral contraction:

$$\varepsilon_{xx}(z) = \varepsilon_{yy}(z) = -\nu \varepsilon_{zz}(z) = -\frac{\nu \rho g}{E}(L - z). \quad (83)$$

Since ε_{zz} varies with z , so does the lateral strain. The rod contracts most strongly near the top (where tension is greatest) and not at all at the free bottom end. Consequently:

- Near the **free bottom** ($z = L$): $\varepsilon_{xx} = 0$, no lateral contraction.
- Near the **fixed top** ($z = 0$): maximum lateral contraction $\varepsilon_{xx} = -\nu \rho g L / E$.
- The rod is therefore **slightly barrel-shaped** (narrower at the top, wider at the bottom) — but the attachment constraint at the ceiling complicates this near $z = 0$.

A full treatment of the lateral displacement $u_x(x, y, z)$ would require solving the complete three-dimensional Lamé–Navier equation (68), which introduces coupling between radial and axial fields. In the present one-dimensional approximation (rod slender, $R \ll L$), the leading lateral displacement is:

$$u_r(r, z) \approx -\nu \frac{\rho g}{E}(L - z)r,$$

which describes a uniform radial contraction of the cross-section at each height z , proportional to the local axial strain.

Summary: Strategy for Solving Elasticity Problems

The worked example illustrates a general procedure for problems where all fields depend on a single coordinate:

1. **Identify the symmetry and the relevant field.** Determine which components of σ_{ij} , ε_{ij} , and u_i are non-zero and on which coordinates they depend.
2. **Write and solve the equilibrium equation for σ .** The divergence condition $\partial_j \sigma_{ij} + \rho g_i = 0$ becomes an ODE; apply traction boundary conditions (zero stress on free surfaces).
3. **Find ε from σ via the inverse Hooke's law.** Use (60) or (61) component by component.
4. **Integrate $\varepsilon = \partial u$ to find u .** Apply displacement boundary conditions (zero displacement at fixed supports).
5. **Compute derived quantities.** Total elongation, stress concentrations, lateral strains, etc.

When the problem has full three-dimensional variation, steps 2–4 merge into solving the Lamé–Navier equation (68) directly for $\vec{u}$, subject to mixed boundary conditions. The one-dimensional problems serve as the essential training ground for developing physical intuition before tackling the full PDE.

7 Lecture 07 — Rotating Cylinder and Elastic Waves in Solids

Overview

This lecture has two main goals. First, we use the field equations of linear isotropic elasticity to solve a concrete two-dimensional problem from Landau–Lifshitz: the deformation of a uniformly rotating cylinder whose length is constrained. The centrifugal body force stretches the cylinder radially, while the free outer boundary fixes the integration constant through the condition of zero radial traction. Second, we add time dependence to the Lamé–Navier equation and show that an elastic solid supports two kinds of sound waves: **longitudinal waves**, in which the displacement is parallel to the direction of propagation, and **transverse waves**, in which the displacement is perpendicular to it. Their different speeds are controlled by different elastic moduli.

How to Read This Lecture

The safest way to follow the calculation is to keep asking three questions.

1. **What is the displacement field allowed to be?** Symmetry is the main simplifier. In the cylinder problem it reduces the vector field $\vec{u}$ to one scalar function $u(r)$. In the wave problem it reduces the discussion to the relation between $\vec{k}$ and the polarization $\vec{u}_0$.
2. **What force is being balanced?** In the rotating cylinder the elastic force balances the centrifugal body force. In the wave problem the elastic force produces acceleration.
3. **Which elastic modulus appears, and why?** Compression-like motion uses $M = \lambda + 2\mu$; shear-like motion uses $\mu = G$. This is the conceptual thread connecting the static cylinder problem to sound waves in solids.

The algebra below is written so that every integration constant is tied to a physical condition: regularity at the axis, zero traction on a free surface, or the polarization constraint of a plane wave.

Part I — Recap of the Elasticity Field Equations

Physical Motivation

Before solving a new problem, it is useful to identify the three ingredients of the theory. The displacement field describes what the material points do, the strain tensor extracts the local deformation from the displacement, and the stress tensor gives the internal force response. Equilibrium, or later dynamics, then tells us how the stress field balances the applied body forces.

The fundamental unknown is the displacement field

$$\vec{u}(\vec{x}) = u_i(\vec{x}) \hat{e}_i,$$

where u_i is the displacement of the material point initially located at $\vec{x}$. For small deformations the strain tensor is the symmetric derivative:

$$\varepsilon_{ij} = \frac{1}{2} (\partial_i u_j + \partial_j u_i). \quad (84)$$

For a homogeneous, isotropic, linear elastic material, Hooke's law can be written in Lamé form:

$$\boxed{\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij}.} \quad (85)$$

Physical significance: the coefficient λ controls the part of the stress coupled to volume change, while $\mu = G$ is the shear modulus controlling shape change.

For use as a problem-solving checklist, read (85) as follows. First compute the strain from the displacement. Then split the response into a trace part, ε_{kk} , and a component-by-component part, ε_{ij} . The trace is the part that knows about local volume change; the remaining information is what distinguishes stretching in one direction from stretching in another.

The static force-balance equation is

$$\partial_j \sigma_{ij} + \rho f_i = 0, \quad (86)$$

where ρf_i is the body force per unit volume. Substituting (84) and (85) gives the Lamé–Navier equation:

$$(\lambda + \mu) \vec{\nabla}(\vec{\nabla} \cdot \vec{u}) + \mu \nabla^2 \vec{u} + \rho \vec{f} = 0. \quad (87)$$

Physical significance: the elastic restoring force density is the sum of a compressional response and a shear response, and in equilibrium it cancels the external body force density.

The divergence of the displacement is especially important:

$$\vec{\nabla} \cdot \vec{u} = \partial_i u_i = \varepsilon_{ii} = \text{Tr } \varepsilon \simeq \frac{\Delta V}{V}. \quad (88)$$

Thus $\vec{\nabla} \cdot \vec{u}$ is the relative volume change to first order in the displacement gradients.

Using

$$\vec{\nabla}(\vec{\nabla} \cdot \vec{u}) = \nabla^2 \vec{u} + \vec{\nabla} \times (\vec{\nabla} \times \vec{u}), \quad (89)$$

the Lamé–Navier equation can also be written as

$$M \vec{\nabla}(\vec{\nabla} \cdot \vec{u}) - \mu \vec{\nabla} \times (\vec{\nabla} \times \vec{u}) + \rho \vec{f} = 0, \quad M = \lambda + 2\mu. \quad (90)$$

Physical significance: this form separates the irrotational, volume-changing part of the deformation from the rotational, shear-like part. The modulus M is the constrained modulus introduced in Lecture 06.

Part II — Rotating Cylinder with Constrained Length

Problem statement

Consider a homogeneous elastic cylinder of radius R , density ρ , Young's modulus E , and Poisson ratio ν . The cylinder rotates uniformly around its symmetry axis with angular velocity ω . Its length is constrained, so there is no axial deformation:

$$u_z = 0, \quad \varepsilon_{zz} = 0.$$

We seek the radial displacement $u(r)$ caused by the centrifugal force.

Physical Motivation

In the rotating frame each material element feels an outward centrifugal body force proportional to its distance from the axis. The center cannot choose a preferred direction, so the radial displacement must vanish there. The outer cylindrical surface is free: it may move, but no external pressure or support applies a radial traction. Therefore the problem is determined by regularity at $r = 0$ and zero radial stress at $r = R$.

Symmetry reduction

By cylindrical symmetry the displacement field has the form

$$\vec{u}(r) = u(r) \hat{r}. \quad (91)$$

It has no ϕ or z dependence, and it has no azimuthal or axial component. This field is irrotational:

$$\vec{\nabla} \times \vec{u} = 0. \quad (92)$$

The centrifugal body force per unit volume is

$$\rho \vec{f} = \rho \omega^2 r \hat{r}. \quad (93)$$

Using (90), the equilibrium equation becomes

$$M \vec{\nabla}(\vec{\nabla} \cdot \vec{u}) + \rho \omega^2 r \hat{r} = 0. \quad (94)$$

Boundary conditions

The free outer boundary has no applied traction:

$$\sigma_{ij} n_j = 0 \quad \text{at } r = R. \quad (95)$$

Since $n_j = \hat{r}_j$, and by symmetry the shear traction vanishes, the relevant scalar condition is

$$\sigma_{rr}(R) = 0. \quad (96)$$

At the axis the displacement must be regular:

$$u(0) = 0 \quad \text{and } u(r) \text{ finite as } r \rightarrow 0. \quad (97)$$

These two boundary conditions play different roles. The condition at $r = 0$ is not an applied constraint; it only rules out unphysical singular solutions. The condition at $r = R$ is a real mechanical statement: the surface is free, so the material cannot transmit a normal force to something outside it.

Physical Motivation

The force equation is second order in $u(r)$, but the centrifugal force is the gradient of a scalar potential proportional to r^2 . Because the displacement is radial and curl-free, the full vector equation collapses to the gradient of a scalar expression. This gives a first integral before we solve the remaining first-order equation.

First integral

For the radial field (91),

$$\vec{\nabla} \cdot \vec{u} = \frac{1}{r} \frac{d}{dr} (ru(r)) = u'(r) + \frac{u(r)}{r}. \quad (98)$$

Also,

$$\rho \omega^2 r \hat{r} = \vec{\nabla} \left(\frac{1}{2} \rho \omega^2 r^2 \right).$$

Therefore (94) is

$$\vec{\nabla} \left[M \left(u'(r) + \frac{u(r)}{r} \right) + \frac{1}{2} \rho \omega^2 r^2 \right] = 0. \quad (99)$$

The bracket must be constant; call it A :

$$M \left(u'(r) + \frac{u(r)}{r} \right) + \frac{1}{2} \rho \omega^2 r^2 = A. \quad (100)$$

Solving for the displacement

Equation (100) gives

$$u'(r) + \frac{u(r)}{r} = \frac{A}{M} - \frac{\rho\omega^2}{2M}r^2.$$

Multiplying by r ,

$$\frac{d}{dr}(ru(r)) = \frac{A}{M}r - \frac{\rho\omega^2}{2M}r^3.$$

Integrating,

$$ru(r) = \frac{A}{2M}r^2 - \frac{\rho\omega^2}{8M}r^4 + B, \quad (101)$$

and therefore

$$u(r) = \frac{A}{2M}r - \frac{\rho\omega^2}{8M}r^3 + \frac{B}{r}. \quad (102)$$

Regularity at the axis requires $B = 0$.

It remains to determine A from $\sigma_{rr}(R) = 0$. With $\varepsilon_{zz} = 0$,

$$\varepsilon_{rr} = u'(r), \quad \varepsilon_{\phi\phi} = \frac{u(r)}{r}, \quad \varepsilon_{kk} = u'(r) + \frac{u(r)}{r}.$$

Hooke's law gives

$$\sigma_{rr} = \lambda \left(u'(r) + \frac{u(r)}{r} \right) + 2\mu u'(r). \quad (103)$$

Using (102) with $B = 0$,

$$u'(r) = \frac{A}{2M} - \frac{3\rho\omega^2}{8M}r^2, \quad u'(r) + \frac{u(r)}{r} = \frac{A}{M} - \frac{\rho\omega^2}{2M}r^2.$$

Thus

$$\sigma_{rr}(R) = \frac{\lambda + \mu}{M}A - \frac{\rho\omega^2 R^2}{M} \left(\frac{\lambda}{2} + \frac{3\mu}{4} \right).$$

The condition $\sigma_{rr}(R) = 0$ fixes

$$A = \frac{3 - 2\nu}{4} \rho\omega^2 R^2. \quad (104)$$

Substituting into (102),

$$\boxed{u(r) = \frac{\rho\omega^2}{8M} [(3 - 2\nu)R^2r - r^3]}. \quad (105)$$

Physical significance: every circular material ring moves outward. The displacement is zero at the axis by symmetry and increases toward the free outer boundary.

The radial displacement of the outer surface is

$$\boxed{u(R) = \frac{\rho\omega^2 R^3}{4M}(1 - \nu)}. \quad (106)$$

Physical significance: the total radial expansion grows with density, angular speed squared, and radius cubed, and is suppressed by the constrained modulus M .

Notice the hierarchy of information in this answer. The dependence $u(R) \propto \omega^2$ comes from the centrifugal acceleration, the dependence $u(R) \propto R^3$ comes from integrating a body force that grows like r , and the factor $1/M$ says that a stiffer material deforms less under the same rotation.

Checks on the result

Dimensions. The factor $\rho\omega^2 R^3/M$ has dimensions

$$\frac{(\text{mass/length}^3)(1/\text{time}^2)\text{length}^3}{\text{force/length}^2} = \text{length},$$

so both (105) and (106) have the correct units.

Sign. For physical materials $-1 < \nu < 1/2$. Since $0 \leq r \leq R$,

$$(3 - 2\nu)R^2 r - r^3 = r [(3 - 2\nu)R^2 - r^2] \geq r [(2 - 2\nu)R^2] \geq 0.$$

Thus $u(r) \geq 0$: the cylinder expands outward, as expected from the centrifugal force.

Free boundary. The constant A was chosen so that $\sigma_{rr}(R) = 0$. The outer surface is therefore not held fixed; it is free to move, but no external radial stress acts on it.

Part III — Elastic Waves and Sound in Solids

Physical Motivation

Equilibrium problems answer what shape a solid takes after forces settle down. To describe sound, vibrations, and seismic waves, we must allow the displacement field to depend on time. Each material element now has acceleration, so the divergence of the stress is no longer zero: it supplies mass times acceleration per unit volume.

Without external body force, Newton's law per unit volume is

$$\partial_j \sigma_{ij} = \rho \ddot{u}_i. \quad (107)$$

Using the same Hooke law and kinematics as before,

$$\boxed{\rho \ddot{\vec{u}} = (\lambda + \mu) \vec{\nabla}(\vec{\nabla} \cdot \vec{u}) + \mu \nabla^2 \vec{u}.} \quad (108)$$

Physical significance: this is the wave equation of an elastic solid. Its vector nature is the new ingredient: the direction of displacement can be either parallel or perpendicular to the direction of propagation.

Using (89), the same equation can be written as

$$\rho \ddot{\vec{u}} = M \vec{\nabla}(\vec{\nabla} \cdot \vec{u}) - \mu \vec{\nabla} \times (\vec{\nabla} \times \vec{u}). \quad (109)$$

Physical Motivation

The material is homogeneous, so the coefficients in the wave equation are constant. As in every linear constant-coefficient wave problem, plane waves diagonalize the equation. The new question is how the polarization vector $\vec{u}_0$ is oriented relative to the wave vector $\vec{k}$.

Take a monochromatic plane wave

$$\vec{u}(\vec{x}, t) = \vec{u}_0 e^{i(\vec{k} \cdot \vec{x} - \omega t)}, \quad (110)$$

where $\vec{k}$ is the wave vector, ω is the angular frequency, and $\vec{u}_0$ is the constant polarization vector. For this ansatz,

$$\vec{\nabla} \rightarrow i\vec{k}, \quad \partial_t^2 \rightarrow -\omega^2.$$

Longitudinal waves

For a longitudinal wave,

$$\vec{u}_0 \parallel \vec{k}. \quad (111)$$

Then $\vec{k} \times \vec{u}_0 = 0$, so the curl term in (109) vanishes. Substitution gives

$$-\rho\omega^2\vec{u} = -Mk^2\vec{u}.$$

Therefore

$$\boxed{\omega^2 = c_L^2 k^2, \quad c_L^2 = \frac{M}{\rho} = \frac{\lambda + 2\mu}{\rho}.} \quad (112)$$

Physical significance: longitudinal waves compress and expand the material as they pass, so their speed is set by the constrained compressional stiffness M divided by the density.

Transverse waves

For a transverse wave,

$$\vec{u}_0 \perp \vec{k}. \quad (113)$$

Then $\vec{k} \cdot \vec{u}_0 = 0$, so $\vec{\nabla} \cdot \vec{u} = 0$ and the compressional term in (108) vanishes. Substitution gives

$$-\rho\omega^2\vec{u} = -\mu k^2\vec{u}.$$

Thus

$$\boxed{\omega^2 = c_T^2 k^2, \quad c_T^2 = \frac{G}{\rho} = \frac{\mu}{\rho}.} \quad (114)$$

Physical significance: transverse waves shear the material without changing its volume to first order, so their speed is controlled only by the shear modulus.

There are three independent polarizations for a given propagation direction in an isotropic solid: one longitudinal polarization and two transverse polarizations. A general small disturbance can be decomposed into a superposition of plane waves with these polarizations.

Comparison with fluids and physical interpretation

Sound in air or water is a longitudinal pressure-density wave. Fluids have no static shear modulus, so they do not support bulk transverse sound waves. Solids do have a shear modulus, and therefore support both longitudinal and transverse elastic waves. This is the same distinction used in seismology: the faster compressional wave arrives first, followed by the slower shear wave.

For ordinary materials with positive Poisson ratio, $M > G$, so

$$c_L > c_T. \quad (115)$$

The lecture emphasized the spring analogy:

$$\omega^2 \sim \frac{\text{stiffness}}{\text{mass}}, \quad c^2 \sim \frac{\text{elastic modulus}}{\text{mass density}}.$$

The modulus in the numerator depends on the kind of deformation: M for longitudinal compression, G for transverse shear.

Part IV — Numerical Example: Sound Speeds in Steel

Physical Motivation

The formulas for c_L and c_T are not only formal. They let us estimate real sound speeds from tabulated material data. Steel is much denser than air, but it is enormously stiffer, and the stiffness wins: sound travels much faster in steel than in air.

Use representative steel values

$$E \simeq 200 \text{ GPa}, \quad \nu \simeq 0.28, \quad \rho \simeq 7.8 \times 10^3 \text{ kg/m}^3. \quad (116)$$

The shear modulus is

$$G = \mu = \frac{E}{2(1 + \nu)} \simeq \frac{200 \text{ GPa}}{2(1.28)} \simeq 78 \text{ GPa}. \quad (117)$$

The transverse speed is therefore

$$c_T = \sqrt{\frac{G}{\rho}} \simeq \sqrt{\frac{78 \times 10^9}{7.8 \times 10^3}} \simeq 3.2 \text{ km/s}. \quad (118)$$

Physical significance: shear waves in steel travel several kilometers per second because steel has a large shear stiffness compared with its mass density.

The constrained modulus is

$$M = \frac{E(1 - \nu)}{(1 + \nu)(1 - 2\nu)} \simeq \frac{200 \text{ GPa} \cdot 0.72}{1.28 \cdot 0.44} \simeq 256 \text{ GPa}. \quad (119)$$

Thus

$$c_L = \sqrt{\frac{M}{\rho}} \simeq \sqrt{\frac{256 \times 10^9}{7.8 \times 10^3}} \simeq 5.7 \text{ km/s}. \quad (120)$$

Physical significance: longitudinal waves travel faster than transverse waves because compression in a constrained solid is stiffer than shear.

Measured sound speeds in steel depend on the exact alloy and elastic constants; values around 3.2 km/s for transverse waves and about 5.9 km/s for longitudinal waves are consistent with the order-of-magnitude estimate above. The small discrepancy is not a new physical correction in this model, but mostly a reminder that “steel” is not a single material with one exact pair of elastic constants.

Summary

The rotating-cylinder problem illustrates the standard elasticity workflow: identify symmetry, reduce the displacement field, write the force-balance equation, impose a free-traction boundary condition, and check the result physically. The wave problem shows what changes when time dependence is added: the same elastic restoring force now produces acceleration, and because $\vec{u}$ is a vector, the solid supports both longitudinal and transverse sound waves with different speeds.

Further Reading

The following references are useful for deepening the material in this lecture.

- L. D. Landau, E. M. Lifshitz, A. M. Kosevich, and L. P. Pitaevskii, *Theory of Elasticity, Course of Theoretical Physics, Vol. 7*. This is the closest textbook match to the lecture style and includes both the rotating-cylinder problem set and elastic waves. URL: [Elsevier book page](#).

- **Martin H. Sadd, *Elasticity: Theory, Applications, and Numerics*.** Good for a more engineering-oriented and pedagogical development of linear elasticity, curvilinear coordinates, and boundary-value problems. URL: [Elsevier latest-edition page](#).
- **B. A. Auld, *Acoustic Fields and Waves in Solids*.** A detailed reference once the elastic-wave part becomes the main topic, especially for polarization, wave modes, and acoustic propagation in solids. URL: [Google Books entry](#).
- **MIT OpenCourseWare 12.005, *Applications of Continuum Mechanics to Earth, Atmospheric, and Planetary Sciences*.** Useful free lecture notes for stress, strain, elasticity, tensors, and geophysical applications of continuum mechanics. URL: [MIT OCW elasticity/moduli note](#).
- **MIT OpenCourseWare 6.641, Lecture 16: *Elastic Waves on a Thin Rod*.** A focused free note on deriving and interpreting elastic waves in a simple one-dimensional setting before moving to full vector waves in solids. URL: [MIT OCW elastic-waves note](#).

8 Lecture 08 — Torsion and Bending of Beams

Overview

This lecture begins the study of **rods and beams** — continuum bodies whose length is much larger than any transverse dimension. Two canonical problems are treated in full. First, we analyze **torsion** of a straight circular rod: a torque T is applied about the long axis, every cross-section rotates by an angle proportional to its position along the axis, and the resulting shear stress grows linearly with radial distance. Integrating the torque from those stresses yields the *torsional stiffness* $C = GJ$, where $J = \pi R^4/2$ is the polar second moment of area. Second, we analyze **bending**: a transverse load forces the beam to curve, the fibers above the neutral axis stretch while those below compress, and the bending stiffness turns out to be EI_y , where $I_y = \int x^2 dA$ is the second moment of area about the bending axis. Both calculations follow the same three-step workflow: write the displacement ansatz, extract the strain and stress, then integrate to find the resultant moment.

Part I — Geometry and Classification of Rods

What is a rod?

Physical intuition: Continuum bodies come in families distinguished by their geometry. An ordinary solid has all three dimensions comparable. A *plate* has two large dimensions and one small one (thickness). A *rod* (or beam) has one large dimension — the length L — and two small transverse dimensions. The small transverse scale is characterized by the cross-sectional radius or half-width a . The rod approximation is valid when

$$a \ll L.$$

Common cross-sectional shapes are the full circle, the rectangle, the hollow circle, the hollow rectangle, and the I-beam. The I-shape is popular in construction because it concentrates material far from the neutral axis, maximizing bending stiffness per unit weight.

Types of structural elements and loading

The name given to a rod depends on how it is loaded:

- **Strut or column** — loaded axially (tension or compression along the long axis). A strut in compression is sometimes called a *column* (vertical) or simply a *compression member*.
- **Beam** — loaded transversely (the external force is perpendicular to the long axis). The weight the beam supports, or its self-weight, causes it to bend.

This lecture focuses on two types of deformation special to rod geometry: **torsion** (twisting about the long axis) and **bending** (curving in a transverse plane).

Part II — Torsion of a Circular Rod

Problem statement

Consider a straight elastic rod of length L and circular cross-section of radius R . The rod is fixed at one end and a torque T is applied about the symmetry axis z at the other end. We want to find:

1. The internal stress distribution.
2. The torsional stiffness $C \equiv T/\kappa$, where $\kappa = d\theta/dz$ is the rate of twist.

Physical intuition and the rate of twist

Physical intuition: By linearity, the total twist angle $\Delta\theta$ between two ends must be proportional to the applied torque T . A free-body cut isolating any segment of the rod reveals that the same torque T acts on both faces of every cross-section — the torque is transmitted uniformly, just as an axial force is transmitted uniformly in a stretched rod. Therefore each infinitesimal slice of length dz twists by the same angle $d\theta$, and the total twist grows linearly with length:

$$\theta(z) = \kappa z, \quad \kappa \equiv \frac{d\theta}{dz} = \text{const.}$$

The quantity κ is the *rate of twist* (angle per unit length), analogous to angular velocity but along a spatial axis rather than in time.

Displacement ansatz

We work in cylindrical coordinates (r, φ, z) . Each cross-section at height z undergoes a rigid rotation by angle $\theta(z) = \kappa z$. A material point at (r, φ, z) is displaced tangentially by $r\theta(z) = \kappa r z$, with no radial or axial movement:

$$u_r = 0, \quad u_\varphi = \kappa r z, \quad u_z = 0. \quad (121)$$

Strain calculation

The relevant off-diagonal strain components in cylindrical coordinates are:

$$2\varepsilon_{\varphi z} = \frac{1}{r}\partial_\varphi u_z + \partial_z u_\varphi, \quad 2\varepsilon_{r\varphi} = r\partial_r\left(\frac{u_\varphi}{r}\right) + \frac{1}{r}\partial_\varphi u_r, \quad 2\varepsilon_{rz} = \partial_r u_z + \partial_z u_r.$$

Substituting (121):

$$\begin{aligned} 2\varepsilon_{\varphi z} &= 0 + \partial_z(\kappa r z) = \kappa r, \\ 2\varepsilon_{r\varphi} &= r\partial_r(\kappa z) + 0 = 0 \quad \left[\text{since } u_\varphi/r = \kappa z \text{ is independent of } r\right], \\ 2\varepsilon_{rz} &= 0. \end{aligned}$$

All diagonal components ε_{rr} , $\varepsilon_{\varphi\varphi}$, ε_{zz} vanish by inspection. The only non-vanishing strain is

$$\boxed{\varepsilon_{\varphi z} = \frac{1}{2}\kappa r.} \quad (122)$$

Physical significance: The deformation is pure shear in the (φ, z) plane. Within each cross-section the rotation is rigid — no inter-fiber stretching occurs inside a slice — so strain lives exclusively across slices. A key cancellation occurs in $\varepsilon_{r\varphi}$: the term from differentiating the $\hat{\varphi}$ basis vector (which produces $-u_\varphi/r$) cancels exactly with $\partial_r u_\varphi$, confirming that a rigid in-plane rotation contributes nothing to strain.

Stress and physical picture

Since the strain is pure shear, only the shear modulus G enters Hooke's law:

$$\sigma_{\varphi z} = 2G\varepsilon_{\varphi z} = G\kappa r. \quad (123)$$

Physical intuition: The shear stress $\sigma_{\varphi z}$ grows linearly with r — zero at the center, maximum at the outer surface. Rotating to the principal axes of this shear reveals equal tension and compression at 45 to the cylinder axis. This explains why a chalk rod breaks along a 45 helical fracture surface when twisted: the principal tensile stress exceeds the tensile strength first, since chalk is much stronger in compression than in tension.

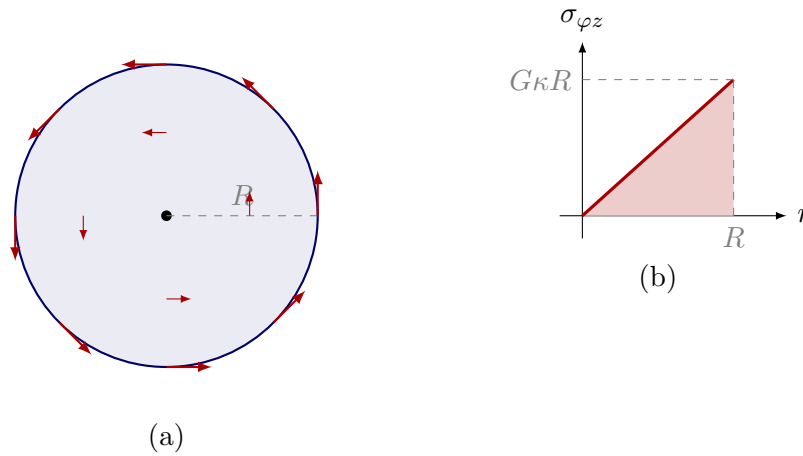


Figure 2: Shear stress in torsion of a circular rod. **(a)** Tangential stress arrows on the cross-section; arrow lengths are proportional to r (the shorter inner arrows at $r \approx R/2$ illustrate the linear growth). **(b)** Linear profile $\sigma_{\phi z} = G\kappa r$: zero at the axis, maximum $G\kappa R$ at the rim.

Torque calculation and torsional stiffness

Integrate the tangential force per unit area times the moment arm r over the cross-section:

$$T = \int_A r \sigma_{\phi z} dA = G\kappa \int_A r^2 dA = G\kappa J, \quad (124)$$

where

$$J \equiv \int_A r^2 dA \quad (125)$$

is the *polar second moment of area*. For a solid circular cross-section of radius R :

$$J = \int_0^R \int_0^{2\pi} r^2 \cdot r d\phi dr = 2\pi \cdot \frac{R^4}{4} = \frac{\pi R^4}{2}. \quad (126)$$

Therefore the *torsional stiffness* is

$$C \equiv \frac{T}{\kappa} = GJ = \frac{\pi GR^4}{2}. \quad (127)$$

Physical significance: Stiffness factorizes into a material part G and a geometric part J . The strong dependence $C \propto R^4$ means a small increase in radius greatly increases torsional stiffness.

Remark 7. Equation (127) holds only for *circular* cross-sections. For non-circular cross-sections the cross-section warps out of its plane and a more general analysis (Saint-Venant torsion) is required. Symmetry forbids warping in the circular case.

Part III — Bending of a Beam

Problem statement and free-body diagram

Consider a beam lying along the z -axis, clamped at the left end, with a transverse force F applied at the right end in the x -direction. A free-body cut at an arbitrary cross-section reveals two internal quantities needed for equilibrium of the right-hand portion:

1. An internal *shear force* $V = F$, opposing the applied load.
2. An internal *bending moment* M_y about the y -axis, preventing the portion from rotating.

Physical intuition: A shear force alone cannot prevent rotation — the finite width of the cross-section is what turns a stress distribution into a net moment. The bending moment M_y is the key quantity that drives the curvature of the beam.

The neutral fiber and radius of curvature

Physical intuition: When the beam curves, fibers on the convex side are stretched and fibers on the concave side are compressed. There is exactly one fiber that neither stretches nor compresses: the *neutral fiber* along the centroid of the cross-section. If the beam curves with local *radius of curvature* R_c , a fiber at signed transverse distance x from the neutral axis traces an arc of radius $R_c + x$. Its fractional change in length relative to the neutral fiber is

$$\frac{(R_c + x) d\theta - R_c d\theta}{R_c d\theta} = \frac{x}{R_c}.$$

Therefore the axial strain is

$$\boxed{\varepsilon_{zz} = \frac{x}{R_c}}. \quad (128)$$

The strain is linear in x : zero on the neutral axis, tensile above, compressive below (for a sagging beam).

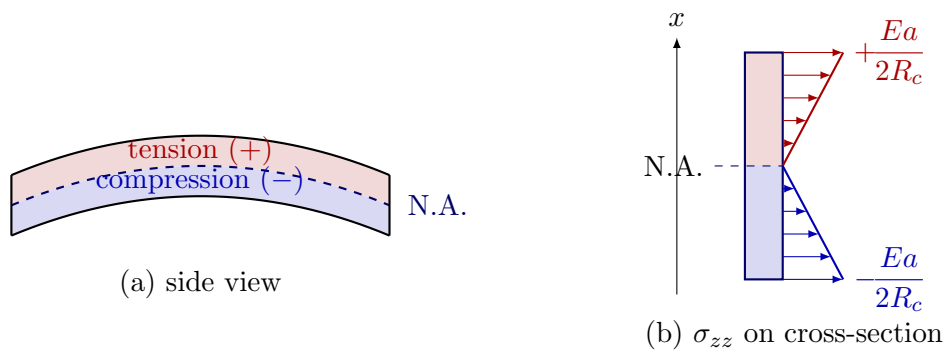


Figure 3: Bending geometry and stress distribution. **(a)** Bent beam (side view): the neutral axis (dashed blue) curves concave-downward; fibres above (pink) are in tension, those below (blue) in compression. **(b)** Axial stress $\sigma_{zz} = Ex/R_c$ varies linearly across the cross-section of height a , reaching $\pm Ea/(2R_c)$ at the extreme fibres.

Stress and bending moment

For a slender beam ($a \ll L$), the dominant stress is axial. Treating each fiber as in simple tension or compression:

$$\sigma_{zz} = E \varepsilon_{zz} = \frac{E x}{R_c}. \quad (129)$$

The internal bending moment about the y -axis is

$$M_y = \int_A x \sigma_{zz} dA = \frac{E}{R_c} \int_A x^2 dA = \frac{E I_y}{R_c}, \quad (130)$$

where

$$I_y \equiv \int_A x^2 dA \quad (131)$$

is the *second moment of area* about the y -axis. Rearranging:

$$\boxed{\frac{1}{R_c} = \frac{M_y}{E I_y}}. \quad (132)$$

The *bending stiffness* is $E I_y$: stiffer material or larger second moment of area means less curvature per unit moment.

Part IV — Second Moment of Area: Calculations

Polar decomposition

By $r^2 = x^2 + y^2$:

$$J = \int_A r^2 dA = I_y + I_x. \quad (133)$$

For any cross-section with two perpendicular axes of symmetry, $I_x = I_y = J/2$.

Circular cross-section

From (126), $J = \pi R^4/2$, so by symmetry:

$$I_y^{(\text{circle})} = \frac{\pi R^4}{4}. \quad (134)$$

Direct verification in polar coordinates.

$$I_y = \int_0^R \int_0^{2\pi} r^2 \cos^2 \varphi \cdot r d\varphi dr = \frac{R^4}{4} \cdot \pi = \frac{\pi R^4}{4},$$

where $\int_0^{2\pi} \cos^2 \varphi d\varphi = \pi$ (mean value of $\cos^2$ is $1/2$ over length 2π).

Rectangular cross-section

For a rectangle of height a (in the bending direction x) and width b :

$$I_y^{(\text{rect})} = b \int_{-a/2}^{a/2} x^2 dx = b \cdot 2 \cdot \frac{(a/2)^3}{3} = \frac{b a^3}{12}. \quad (135)$$

Physical intuition: Height a enters to the *third* power while width b is linear. This is why deep beams are far stiffer in bending than shallow wide ones with the same cross-sectional area, and why I-beams concentrate material in flanges far from the neutral axis.

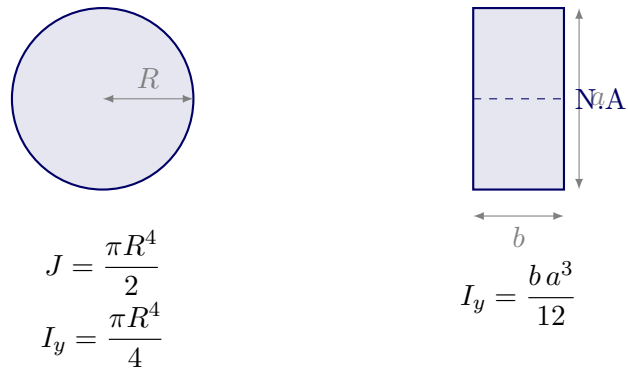


Figure 4: Cross-sectional shapes and second moments of area. **Left:** circular cross-section; by symmetry $I_x = I_y = J/2$. **Right:** rectangular cross-section of height a (bending direction) and width b ; the a^3 dependence means doubling the height increases bending stiffness eightfold.

Analogy with the mass moment of inertia

The second moment of area $I_y = \int x^2 dA$ is structurally identical to the mass moment of inertia $\mathcal{I} = \int r^2 dm$ from rigid-body mechanics, with area replacing mass. Dimensions differ:

$$[I_y] = \text{length}^4, \quad [\mathcal{I}] = \text{mass} \times \text{length}^2.$$

For a uniform-density cross-section the two are proportional. The Steiner (parallel-axis) theorem $\mathcal{I} = \mathcal{I}_{\text{cm}} + Md^2$ has a direct area analogue $I = I_{\text{centroid}} + Ad^2$.

Part V — Displacement Field for Bending (Brief)

Integrating $\varepsilon_{zz} = x/R_c$ with Poisson contraction in the transverse directions gives the full displacement field:

$$u_z = -\frac{xz}{R_c}, \quad u_x = \frac{z^2 + \nu(x^2 - y^2)}{2R_c}, \quad u_y = -\frac{\nu xy}{R_c}, \quad (136)$$

where ν is the Poisson ratio. The transverse displacements u_x, u_y arise from Poisson contraction of the axially stressed fibers. The next lecture will use $1/R_c \approx d^2u_x/dz^2$ (valid for small deflections) to derive the **Euler–Bernoulli beam equation**.

Summary

Both torsion and bending follow the same pattern: displacement ansatz $\rightarrow$ strain $\rightarrow$ stress $\rightarrow$ integrate for resultant moment $\rightarrow$ read off stiffness.

Problem	Stiffness	Geometric factor
Torsion (circle)	$C = GJ$	$J = \frac{\pi R^4}{2}$
Bending	EI_y	$I_y = \int_A x^2 dA$

In both cases stiffness factorizes into an elastic modulus (G for torsion, E for bending) and a purely geometric moment of the cross-section.

Further Reading

- **L. D. Landau, E. M. Lifshitz, *Theory of Elasticity*, Vol. 7, §§16–17.** Primary reference for this lecture: §16 covers torsion (including Saint-Venant’s general solution), §17 covers bending.
- **S. Timoshenko and J. N. Goodier, *Theory of Elasticity*.** Engineering treatment of torsion (Ch. 10) and bending with many cross-sectional examples.
- **S. Timoshenko and S. Woinowsky-Krieger, *Theory of Plates and Shells*.** Extends beam results to plates (two large dimensions) — the family mentioned but not covered in this course.

9 Lecture 09 — Euler–Bernoulli Beam Equation

Overview

This lecture derives the **Euler–Bernoulli beam equation** — the capstone of the elasticity section. Starting from the bending results of Lecture 08, we first compare torsion and bending stiffness, then settle two bookkeeping issues: the location of the neutral axis and the motivation for I-beam cross-sections. We then derive a differential-geometric formula for the beam curvature κ in terms of the deflection $x(z)$. Combining this with force and moment equilibrium of a beam element yields a fourth-order ODE for the deflection:

$$EI_y \frac{d^4 x}{dz^4} = q(z),$$

where $q(z)$ is the distributed load per unit length and EI_y is the *bending stiffness*. The lecture uses the same physical pattern as torsion: a moment is linearly related to a geometric rate of deformation, but torsion uses the shear modulus and twist rate, while bending uses Young’s modulus and curvature. Each successive derivative of x carries a distinct physical meaning (slope, curvature/moment, shear force, load density). We close with a complete worked example: a beam clamped at both ends deflecting under its own weight.

Opening Reminder — Torsion Versus Bending Stiffness

Physical Motivation

Before deriving the beam equation, it is useful to recall why bending and torsion look mathematically similar but use different material constants. In both cases an internal moment balances an imposed deformation; the distinction is the kind of stress carried by the cross-section.

For a circular rod in torsion, the moment acts along the rod axis and the deformation is the twist rate $d\phi/dz$. The stresses are shear stresses, so the relevant material constant is the shear modulus G . The torsional stiffness has the form

$$C_{\text{torsion}} = GI_c, \quad I_c = \int_A r^2 dA, \quad (137)$$

where I_c is the polar second moment of area of the cross-section.

For bending, the moment is perpendicular to the beam axis, here about the y -axis. The upper and lower fibers are stretched or compressed along z , so the stresses are normal stresses and the material constant is Young’s modulus E . The bending stiffness is

$$C_{\text{bend}} = EI_y, \quad I_y = \int_A x^2 dA. \quad (138)$$

Physical significance: Both stiffnesses are “elastic modulus times area moment”. The modulus tells us which stress mode is active; the area moment tells us how effectively the cross-section places material away from the relevant axis.

Part I — Neutral Axis = Centroid of Cross-Section

Physical Motivation

When a beam bends, fibers above the neutral axis stretch and fibers below compress. The neutral axis is the surface where $\varepsilon_{zz} = 0$. Its location is not assumed — it follows from the condition that the net axial force across any cross-section is zero (the beam carries only transverse loads, not axial ones).

Derivation

From Lecture 08, the axial strain in a bent beam at transverse distance x from the neutral axis with radius of curvature R_c is $\varepsilon_{zz} = x/R_c$, giving the axial stress

$$\sigma_{zz} = \frac{E x}{R_c}. \quad (139)$$

Requiring zero net axial force on the cross-section:

$$F_z = \int_A \sigma_{zz} dA = \frac{E}{R_c} \int_A x dA = 0 \quad \Rightarrow \quad \boxed{\int_A x dA = 0}. \quad (140)$$

This is precisely the condition that $x = 0$ passes through the *centroid* of the cross-sectional area. The neutral axis coincides with the centroid. More generally, if an arbitrary origin is used, the centroid coordinate is

$$\bar{x} = \frac{1}{A} \int_A x dA, \quad (141)$$

and the neutral-axis coordinate system is the one for which $\bar{x} = 0$.

Physical significance: The result is analogous to placing the origin at the center of mass, with area replacing mass. For cross-sections with a reflection symmetry about the bending axis (circles, rectangles, I-beams), the centroid lies on that symmetry axis by inspection. For fully asymmetric sections, bending about one axis can couple to bending about another; this course restricts the beam discussion to the common symmetric cases where the scalar formula $M_y = EI_y \kappa$ is enough.

Part II — I-Beam Cross-Section

Physical Motivation

The bending stiffness is EI_y with $I_y = \int_A x^2 dA$. To maximize stiffness for a fixed amount of material (fixed cross-sectional area A), one must maximize I_y/A , which means pushing material as far as possible from the neutral axis.

I-Beam Geometry

An **I-beam** concentrates material in two horizontal *flanges* at the top and bottom, connected by a thin vertical *web*. The flanges contribute large $x^2 dA$ terms because $|x|$ is large; the web keeps the flanges separated and carries much of the shear. For a given cross-sectional area, the I-shape yields far greater bending stiffness than a solid rectangle.

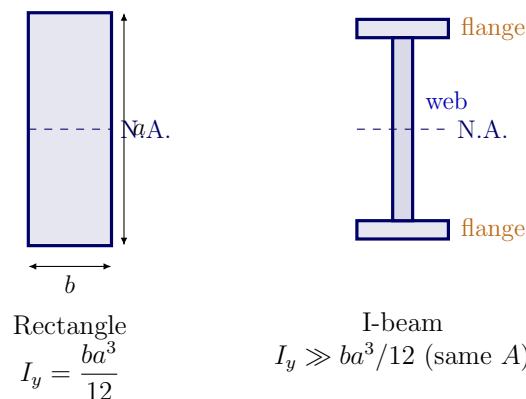


Figure 5: Comparison of cross-sections for equal material (A fixed). The I-beam places flanges far from the neutral axis, giving a much larger $I_y = \int_A x^2 dA$ and hence greater bending stiffness EI_y .

Part III — Curvature of a Deflected Beam

Physical Motivation

The bending constitutive law from Lecture 08 is $M_y = EI_y \kappa$, where $\kappa = 1/R_c$ is the curvature of the neutral axis. To write a differential equation for the deflection $x(z)$, we need κ as a function of $x(z)$.

Differential-Geometric Derivation

Let $\theta(z)$ be the angle the tangent to the neutral axis makes with the z -axis, and let s be the arc length along the neutral axis. The exact curvature is

$$\kappa = \frac{d\theta}{ds}.$$

In the **small-deflection approximation** ($|x'| \ll 1$):

1. The slope is small: $\theta \approx \tan \theta = dx/dz \equiv x'$.
2. The arc element $ds = dz\sqrt{1 + (x')^2} \approx dz$.

Substituting both into the curvature formula:

$$\boxed{\kappa \approx \frac{d\theta}{dz} \approx \frac{d^2x}{dz^2} = x''} \quad (142)$$

Physical significance: The exact expression is $\kappa = |x''|/(1 + x'^2)^{3/2}$; the linearization discards the denominator. Equation (142) is the beam-theory specialisation of the general differential-geometric relation $\vec{\kappa} = d\hat{T}/ds$ (rate of change of unit tangent with arc length), which is studied in the full theory of curves.

Part IV — Equilibrium of a Beam Element

Physical Motivation

We cut out a differential element of length dz and balance forces and moments. Two internal resultants characterize the state at any cross-section:

- **Shear force** $V(z)$: net transverse force ($+x$ direction) exerted by the right portion on the left portion.
- **Bending moment** $M(z)$: net moment about the y -axis exerted by the right portion on the left portion.

The distributed load $q(z)$ (force per unit length in the $+x$ direction) is the external load density. Two useful examples are:

$$q(z) = \rho Ag, \quad \text{uniform self-weight for a beam of constant cross-section,} \quad (143)$$

$$q(z) = F_0 \delta(z - z_0), \quad \text{a concentrated force } F_0 \text{ applied at } z = z_0. \quad (144)$$

The notation q is chosen to avoid confusion with the curvature κ ; the transcript notation sometimes calls the load density K , but the physical quantity is the same: force per unit length.

Force Equilibrium

Summing forces in the $+x$ direction (down positive):

$$-V(z) + V(z + dz) + q dz = 0$$

$$\boxed{\frac{dV}{dz} = -q} \quad (145)$$

Physical significance: If a downward load is present, the shear force cannot be constant along the beam. Its spatial change is precisely what balances the load carried by the small element.

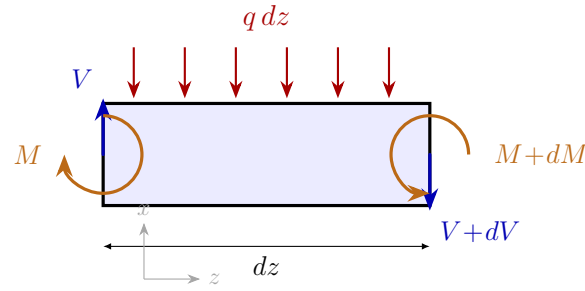


Figure 6: Free-body diagram of a beam element of length dz . The distributed load $q dz$ acts in the $+x$ direction (downward). V and M are the internal shear force and bending moment; Newton's third law reverses their sense between the two faces.

Moment Equilibrium

Taking moments about the right face and neglecting the second-order term $q(dz)^2/2$:

$$M(z + dz) - M(z) + V(z) dz \approx 0$$

$$\boxed{\frac{dM}{dz} = -V.} \quad (146)$$

Physical significance: A varying bending moment is what produces the shear force. Force balance and moment balance therefore form a chain: load changes shear, and shear changes moment.

Differentiating (146) and using (145):

$$\boxed{\frac{d^2M}{dz^2} = q.} \quad (147)$$

Physical significance: The load density is the second derivative of the bending moment. This is the equilibrium half of the Euler–Bernoulli equation; the material law will now convert M into the deflection $x(z)$.

Part V — The Euler–Bernoulli Beam Equation

Physical Motivation

The equilibrium equations know nothing about the material; they only say how q , V , and M must vary for a static beam element. To predict the shape of the beam, we must add the elastic bending law from Lecture 08, which tells us how much curvature is produced by a given moment.

Derivation

From Lecture 08 and equation (142):

$$\boxed{M = EI_y \kappa = EI_y x''} \quad (148)$$

Physical significance: The bending stiffness EI_y is the proportionality factor between geometry and mechanics: for the same curvature, a stiffer material or a better-shaped cross-section carries a larger moment.

Substituting into (147):

$$\boxed{EI_y \frac{d^4x}{dz^4} = q(z).} \quad (149)$$

Physical significance: This fourth-order linear ODE is the **Euler–Bernoulli beam equation**. Unlike Newton's second law (order 2) or the wave equation (order 2), it requires *four* boundary

conditions — two per end. It is a purely static equation (no time dependence). The strong R^4 or a^3 stiffness scaling from Lectures 07–08 is encoded entirely in I_y ; changing I_y rescales the right-hand side of the equation for x without altering its structure.

Physical Interpretation of the Derivatives

Each successive derivative of x with respect to z carries an independent physical meaning:

Quantity	Expression	Physical meaning
x	deflection	transverse displacement of neutral axis
x'	slope	tangent angle (small-deflection)
$EI_y x''$	$M(z)$	bending moment about y -axis
$-EI_y x'''$	$V(z)$	shear force on cross-section
$EI_y x^{(4)}$	$q(z)$	applied load per unit length

Part VI — Boundary Conditions

A beam has two ends; each end supplies two boundary conditions, giving the required four. The three standard end types are:

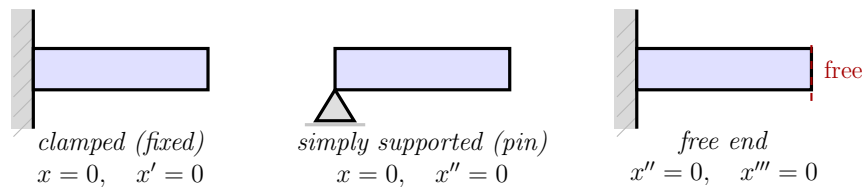


Figure 7: The three standard beam boundary conditions. **Left:** clamped end — zero deflection and zero slope. **Centre:** simply-supported (pin) — zero deflection and zero moment ($x'' = 0$ because $M = EI_y x''$ and no external moment applied). **Right:** free end — zero moment and zero shear force ($x'' = x''' = 0$).

Boundary Conditions as External Actions

The same conditions can be read directly from the internal resultants. Since

$$M = EI_y x'', \quad V = -EI_y x''',$$

prescribing x or x' fixes the geometry at an end, while prescribing x'' or x''' fixes the applied moment or transverse force at that end. With the sign convention used above, the right end and left end have opposite outward normals, so the external-action signs differ. Schematically,

$$\text{right end:} \quad M_{\text{ext}} = EI_y x'', \quad F_{\text{ext}} = -EI_y x''', \quad (150)$$

$$\text{left end:} \quad M_{\text{ext}} = -EI_y x'', \quad F_{\text{ext}} = EI_y x'''. \quad (151)$$

Physical significance: A free end has no externally applied moment or force, so $M = V = 0$ there. A support or clamp can exert reaction forces and moments, so the corresponding derivatives need not vanish; instead they encode the reactions.

Part VII — Worked Example: Clamped Beam Under Its Own Weight

Problem Statement

A uniform beam of length L , cross-section A , and density ρ is clamped at both ends. Find the deflection $x(z)$, the maximum sag, the bending moment, and the reaction forces.

The distributed load (weight per unit length, $+x = \text{downward}$) is

$$q = \rho Ag = \text{const.} \quad (152)$$

Boundary conditions (both ends clamped, beam axis along z , $z \in [0, L]$):

$$x(0) = 0, \quad x'(0) = 0, \quad x(L) = 0, \quad x'(L) = 0. \quad (153)$$

In a more careful support model one may allow a tiny longitudinal sliding of one end as the beam sags, to avoid adding an axial-stretching problem on top of the transverse bending problem. This does not change the transverse boundary conditions in (153): the vertical deflection and slope are still fixed at both ends.

Solution of the ODE

The governing equation $EI_y x^{(4)} = q$ (constant q) has the general solution

$$x(z) = \frac{q}{24 EI_y} z^4 + c_3 z^3 + c_2 z^2 + c_1 z + c_0.$$

Applying the four conditions (153): $c_0 = 0$, $c_1 = 0$, and solving the two remaining equations yields $c_2 = qL^2/(24EI_y)$, $c_3 = -qL/(12EI_y)$. The deflection simplifies to

$$x(z) = \frac{q}{24 EI_y} z^2 (L - z)^2. \quad (154)$$

Verification: $x^{(4)} = q/(24EI_y) \cdot 24 = q/EI_y \checkmark$. At $z = 0$: $x = 0$, $x' = \frac{q}{12EI_y} \cdot z(L-z)(L-2z)|_{z=0} = 0 \checkmark$. By symmetry the same holds at $z = L$. The solution is non-negative everywhere and symmetric about $z = L/2$, as expected physically.

The derivatives that will be used below are

$$x'(z) = \frac{q}{12EI_y} z(L-z)(L-2z), \quad (155)$$

$$x''(z) = \frac{q}{12EI_y} (L^2 - 6Lz + 6z^2), \quad (156)$$

$$x'''(z) = \frac{q}{2EI_y} (2z - L). \quad (157)$$

These formulas make the physics transparent: x' vanishes at both clamps and at the midpoint, x'' gives curvature and moment, and x''' gives shear and support reactions.

Maximum Deflection

The maximum occurs at midspan ($z = L/2$, where $x' = 0$ by symmetry):

$$x_{\max} = x\left(\frac{L}{2}\right) = \frac{q L^4}{384 EI_y}. \quad (158)$$

Bending Moment Distribution and Inflection Points

From $M = EI_y x''$:

$$M(z) = \frac{q}{12}(L^2 - 6Lz + 6z^2). \quad (159)$$

Key values:

$$\boxed{M(0) = M(L) = \frac{qL^2}{12}, \quad M\left(\frac{L}{2}\right) = -\frac{qL^2}{24}.} \quad (160)$$

Physical significance: The curvature is positive near the clamps and negative near the midpoint. Thus a fixed-fixed beam under uniform load does not have one curvature sign everywhere; the bending moment must change sign.

The curvature changes sign at the *inflection points* where $M = 0$:

$$\boxed{z_{\pm} = \frac{L}{2} \pm \frac{L}{2\sqrt{3}} = \frac{L(3 \pm \sqrt{3})}{6} \approx \{0.211 L, 0.789 L\}.} \quad (161)$$

Shear Force and Reaction Forces

From $V = -EI_y x'''$:

$$V(z) = \frac{q}{2}(L - 2z). \quad (162)$$

At the two ends,

$$\boxed{V(0) = \frac{qL}{2}, \quad V(L) = -\frac{qL}{2}.} \quad (163)$$

Physical significance: The magnitudes are equal because the loading and supports are symmetric. Each support carries half the total weight qL ; the sign flip is a convention effect caused by the two opposite end normals.

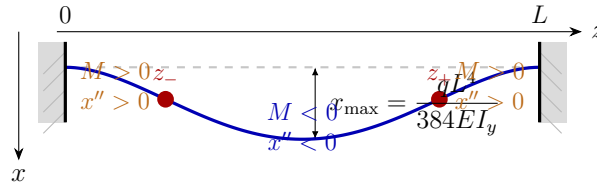


Figure 8: Deflection curve $x(z) = qz^2(L - z)^2/(24EI_y)$ for a clamped-clamped beam under uniform load. Maximum sag $qL^4/(384EI_y)$ at midspan. Red dots mark the inflection points $z_{\pm} = L(3 \pm \sqrt{3})/6$ where the bending moment changes sign.

Summary

The Euler–Bernoulli beam equation results from combining three steps:

1. **Kinematics** (small deflection): $\kappa \approx x''$.
2. **Bending constitutive law** (Lec 08): $M = EI_y x''$.
3. **Equilibrium**: $d^2M/dz^2 = q$, derived from $dV/dz = -q$ and $dM/dz = -V$.

$$x \xrightarrow{d/dz} x' \xrightarrow{\cdot EI_y} M = EI_y x'' \xrightarrow{-d/dz} V \xrightarrow{-d/dz} q = EI_y x^{(4)}.$$

Clamped-clamped beam under UDL q :

$$x(z) = \frac{qz^2(L - z)^2}{24EI_y}, \quad x_{\max} = \frac{qL^4}{384EI_y}, \quad M_{\text{ends}} = \frac{qL^2}{12}, \quad V_{\text{supports}} = \frac{qL}{2}.$$

Part II

Mechanics of Fluids

10 Lecture 10 – From Elasticity to Hydrostatics

Overview

This lecture closes the elasticity part of the course and opens the fluid-mechanics part. The transition is conceptually important: in elasticity we learned that a solid can sustain a general stress tensor, including shear stresses, because a deformed solid has restoring forces. A fluid at rest is more restrictive. It cannot sustain a static shear stress; otherwise the material would start moving. Therefore the stress tensor of a fluid in hydrostatic equilibrium is completely determined by one scalar field, the pressure $p(\mathbf{x})$.

The first main goal is to derive the hydrostatic equilibrium equation

$$\boxed{\nabla p = \rho \mathbf{g}.} \quad (164)$$

Here $\rho(\mathbf{x})$ is the mass density and $\mathbf{g}$ is the gravitational acceleration vector. This equation is the fluid analogue of the static force-balance equation from elasticity, but with the elastic stress tensor replaced by the simple hydrostatic form $\boldsymbol{\sigma} = -p \mathbf{I}$.

The second main goal is to solve this equation in two regimes. For an incompressible fluid, such as water to a good approximation, ρ is constant and the pressure grows linearly with depth. For a compressible fluid, such as air, ρ depends on p through an equation of state; the same equilibrium equation then leads to the Gibbs potential per unit mass and, for an isothermal ideal gas atmosphere, to an exponential pressure profile.

10.1 Final Recap of the Elasticity Chapter

Physical Motivation

Before starting fluids, it is useful to identify what the elasticity chapter really gave us. The formulas for torsion and bending were important, but the deeper lesson was broader: continua are described by fields, and their static configurations are governed by field equations.

Fields, Tensors, and Equilibrium

In the elastic-solid part of the course we introduced the displacement field

$$\mathbf{u}(\mathbf{x}) = \text{displacement of the material point originally near } \mathbf{x}. \quad (165)$$

From $\mathbf{u}$ we built the small-strain tensor

$$\boxed{\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right).} \quad (166)$$

Physical significance: The strain tensor measures local stretching and shearing, after removing pure rigid translations and rotations.

The stress tensor σ_{ij} encodes internal forces. If $d\mathbf{S}$ is an oriented area element, the force exerted across it is

$$\boxed{dF_i = \sigma_{ij} dS_j.} \quad (167)$$

Physical significance: A stress tensor is a rule that turns a surface orientation into a force. This is why stress is naturally a rank-two tensor, not a vector.

Static equilibrium required

$$\boxed{\frac{\partial \sigma_{ij}}{\partial x_j} + f_i = 0,} \quad (168)$$

where f_i is an external force density, i.e. force per unit volume. In an isotropic linear elastic solid, Hooke's law relates σ_{ij} to ε_{ij} through Young's modulus E and Poisson's ratio ν .

Physical significance: Elasticity gave us an explicit example of a field theory in mechanics: the unknowns are functions of position, and the laws are differential equations for those functions.

Rods and Beams

After the general tensor formulation, we specialized to one-dimensional structures:

- **Torsion:** a torque is proportional to the twist rate.
- **Bending:** a bending moment is proportional to the curvature.
- **Euler–Bernoulli beams:** equilibrium plus bending elasticity gives a fourth-order equation.

The beam equation derived in Lecture 09 was

$$EI_y \frac{d^4 x}{dz^4} = q(z). \quad (169)$$

Physical significance: This equation is the endpoint of the solid-mechanics part of the course: it combines geometry, material response, and static equilibrium in one ordinary differential equation.

Problems in L. D. Landau and E. M. Lifshitz

- §7 problems 1-5
- §20 problems 1-5

10.2 What Is a Fluid?

Physical Motivation

The word “fluid” groups liquids and gases together. This is not because they are microscopically identical, but because their mechanical response has a common feature: at rest, they cannot support shear stresses. A solid can sit in a deformed state; a fluid under shear keeps deforming, which means it flows.

Stable Versus Neutral Equilibrium

A solid at rest is typically in **stable equilibrium**. If it is slightly displaced or deformed, restoring forces try to bring it back. Microscopically, one may imagine atoms connected by effective springs, and small disturbances produce elastic waves or vibrations.

A fluid at rest is in **neutral equilibrium**. A small rearrangement of neighboring fluid elements does not produce the same kind of static restoring shear force. This is why water in a bowl can begin to flow when disturbed, whereas a solid object mainly vibrates.

Liquids and Gases

The two most important classes of fluids are:

- **Liquids**, such as water, often modeled as incompressible:

$$\rho(\mathbf{x}) = \rho_0 = \text{const.} \quad (170)$$

- **Gases**, such as air, usually compressible:

$$\rho = \rho(p, T). \quad (171)$$

Physical significance: Liquids and gases differ strongly in compressibility, but many of their force-balance and motion equations have the same structure. This shared structure is the reason for treating them together under fluid mechanics.

10.3 Hydrostatic Stress: Only Pressure Survives

Physical Motivation

In a solid, a surface element can experience both normal and tangential components of force. In a fluid at rest, a tangential force would be a shear stress, and a shear stress would start a flow. Therefore only normal compressive forces can remain in hydrostatic equilibrium.

Force on a Surface Element

Let $d\mathbf{S}$ be an oriented area element whose direction is the outward normal. In a fluid at rest, the force exerted by the surrounding fluid on the element is normal to the surface and points inward:

$$\boxed{d\mathbf{F} = -p d\mathbf{S}.} \quad (172)$$

Physical significance: Pressure is a scalar. It has no preferred direction; the direction of the force comes entirely from the orientation of the surface.

Comparing the general traction formula $dF_i = \sigma_{ij}dS_j$ with equation (172), we identify the hydrostatic stress tensor:

$$\boxed{\sigma_{ij} = -p \delta_{ij}.} \quad (173)$$

Physical significance: A fluid at rest has no deviatoric part of the stress tensor. The entire stress state is encoded in one scalar field $p(\mathbf{x})$.

In tensor language, a general stress tensor can be decomposed as

$$\sigma_{ij} = -p \delta_{ij} + \tau_{ij}, \quad \tau_{ii} = 0, \quad (174)$$

where τ_{ij} is the shear/deviatoric part. Hydrostatics is the special case

$$\boxed{\tau_{ij} = 0.} \quad (175)$$

Physical significance: The vanishing of τ_{ij} is the mathematical statement that a resting fluid cannot hold a static shear stress.

10.4 Units and Scales of Pressure

Physical Motivation

Pressure is force per unit area. In fluids, numerical pressure scales become physically intuitive only after comparing them with everyday columns of water or air.

Standard Units

The SI unit is the pascal:

$$\boxed{1 \text{ Pa} = 1 \text{ N m}^{-2}.} \quad (176)$$

Atmospheric pressure at sea level is approximately

$$\boxed{1 \text{ atm} \simeq 1.01 \times 10^5 \text{ Pa} \simeq 10^5 \text{ Pa}.} \quad (177)$$

Physical significance: The atmosphere exerts a force of order one kilogram-force on each square centimeter. We do not usually notice it because our bodies and surroundings are nearly pressure-balanced.

Another common pressure unit is millimeters of mercury:

$$\boxed{1 \text{ atm} = 760 \text{ mmHg}.} \quad (178)$$

In engineering contexts, one also meets PSI, pounds-force per square inch.

10.5 Hydrostatic Equilibrium Equation

Physical Motivation

We now derive the central equation of hydrostatics. The force balance is the same continuum idea used in elasticity: internal forces plus external body forces must sum to zero for every small volume element.

Derivation from Static Force Balance

The general static equilibrium equation is

$$\frac{\partial \sigma_{ij}}{\partial x_j} + f_i = 0. \quad (179)$$

For gravity, the external force density is

$$f_i = \rho g_i, \quad (180)$$

where $\mathbf{g}$ is the gravitational acceleration vector. Substituting the hydrostatic stress tensor $\sigma_{ij} = -p\delta_{ij}$ gives

$$\frac{\partial}{\partial x_j}(-p\delta_{ij}) + \rho g_i = 0, \quad (181)$$

$$-\delta_{ij} \frac{\partial p}{\partial x_j} + \rho g_i = 0, \quad (182)$$

$$-\frac{\partial p}{\partial x_i} + \rho g_i = 0. \quad (183)$$

Therefore, in vector form,

$$\boxed{\nabla p = \rho \mathbf{g}.} \quad (184)$$

Physical significance: Pressure changes in the direction of the body force. In a gravitational field, the pressure is larger lower down because the lower fluid must support the weight of the fluid above it.

10.6 Incompressible Fluid: Pressure in Water

Physical Motivation

Water is not perfectly incompressible, but for many hydrostatic problems its density changes so little that we can treat ρ as constant. Then the hydrostatic equation can be integrated immediately.

Linear Pressure Profile

Let h denote depth measured downward from a reference surface. Then $\mathbf{g} = g \hat{\mathbf{h}}$ and equation (184) becomes

$$\frac{dp}{dh} = \rho_0 g. \quad (185)$$

Integrating from the surface $h = 0$, where $p = p_0$, to depth h gives

$$\boxed{p(h) = p_0 + \rho_0 g h.} \quad (186)$$

Physical significance: The pressure grows linearly with depth because each additional layer must support an additional slab of water of weight $\rho_0 g dh$ per unit area.

One Atmosphere per Ten Meters

For water,

$$\rho_{\text{water}} \simeq 10^3 \text{ kg m}^{-3}, \quad g \simeq 10 \text{ m s}^{-2}. \quad (187)$$

Hence

$$\frac{dp}{dh} = \rho_{\text{water}} g \simeq 10^4 \text{ Pa m}^{-1}, \quad (188)$$

so over 10 m,

$$\boxed{\Delta p \simeq 10^5 \text{ Pa} \simeq 1 \text{ atm.}} \quad (189)$$

Physical significance: A diver experiences roughly one additional atmosphere of pressure for every ten meters of depth. This simple estimate is one of the most useful hydrostatic scales.

Pascal's Principle

If the height variation in a container is small compared with the scale over which pressure changes appreciably, then $\rho g \Delta h$ is negligible compared with the imposed pressure scale. In that limit

$$\boxed{\nabla p \simeq 0, \quad p \simeq \text{const throughout the connected fluid.}} \quad (190)$$

Physical significance: This is Pascal's principle. A pressure applied to a nearly incompressible confined fluid is transmitted throughout the fluid, which is the basis for hydraulic machines.

10.7 Compressible Hydrostatics and the Equation of State

Physical Motivation

For a gas, the density is not fixed. Air near the ground is denser than air high in the atmosphere. To determine $p(z)$ we need, in addition to force balance, a relation between pressure, density, and temperature: an equation of state.

Ideal Gas Reminder

For an ideal gas,

$$\boxed{pV = nRT = Nk_B T}, \quad (191)$$

where n is the number of moles, R is the gas constant, N is the number of molecules, k_B is Boltzmann's constant, and T is the absolute temperature measured in kelvin.

If the gas molecules have mass m , then $\rho = mN/V$. The ideal gas law can be written as

$$\boxed{p = \frac{\rho}{m} k_B T.} \quad (192)$$

Equivalently, using molar mass M ,

$$\boxed{p = \rho \frac{R}{M} T.} \quad (193)$$

Physical significance: At fixed temperature, pressure is proportional to density. Therefore, if pressure decreases with height, density decreases with height as well.

Thermal Equilibrium

In hydrostatic equilibrium with no heat currents, the temperature is spatially uniform:

$$\boxed{\nabla T = 0.} \quad (194)$$

Physical significance: The hydrostatic part of the course will mostly use cases with no heat flow. More general fluid mechanics can include heat currents, but that requires additional equations.

10.8 The Gibbs Potential Per Unit Mass

Physical Motivation

For a compressible fluid, ρ depends on p , so we cannot integrate $\nabla p = \rho \mathbf{g}$ as simply as in water. The useful trick is to divide by $\rho(p)$ and recognize the result as the gradient of a new scalar function.

Mathematical Definition

Assume the temperature is fixed, so the equation of state gives $\rho = \rho(p)$. Divide equation (184) by $\rho(p)$:

$$\frac{1}{\rho(p)} \nabla p = \mathbf{g}. \quad (195)$$

Define

$$g_T(p) = \int^p \frac{dp'}{\rho(p')}. \quad (196)$$

By the chain rule,

$$\nabla g_T(p) = \frac{dg_T}{dp} \nabla p \quad (197)$$

$$= \frac{1}{\rho(p)} \nabla p. \quad (198)$$

Therefore the hydrostatic equation becomes

$$\nabla g_T = \mathbf{g}. \quad (199)$$

Physical significance: The function $g_T(p)$ packages the compressibility of the fluid into one scalar potential. For incompressible water it reduces to p/ρ_0 ; for a gas it becomes a logarithmic function of pressure.

Connection to Thermodynamics

The notation g_T is not accidental. The Gibbs free energy satisfies

$$dG = -S dT + V dp. \quad (200)$$

At fixed temperature,

$$\left. \frac{\partial G}{\partial p} \right|_T = V. \quad (201)$$

For the Gibbs free energy per unit mass, $g = G/M_{\text{mass}}$, the volume per unit mass is $V/M_{\text{mass}} = 1/\rho$. Hence

$$\left. \frac{\partial g}{\partial p} \right|_T = \frac{1}{\rho}. \quad (202)$$

Physical significance: The same integral that solves compressible hydrostatics is the Gibbs potential per unit mass at fixed temperature. Hydrostatics and thermodynamics meet naturally here.

Including the Gravitational Potential

If gravity derives from a potential Φ ,

$$\mathbf{g} = -\nabla \Phi, \quad (203)$$

then equation (199) gives

$$\nabla g_T = -\nabla \Phi. \quad (204)$$

Therefore

$$\boxed{g_T + \Phi = \text{const.}} \quad (205)$$

Physical significance: In a static compressible fluid, the sum of thermodynamic Gibbs potential per unit mass and gravitational potential energy per unit mass is constant. This is the compressible analogue of $p + \rho g z = \text{const}$ for an incompressible fluid.

For an incompressible fluid, $\rho = \rho_0$, so equation (196) gives

$$\boxed{g_T(p) = \frac{p}{\rho_0} + \text{const.}} \quad (206)$$

Substituting into equation (205) recovers the usual hydrostatic pressure law.

10.9 Example: Isothermal Atmosphere

Physical Motivation

The atmosphere is the canonical compressible hydrostatic example. Unlike water, air has a density that decreases strongly with height. We simplify the real atmosphere by assuming constant temperature; this is not exact, but it captures the central exponential scale.

Direct Derivation

Let z point upward. Then $\mathbf{g} = -g \hat{\mathbf{z}}$ and the hydrostatic equation becomes

$$\frac{dp}{dz} = -\rho g. \quad (207)$$

For an isothermal ideal gas,

$$p = \frac{\rho}{m} k_B T, \quad \rho = \frac{m}{k_B T} p. \quad (208)$$

Substitute this into equation (207):

$$\frac{dp}{dz} = -\frac{mg}{k_B T} p. \quad (209)$$

Separate variables:

$$\frac{1}{p} dp = -\frac{mg}{k_B T} dz, \quad (210)$$

$$\ln \frac{p(z)}{p_0} = -\frac{z}{H}, \quad (211)$$

where the scale height is

$$\boxed{H = \frac{k_B T}{mg} = \frac{RT}{Mg}.} \quad (212)$$

Thus

$$\boxed{p(z) = p_0 e^{-z/H}, \quad \rho(z) = \rho_0 e^{-z/H}.} \quad (213)$$

Physical significance: In an isothermal atmosphere, pressure and density fall by a factor of e every scale height H . The exponential appears because the pressure gradient is proportional to the pressure itself.

Numerical Estimate of the Scale Height

A quick estimate can be made from sea-level values:

$$p_0 \simeq 10^5 \text{ Pa}, \quad \rho_0 \simeq 1.3 \text{ kg m}^{-3}, \quad g \simeq 10 \text{ m s}^{-2}. \quad (214)$$

Since $H = p_0/(\rho_0 g)$ for an isothermal ideal gas at $z = 0$,

$$H \simeq \frac{10^5}{(1.3)(10)} \text{ m} \simeq 8 \times 10^3 \text{ m}. \quad (215)$$

Physical significance: The atmospheric scale height is of order several kilometers, much larger than the 10 m water scale for one atmosphere. The reason is simple: air is roughly a thousand times less dense than water.

The density of air can be estimated from its composition. Dry air is mostly nitrogen and oxygen, with a typical molar mass

$$M_{\text{air}} \simeq 29 \text{ g mol}^{-1}. \quad (216)$$

Using the molar volume near standard conditions, about 22.4 L mol^{-1} , gives a density of order

$$\rho_{\text{air}} \simeq 1.3 \text{ kg m}^{-3}, \quad (217)$$

consistent with the estimate above.

10.10 Summary and Bridge to the Next Lecture

The elasticity chapter ended with fields, tensors, and equilibrium equations. Hydrostatics keeps the same continuum logic but uses a much simpler stress tensor:

$$\sigma_{ij} = -p\delta_{ij}. \quad (218)$$

Substituting this into static force balance gives the central equation:

$$\nabla p = \rho \mathbf{g}. \quad (219)$$

For incompressible fluids:

$$p(h) = p_0 + \rho_0 gh, \quad \Delta p \simeq 1 \text{ atm per 10 m of water}. \quad (220)$$

For compressible fluids at fixed temperature:

$$g_T(p) = \int^p \frac{dp'}{\rho(p')}, \quad g_T + \Phi = \text{const.} \quad (221)$$

For an isothermal ideal-gas atmosphere:

$$p(z) = p_0 e^{-z/H}, \quad \rho(z) = \rho_0 e^{-z/H}, \quad H = \frac{k_B T}{mg}. \quad (222)$$

Further Reading

- L. D. Landau and E. M. Lifshitz, *Fluid Mechanics*, Secs. 1–2: hydrostatics, stress in a fluid, and equilibrium.
- G. K. Batchelor, *An Introduction to Fluid Dynamics*, Ch. 1: continuum description and pressure in a fluid.

- D. J. Acheson, *Elementary Fluid Dynamics*, Ch. 1: basic ideas of pressure, hydrostatics, and scaling.
- MIT OpenCourseWare, [2.06 Fluid Dynamics](#): lecture notes for a broader engineering treatment of hydrostatics and fluid motion.
- HyperPhysics, [Fluid Pressure](#): quick reference for pressure units, hydrostatic pressure, and atmospheric pressure.

11 Lecture 11 – Hydrostatics Applications and Buoyancy

Date: May 13, 2026

Overview

Lecture 10 developed the local law of hydrostatic equilibrium. For an incompressible liquid, with depth h measured downward from the free surface, the result is

$$p(h) = p_0 + \rho gh. \quad (223)$$

Physical significance: In water or hydraulic oil, pressure changes by about one atmosphere only over a height of order 10 m, so in small hydraulic devices the pressure can often be treated as spatially uniform.

This lecture uses that simple pressure law in three ways:

- hydraulic machines, where Pascal's principle transmits pressure through a liquid;
- buoyancy, where the vertical pressure gradient produces a net upward force;
- hydrostatic exam problems, where one integrates pressure over a surface and locates the effective line of action by moments.

The important conceptual shift is that pressure is local and normal to surfaces. Once we know $p(\mathbf{x})$, every force on a boundary is obtained from

$$d\mathbf{F} = -p d\mathbf{S}. \quad (224)$$

Physical significance: Hydrostatics problems are mostly geometry problems after the pressure field has been found.

Note:

The working recipe for this lecture is always the same: find the pressure field, multiply pressure by the appropriate surface element, then integrate forces or moments. The physics is local pressure; the rest is geometry and torque balance.

11.1 Pascal's Principle and Hydraulic Machines

Physical Motivation

Hydraulic machines use a liquid not because the liquid creates energy, but because it transmits pressure efficiently. If the device is much smaller than the depth scale $\Delta h \sim p/(\rho g)$, gravity produces only a small pressure variation inside the machine. The liquid can then be approximated as being at one pressure everywhere.

11.1.1 Uniform Pressure in a Small Hydraulic Device

Suppose a sealed incompressible liquid is in quasistatic equilibrium and height differences are negligible. From equation (223),

$$\Delta p = \rho g \Delta h. \quad (225)$$

For a device whose height is small compared with 10 m of water, or compared with the relevant scale for hydraulic oil, this pressure difference is negligible compared with atmospheric or working pressures. Therefore

$$p(\mathbf{x}) \simeq p = \text{const.} \quad (226)$$

Physical significance: Pascal's principle says that an externally applied pressure change is transmitted through the confined liquid. This is the basic reason a force can be applied at one place and received at another.

Physical Concept 11.1: Pascal's Principle

In a small quasistatic hydraulic device, an imposed pressure change is transmitted throughout the confined fluid. The statement is approximate when height differences are not negligible, because gravity then adds the hydrostatic correction $\rho g \Delta h$.

11.1.2 Force Transmission Without Amplification

Consider a piston of area A pushed with an input force F_{in} . The pressure increase in the fluid is

$$p = \frac{F_{\text{in}}}{A}. \quad (227)$$

If the output piston has the same area A , then

$$F_{\text{out}} = pA = \frac{F_{\text{in}}}{A}A = \boxed{F_{\text{in}}}. \quad (228)$$

Physical significance: A hydraulic fluid can carry a force to a different location and redirect it, even when it does not amplify the force.

11.1.3 The Hydraulic Press

Now let the input piston have area A_1 and the output piston have a larger area A_2 . The input pressure is

$$p = \frac{F_1}{A_1}. \quad (229)$$

The output force is the same pressure times the larger area:

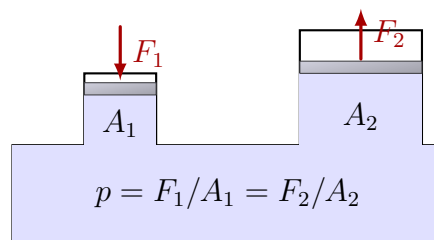
$$F_2 = pA_2, \quad (230)$$

$$= \frac{F_1}{A_1}A_2. \quad (231)$$

Therefore

$$\boxed{\frac{F_2}{F_1} = \frac{A_2}{A_1}}. \quad (232)$$

Physical significance: The force gain is exactly the area ratio. A small force on a small piston can produce a large force on a large piston.



same pressure, different piston areas

This result does not violate energy conservation. If the input piston moves a distance Δx_1 and the output piston moves Δx_2 , incompressibility gives conservation of displaced volume:

$$A_1 \Delta x_1 = A_2 \Delta x_2. \quad (233)$$

Multiplying by the common pressure,

$$pA_1\Delta x_1 = pA_2\Delta x_2. \quad (234)$$

Using $F_1 = pA_1$ and $F_2 = pA_2$ gives

$$F_1\Delta x_1 = F_2\Delta x_2. \quad (235)$$

Physical significance: Hydraulic machines trade distance for force. The larger output force moves through a shorter distance.

Note:

A hydraulic press amplifies force, not work. The larger piston gains force by moving a smaller distance, exactly as required by incompressibility and energy conservation.

11.1.4 Quasistatic Assumption and Boundary Forces

During operation, the fluid is not literally motionless at every instant. A piston is moved, fluid flows, and then the system settles. Hydrostatics applies when this process is slow enough that the pressure field is well approximated by a sequence of static equilibria.

The applied piston forces are boundary forces. In continuum language, this is the same logic as in elasticity: one imposes boundary tractions, solves the equilibrium equations, and obtains the transmitted stress or pressure field.

11.2 Archimedes' Principle

Physical Motivation

A submerged body feels pressure from every side. If pressure were the same everywhere, these forces would cancel. In a gravitational field, however, the pressure is larger lower down than higher up. The lower surface is pushed upward more strongly than the upper surface is pushed downward, so a net upward force remains.

11.2.1 Statement of the Buoyant Force

Let a body displace a volume V_{disp} of a fluid of density ρ_f . The buoyant force is upward and has magnitude

$$B = \rho_f g V_{\text{disp}}. \quad (236)$$

Physical significance: A body is pushed upward by the weight of the fluid it displaces, not by its own weight or its own density directly.

In vector form, if $\hat{\mathbf{z}}$ points upward,

$$\mathbf{B} = \rho_f g V_{\text{disp}} \hat{\mathbf{z}}. \quad (237)$$

Physical significance: The buoyant force is vertical because the hydrostatic pressure gradient is vertical.

Physical Concept 11.2: Center of Buoyancy

The center of buoyancy is the centroid of the displaced fluid volume. It is the point through which the resultant buoyant force acts.

11.2.2 Displaced-Fluid Proof

The most economical proof is to replace the body by the same volume of fluid. Imagine the region occupied by the body is instead filled with the surrounding fluid. The pressure field outside that region is unchanged, because in hydrostatics

$$p = p_0 + \rho_f g h \quad (238)$$

depends only on depth h , not on the material placed inside the imagined boundary.

Therefore the surface pressure forces on the original body are the same as the surface pressure forces on the imagined displaced fluid. But the imagined displaced fluid is in equilibrium. It has weight

$$W_{\text{disp}} = \rho_f V_{\text{disp}} g \quad (239)$$

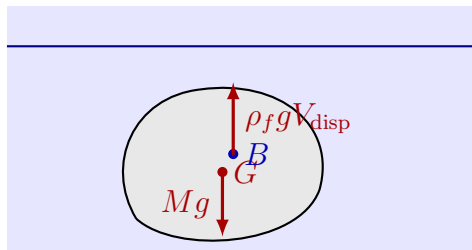
downward, so the total pressure force on it must be the same magnitude upward:

$$B - W_{\text{disp}} = 0. \quad (240)$$

Thus

$$\boxed{B = W_{\text{disp}} = \rho_f g V_{\text{disp}}.} \quad (241)$$

Physical significance: The theorem works for any body shape because it avoids doing a complicated surface integral; the integral is evaluated indirectly by equilibrium of the displaced fluid.



pressure resultant equals displaced-fluid weight

11.2.3 Point of Application

The buoyant force acts through the center of mass of the displaced fluid. For a homogeneous fluid this is the center of volume of the displaced region, called the center of buoyancy:

$$\boxed{\mathbf{r}_B = \frac{1}{V_{\text{disp}}} \int_{V_{\text{disp}}} \mathbf{r} dV.} \quad (242)$$

Physical significance: The center of buoyancy is determined by the geometry of the submerged volume, not by how mass is distributed inside the solid object.

For a completely submerged homogeneous body, the center of buoyancy may coincide with the body's center of mass. For an inhomogeneous body, or for a floating body only partly submerged, the two points need not coincide.

11.3 Floating Bodies and Ship Stability

Physical Motivation

Floating is not only about balancing vertical forces. A ship must also be stable against rotations. If a small tilt creates a restoring torque, the ship returns toward upright; if it creates an overturning torque, the ship is unstable.

11.3.1 Force Balance for Floating

For a floating body in vertical equilibrium,

$$B = W, \quad (243)$$

or

$$\boxed{\rho_f g V_{\text{disp}} = Mg}, \quad (244)$$

where M is the total mass of the body. *Physical significance:* A ship floats by sinking until it displaces a weight of water equal to its own weight.

The weight acts through the center of mass G , while buoyancy acts through the center of buoyancy B . In the untilted symmetric state, their lines of action are vertical and usually coincide.

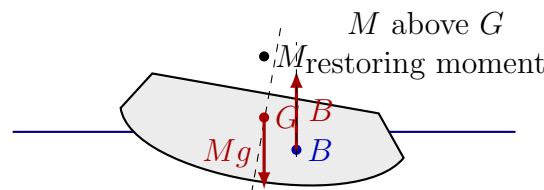
11.3.2 Small Tilt and the Restoring Moment

When the ship tilts, the submerged shape changes. The center of mass G stays fixed in the ship, but the center of buoyancy moves toward the side that sinks deeper. The upward buoyant force now acts along a shifted vertical line. The intersection of this line with the ship's symmetry axis is called the metacenter M .

The lecture used the following stability criterion:

$$\boxed{\text{stable floating equilibrium} \iff M \text{ lies above } G.} \quad (245)$$

Physical significance: If the metacenter is above the center of mass, the buoyancy and weight form a restoring couple; if it is below, the couple increases the tilt.



Equivalently, the metacentric height

$$\boxed{GM > 0} \quad (246)$$

is the sign criterion for small-angle stability. *Physical significance:* Increasing stability means lowering the center of mass or designing the hull so that the center of buoyancy shifts sideways enough when the ship tilts.

The historical example discussed in class was the Swedish warship *Vasa*, launched in 1628. It was tall and heavily armed, and it capsized shortly after leaving harbor. The pedagogical lesson is mechanical: impressive size is irrelevant if the center of mass and hull geometry do not give a restoring moment.

11.4 Resultant Hydrostatic Force on a Dam

Physical Motivation

Hydrostatic forces on walls and dams are distributed forces. To replace them by a single resultant force, we must find two things: the total force and the point where it acts. The point of application matters because it determines the torque.

11.4.1 Pressure Distribution

Consider a vertical rectangular dam of width w holding water of height H . Let y be depth measured downward from the free surface. The pressure on the water side is

$$p_{\text{water}}(y) = p_0 + \rho g y. \quad (247)$$

The outside air exerts atmospheric pressure p_0 , so the net pressure difference is

$$\Delta p(y) = p_{\text{water}}(y) - p_0 = \rho g y. \quad (248)$$

Physical significance: Atmospheric pressure cancels in this problem because it acts on both sides. Only the gauge pressure $\rho g y$ pushes the dam.

An area strip of height dy has area

$$dA = w dy. \quad (249)$$

The horizontal force on that strip is

$$dF = \Delta p(y) dA = \rho g y w dy. \quad (250)$$

11.4.2 Total Force

Integrating from the surface to the bottom,

$$F_{\text{tot}} = \int_0^H dF, \quad (251)$$

$$= \rho g w \int_0^H y dy, \quad (252)$$

$$= \rho g w \left[\frac{y^2}{2} \right]_0^H. \quad (253)$$

Thus

$$\boxed{F_{\text{tot}} = \frac{1}{2} \rho g w H^2.} \quad (254)$$

Physical significance: The pressure distribution is triangular, so the total force is the area of a triangle with height $\rho g H$ and base H , multiplied by the width w .

11.4.3 Center of Pressure

Let Y be the depth at which the single resultant force acts. It is defined by matching the torque about the free surface:

$$Y F_{\text{tot}} = \tau_{\text{tot}}, \quad (255)$$

where

$$\tau_{\text{tot}} = \int_0^H y dF, \quad (256)$$

$$= \rho g w \int_0^H y^2 dy, \quad (257)$$

$$= \rho g w \left[\frac{y^3}{3} \right]_0^H = \frac{1}{3} \rho g w H^3. \quad (258)$$

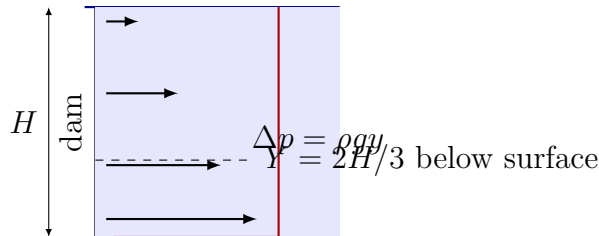
Therefore

$$Y = \frac{\tau_{\text{tot}}}{F_{\text{tot}}}, \quad (259)$$

$$= \frac{(1/3)\rho g w H^3}{(1/2)\rho g w H^2}, \quad (260)$$

$$= \boxed{\frac{2H}{3}}. \quad (261)$$

Physical significance: The resultant force acts below the midpoint because the pressure is larger at greater depth. Measured from the bottom, the line of action is at height $H/3$.



11.5 Lift on a Truncated Cone Filled with Water

Physical Motivation

The final problem asks why a water-filled vessel resting on a table might need its own weight to keep it in contact with the table. The key is that pressure acts normal to the walls. On slanted walls, the pressure force has a vertical component. If that component points upward, the water tries to lift the vessel.

11.5.1 Setting Up the Integral

Consider an open truncated cone resting on its larger base, filled with water to height H . Let y be depth below the free surface. Let α be the angle between the side wall and the vertical direction, and let R be the radius at the free surface. Then the radius of a horizontal ring at depth y is

$$r(y) = R + y \tan \alpha. \quad (262)$$

The gauge pressure is

$$\Delta p(y) = \rho g y. \quad (263)$$

The slanted area of a thin ring is its circumference times its slant height:

$$dS = 2\pi r(y) \frac{dy}{\cos \alpha}. \quad (264)$$

The vertical component of the pressure force contains a factor $\sin \alpha$, since for a vertical wall $\alpha = 0$ there should be no vertical pressure component. Hence

$$dF_{\uparrow} = \Delta p(y) dS \sin \alpha. \quad (265)$$

11.5.2 Total Upward Force

Substituting equations (262)–(264) gives

$$F_{\uparrow} = \int_0^H \rho g y 2\pi (R + y \tan \alpha) \frac{\sin \alpha}{\cos \alpha} dy, \quad (266)$$

$$= 2\pi \rho g \tan \alpha \int_0^H (Ry + y^2 \tan \alpha) dy, \quad (267)$$

$$= 2\pi \rho g \tan \alpha \left(\frac{RH^2}{2} + \frac{H^3 \tan \alpha}{3} \right). \quad (268)$$

Therefore

$$F_{\uparrow} = \pi \rho g R H^2 \tan \alpha + \frac{2\pi}{3} \rho g H^3 \tan^2 \alpha. \quad (269)$$

Physical significance: A slanted wall converts part of the normal pressure force into a vertical lifting force. The effect vanishes for a vertical wall, $\alpha = 0$, as it should.

For the cone to remain in contact with the table, its weight W must be at least this upward hydrostatic lift:

$$W_{\min} = F_{\uparrow}. \quad (270)$$

Physical significance: Contact is lost when the upward force from water pressure exceeds the downward weight of the container.

Worked Example 11.1 (How to Read Slanted-Wall Pressure)

Pressure is normal to the wall, not vertical. The upward force in the truncated-cone problem appears only because the normal to a slanted wall has a vertical component. When $\alpha = 0$, the wall is vertical, the normal is horizontal, and the lifting contribution vanishes.

11.6 Summary

The lecture's main tools are all consequences of hydrostatic pressure:

$$p = p_0 + \rho g h, \quad d\mathbf{F} = -p d\mathbf{S}. \quad (271)$$

For hydraulic devices,

$$\frac{F_2}{F_1} = \frac{A_2}{A_1}, \quad F_1 \Delta x_1 = F_2 \Delta x_2. \quad (272)$$

For buoyancy,

$$B = \rho_f g V_{\text{disp}}, \quad \mathbf{r}_B = \frac{1}{V_{\text{disp}}} \int_{V_{\text{disp}}} \mathbf{r} dV. \quad (273)$$

For distributed hydrostatic loads, the point of application is found by the same moment rule used in mechanics:

12 Lecture 12 – Dynamics of an Ideal Fluid

Date: May 18, 2026

Overview

The previous lectures treated fluids at rest. In hydrostatics the velocity field vanishes, and the central equation is the force balance

$$\nabla p = \rho \mathbf{g}. \quad (274)$$

Physical significance: Hydrostatics is the special case in which pressure gradients only balance body forces; there is no acceleration of fluid elements.

This lecture begins fluid dynamics. The unknowns are now fields depending on position $\mathbf{x}$ and time t :

$$\rho(\mathbf{x}, t), \quad p(\mathbf{x}, t), \quad \mathbf{v}(\mathbf{x}, t). \quad (275)$$

Physical significance: Once the fluid moves, density and pressure are not enough. We must also know the velocity of the material passing through each point.

The first dynamical model is the *ideal fluid*. It keeps pressure forces but neglects viscosity, that is, internal friction between neighboring layers of the fluid. This is not the most general model, but it is the simplest model that already contains mass conservation, Newton's second law, equations of state, boundary conditions, and the first conservation law: Bernoulli's equation.

Note:

The logical order of the lecture is important: first identify the fields, then write conservation of mass, then write Newton's law for a moving fluid element, then close the system with an equation of state. Bernoulli's equation is not an extra postulate; it is a consequence of Euler's equation under additional assumptions.

12.1 Fields and the Ideal-Fluid Approximation

Physical Motivation

A continuum description replaces many molecules by smooth fields. At each point we ask: how much mass is there, how strongly is it compressed, and with what velocity is it moving? The ideal-fluid approximation then says that the local stress is still isotropic, as in hydrostatics, even though the fluid is moving.

12.1.1 Pressure Versus General Stress

In a general continuum the force transmitted through surfaces is described by the stress tensor $\boldsymbol{\sigma}$. In a static fluid the stress is isotropic:

$$\boldsymbol{\sigma} = -p\mathbf{I}, \quad (276)$$

or in components,

$$\sigma_{ij} = -p\delta_{ij}. \quad (277)$$

The ideal-fluid approximation assumes that this remains true during motion:

$$\sigma_{ij}(\mathbf{x}, t) = -p(\mathbf{x}, t)\delta_{ij}. \quad (278)$$

Physical significance: An ideal fluid can push normally on surfaces through pressure, but it cannot exert tangential viscous shear stresses.

Physically, this means that we neglect viscosity. In a real moving fluid, neighboring fluid elements with different velocities rub against one another and exchange momentum through viscous stresses. Those terms will be added later. For now, pressure is the only internal mechanical stress.

Physical Concept 12.1: Ideal Fluid

An ideal fluid is a continuum model in which the stress tensor is purely isotropic,

$$\sigma_{ij} = -p\delta_{ij},$$

so the only internal force is pressure. It is also called an inviscid fluid because viscous shear stresses are neglected.

12.2 Mass Conservation: The Continuity Equation

Physical Motivation

Mass cannot appear or disappear inside a small volume. If the mass in that volume changes, the reason must be that mass flowed through the boundary. This is the same local conservation idea as charge conservation in electromagnetism.

12.2.1 Mass Current

The mass current density is defined by

$$\mathbf{j}_m = \rho \mathbf{v}. \quad (279)$$

Physical significance: The amount of mass crossing a unit area per unit time is density times the velocity normal to that area.

For a fixed control volume V with boundary ∂V , conservation of mass says

$$\frac{d}{dt} \int_V \rho dV = - \oint_{\partial V} \mathbf{j}_m \cdot d\mathbf{S}. \quad (280)$$

Using the divergence theorem,

$$\frac{d}{dt} \int_V \rho dV = - \int_V \nabla \cdot \mathbf{j}_m dV, \quad (281)$$

$$\int_V \left(\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j}_m \right) dV = 0. \quad (282)$$

Since this holds for every small volume V , the integrand must vanish:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (283)$$

Physical significance: Local density increases only where more mass flows in than flows out.

12.2.2 Steady Flow Through a Tube

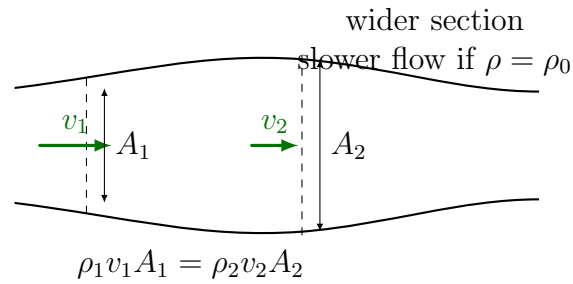
Consider steady flow through a tube whose cross-sectional area is $A(z)$. If the velocity is approximately uniform across each section, the mass flux is

$$\dot{M}(z) = \rho(z)v(z)A(z). \quad (284)$$

For a steady flow no mass accumulates between two cross sections, so

$$\rho(z)v(z)A(z) = \text{const.} \quad (285)$$

Physical significance: In steady flow the same mass per unit time must pass through every section of the tube.



For an incompressible fluid, $\rho = \rho_0$ is constant, and therefore

$$\boxed{v(z)A(z) = \text{const.}} \quad (286)$$

Physical significance: Where the tube widens the flow slows down; where the tube narrows the flow speeds up.

The qualification “steady” matters. If the flow depends on time, mass may temporarily accumulate inside a section of the tube, so the incoming and outgoing fluxes need not be equal at the same instant.

12.3 The Material Derivative

Physical Motivation

A field such as temperature, density, or velocity can change at a fixed point. But Newton’s second law follows a material element. We therefore need a derivative that moves with the fluid particle, not one that watches a fixed point in space.

12.3.1 Definition

Let $\phi(\mathbf{x}, t)$ be any scalar field. The material derivative is

$$\boxed{\frac{D}{Dt}\phi = \frac{\partial\phi}{\partial t} + \mathbf{v} \cdot \nabla\phi.} \quad (287)$$

Physical significance: The first term is the explicit time change at a fixed point; the second term is the change seen because the fluid element moves into a neighboring region where the field has a different value.

Equivalently, as an operator,

$$\boxed{\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla.} \quad (288)$$

Physical significance: The material derivative is the time derivative in the frame that accompanies the moving fluid element.

Physical Concept 12.2: Material Derivative

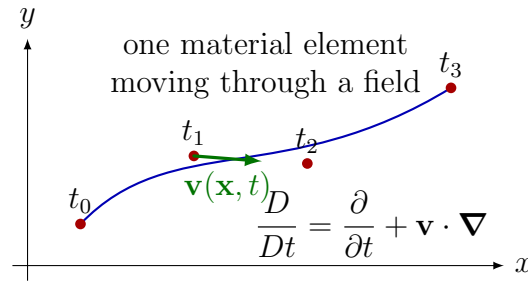
The material derivative $\frac{D}{Dt}$ means “differentiate while riding with the same fluid element.” It is the correct derivative for acceleration because Newton’s law follows matter, not a fixed point in space.

12.3.2 Derivation from a Particle Trajectory

Let a fluid element be labeled by its initial position $\mathbf{X}$, and let its position at time t be

$$\mathbf{x} = \mathbf{x}(\mathbf{X}, t). \quad (289)$$

Along this particular element, the scalar field becomes the single-variable function



$$\Phi(t) = \phi(\mathbf{x}(\mathbf{X}, t), t). \quad (290)$$

Applying the chain rule,

$$\frac{d\Phi}{dt} = \frac{\partial\phi}{\partial t} + \frac{\partial\phi}{\partial x_i} \frac{dx_i}{dt}, \quad (291)$$

$$= \frac{\partial\phi}{\partial t} + v_i \partial_i \phi, \quad (292)$$

$$= \frac{\partial\phi}{\partial t} + \mathbf{v} \cdot \nabla \phi. \quad (293)$$

This is exactly equation (287).

The same operator applies component by component to vector fields. In particular, the acceleration of a fluid element is

$$\mathbf{a} = \frac{D}{Dt} \mathbf{v} = \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v}. \quad (294)$$

Physical significance: A flow can accelerate a fluid element even when the velocity field is steady in time, because the element moves from one location to another where the velocity is different.

In index notation,

$$a_i = \frac{\partial v_i}{\partial t} + v_j \partial_j v_i. \quad (295)$$

Physical significance: The index i labels the component of acceleration, while j is summed over and describes the direction along which the fluid element is advected.

This is the fluid-mechanics analogue of the total derivative used in analytical mechanics. There, one follows a trajectory in phase space; here, one follows a continuum of material elements carried by the flow.

12.4 Momentum Balance and Euler's Equation

Physical Motivation

Once the acceleration of a material element is known, the equation of motion is still Newton's second law: mass times acceleration equals force. The only new point is that the forces are continuum forces, produced by stresses on surfaces and by body forces such as gravity.

12.4.1 General Momentum Balance

Consider a small material volume element ΔV with mass

$$\Delta m = \rho \Delta V. \quad (296)$$

Newton's second law is

$$\Delta m \frac{D}{Dt} \mathbf{v} = \Delta \mathbf{F}. \quad (297)$$

The force contains an internal stress contribution and an external body-force contribution:

$$\Delta \mathbf{F} = (\nabla \cdot \boldsymbol{\sigma}) \Delta V + \rho \mathbf{g} \Delta V, \quad (298)$$

where $\mathbf{g}$ denotes the acceleration due to the external force field. Dividing by $\Delta m = \rho \Delta V$ gives

$$\boxed{\frac{D}{Dt} \mathbf{v} = \frac{1}{\rho} \nabla \cdot \boldsymbol{\sigma} + \mathbf{g}.} \quad (299)$$

Physical significance: Fluid acceleration is driven by stress gradients per unit mass and by external acceleration fields.

In components this is

$$\boxed{\frac{D}{Dt} v_i = \frac{1}{\rho} \partial_j \sigma_{ij} + g_i.} \quad (300)$$

Physical significance: The divergence of the stress tensor is a force density; dividing by ρ converts it into acceleration.

12.4.2 Euler Equation for an Ideal Fluid

Substituting the ideal-fluid stress (278) into the stress divergence gives

$$\partial_j \sigma_{ij} = \partial_j (-p \delta_{ij}), \quad (301)$$

$$= -\delta_{ij} \partial_j p, \quad (302)$$

$$= -\partial_i p. \quad (303)$$

Therefore

$$\boxed{\frac{D}{Dt} \mathbf{v} = -\frac{1}{\rho} \nabla p + \mathbf{g}.} \quad (304)$$

Physical significance: An ideal fluid accelerates from high pressure toward low pressure, with body forces added in the usual way.

Writing out the material derivative,

$$\boxed{\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \mathbf{g}.} \quad (305)$$

Physical significance: Euler's equation is Newton's second law for an inviscid continuum.

This equation is more general than hydrostatics. If $\mathbf{v} = 0$ and the fields are time independent, the left side vanishes and equation (305) reduces to

$$\nabla p = \rho \mathbf{g}, \quad (306)$$

which is the hydrostatic equation recalled at the beginning of the lecture.

The unknowns in the ideal-fluid problem are ρ , p , and the three components of $\mathbf{v}$. Continuity gives one scalar equation, Euler's equation gives three scalar equations, and one more scalar relation is still needed. That last relation is supplied by the equation of state.

12.5 Mechanical Closure and Equations of State

Physical Motivation

A gas or liquid is not only mechanical; it also has temperature, entropy, and internal energy. A fully general fluid theory must follow those thermal variables. In this lecture we restrict to mechanically closed situations, where the thermal behavior can be summarized by a relation between pressure and density.

12.5.1 Barotropic Closure

The closure assumption is that pressure and density are connected by an algebraic equation of state

$$\boxed{f(p, \rho) = 0.} \quad (307)$$

Physical significance: If this relation holds, knowing p determines ρ or knowing ρ determines p , so no independent thermal field is needed for the mechanical problem.

Equivalently, when the relation can be solved explicitly, one may write

$$\boxed{p = p(\rho).} \quad (308)$$

Physical significance: A fluid with $p = p(\rho)$ is called barotropic; pressure surfaces and density surfaces are locked together.

Note:

The word “closure” means that the mechanical equations now have enough relations to determine the mechanical unknowns. Without an equation of state, continuity and Euler’s equation do not know how pressure responds when a fluid element is compressed.

A useful diagnostic follows by taking the gradient of $f(p, \rho) = 0$:

$$f_p \nabla p + f_\rho \nabla \rho = 0. \quad (309)$$

Thus ∇p and $\nabla \rho$ are parallel, and

$$\boxed{\nabla p \times \nabla \rho = \mathbf{0}.} \quad (310)$$

Physical significance: If measured pressure and density gradients are not parallel, the flow cannot be described by a single equation of state $f(p, \rho) = 0$.

12.5.2 Incompressible Fluids

The simplest example is an incompressible fluid:

$$\boxed{\rho = \rho_0 = \text{const.}} \quad (311)$$

Physical significance: The density is fixed independently of pressure, so volume elements do not compress or expand.

Substituting $\rho = \rho_0$ into the continuity equation gives

$$\frac{\partial \rho_0}{\partial t} + \nabla \cdot (\rho_0 \mathbf{v}) = 0, \quad (312)$$

$$\rho_0 \nabla \cdot \mathbf{v} = 0. \quad (313)$$

Since $\rho_0 \neq 0$,

$$\boxed{\nabla \cdot \mathbf{v} = 0.} \quad (314)$$

Physical significance: In an incompressible flow, the velocity field has no local source or sink of volume.

12.5.3 Isentropic Ideal Gases

Another important class is isentropic flow. The entropy carried by each fluid element is constant:

$$\boxed{\frac{D}{Dt} s = 0,} \quad (315)$$

where s is entropy per unit mass. *Physical significance:* No heat flows between neighboring elements and no heat is produced inside an element, so the thermal state is carried along with the motion.

For an isentropic ideal gas the equation of state has the adiabatic form

$$p = K \rho^\gamma, \quad (316)$$

where K is constant along the chosen isentrope and γ is the adiabatic index. *Physical significance:* Compression raises the pressure more strongly than in an isothermal gas because the temperature also rises.

For an ideal gas with f active molecular degrees of freedom,

$$\gamma = 1 + \frac{2}{f}. \quad (317)$$

Physical significance: More internal degrees of freedom store energy during compression, so they reduce the pressure response and lower γ .

Typical values are

$$\gamma = \frac{5}{3} \quad \text{monatomic}, \quad \gamma = \frac{7}{5} \quad \text{diatomic}, \quad \gamma = \frac{4}{3} \quad \text{polyatomic with } f = 6. \quad (318)$$

Physical significance: The equation of state depends on microscopic structure, but once it is known it closes the macroscopic mechanical equations.

12.5.4 Enthalpy per Unit Mass

For isentropic motion it is useful to introduce the enthalpy per unit mass, denoted here by h_{th} . Thermodynamics gives

$$dh_{\text{th}} = T ds + \frac{1}{\rho} dp. \quad (319)$$

Along an isentrope $ds = 0$, so

$$dh_{\text{th}} = \frac{1}{\rho} dp. \quad (320)$$

Physical significance: In an isentropic ideal fluid, pressure work can be represented as a gradient of enthalpy per unit mass.

Taking a spatial gradient,

$$\nabla h_{\text{th}} = \frac{1}{\rho} \nabla p. \quad (321)$$

Physical significance: This identity makes the pressure force behave like a conservative force.

12.6 The Closed Ideal-Fluid System and Boundary Conditions

Physical Motivation

Differential equations do not define a physical flow by themselves. We also need initial conditions and boundary conditions. The most basic boundary condition says that a fluid cannot pass through an impermeable wall.

12.6.1 Counting Unknowns and Equations

The unknowns are

$$\boxed{\rho, \quad p, \quad \mathbf{v}.} \quad (322)$$

Physical significance: In three dimensions this is five scalar unknowns: two scalars and one vector.

The equations are

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0,} \quad (323)$$

Physical significance: This is one scalar equation expressing mass conservation.

$$\boxed{\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \mathbf{g},} \quad (324)$$

Physical significance: This is a vector equation, giving three scalar momentum equations.

$$\boxed{f(p, \rho) = 0.} \quad (325)$$

Physical significance: This final scalar equation supplies the mechanical closure.

Thus the number of scalar equations matches the number of scalar unknowns. The system is mechanically closed, but a particular solution still requires initial and boundary data.

12.6.2 Impermeable Boundary

Let $\hat{\mathbf{n}}$ be the unit normal to a fixed impermeable wall. The normal component of the fluid velocity at the wall must vanish:

$$\boxed{\mathbf{v} \cdot \hat{\mathbf{n}} = 0.} \quad (326)$$

Physical significance: The fluid may slide along an ideal wall, but it cannot flow through the wall.

For an ideal fluid there is no no-slip condition. The tangential velocity at the wall is not forced to vanish:

$$\boxed{\mathbf{v}_{\parallel} \text{ may be nonzero at an ideal boundary.}} \quad (327)$$

Physical significance: No viscosity means no tangential friction force that would glue the fluid to the wall.

For a viscous fluid, by contrast, the tangential velocity at a stationary wall will also vanish. That stronger boundary condition belongs to the later, viscous theory.

12.7 How These Field Equations Are Solved

Physical Motivation

The equations above are field equations: they must hold at every point in space and time. This makes them partial differential equations. Sometimes symmetry gives analytic solutions; in general, one solves them numerically.

12.7.1 Analytic Versus Numerical Solutions

An analytic solution gives formulae such as

$$\rho = \rho(\mathbf{x}, t), \quad p = p(\mathbf{x}, t), \quad \mathbf{v} = \mathbf{v}(\mathbf{x}, t). \quad (328)$$

Such solutions are especially valuable because they expose parameter dependence and physical structure.

For general geometries one discretizes space and time:

$$t \rightarrow t_n, \quad \mathbf{x} \rightarrow \mathbf{x}_j. \quad (329)$$

The fields become a finite set of values,

$$\rho(\mathbf{x}, t) \rightarrow \rho_j^n, \quad p(\mathbf{x}, t) \rightarrow p_j^n, \quad \mathbf{v}(\mathbf{x}, t) \rightarrow \mathbf{v}_j^n. \quad (330)$$

Derivatives are replaced by difference approximations. For example,

$$\boxed{\frac{\partial f}{\partial x} \simeq \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}}. \quad (331)$$

Physical significance: A continuum field problem is converted into a large algebraic or time-stepping problem for finitely many numbers.

Worked Example 12.1 (Finite-Difference Mindset)

The computer does not store a whole continuous function. It stores numbers such as p_j^n and ρ_j^n on grid points, then replaces derivatives by algebraic differences. The physical equation is still the same conservation law, but its numerical form is a rule for updating finitely many stored values.

Computational fluid dynamics therefore sits between theory and experiment. It uses theoretical equations, but the outcome of a complex simulation is often not obvious before the computation is performed.

12.8 Bernoulli's Equation as Energy Conservation

Physical Motivation

After writing Newton's law, the next useful step in mechanics is usually to find conserved quantities. Bernoulli's equation is the corresponding energy conservation statement for an ideal flow, under additional assumptions.

12.8.1 Mechanical Analogy

For a particle in a conservative potential $U(\mathbf{x})$,

$$m\ddot{\mathbf{x}} = -\nabla U. \quad (332)$$

Taking the scalar product with $\dot{\mathbf{x}}$ gives

$$m\ddot{\mathbf{x}} \cdot \dot{\mathbf{x}} = -\nabla U \cdot \dot{\mathbf{x}}. \quad (333)$$

The two sides are total time derivatives:

$$\frac{d}{dt} \left(\frac{1}{2} m \dot{\mathbf{x}}^2 \right) = -\frac{dU}{dt}. \quad (334)$$

Therefore

$$\boxed{\frac{1}{2} m \dot{\mathbf{x}}^2 + U = \text{const.}} \quad (335)$$

Physical significance: Multiplying Newton's law by velocity converts a force law into an energy law.

12.8.2 Derivation for an Ideal Isentropic Flow

Now take Euler's equation with no external body force:

$$\frac{D}{Dt}\mathbf{v} = -\frac{1}{\rho}\nabla p. \quad (336)$$

Assume the flow is isentropic, so equation (321) applies:

$$\frac{1}{\rho}\nabla p = \nabla h_{\text{th}}. \quad (337)$$

Then

$$\frac{D}{Dt}\mathbf{v} = -\nabla h_{\text{th}}. \quad (338)$$

Take the scalar product with $\mathbf{v}$:

$$\mathbf{v} \cdot \frac{D}{Dt}\mathbf{v} = -\mathbf{v} \cdot \nabla h_{\text{th}}. \quad (339)$$

The left side is

$$\mathbf{v} \cdot \frac{D}{Dt}\mathbf{v} = \frac{D}{Dt}\left(\frac{1}{2}v^2\right). \quad (340)$$

For steady flow, $\partial h_{\text{th}}/\partial t = 0$, so

$$\frac{D}{Dt}h_{\text{th}} = \frac{\partial h_{\text{th}}}{\partial t} + \mathbf{v} \cdot \nabla h_{\text{th}} = \mathbf{v} \cdot \nabla h_{\text{th}}. \quad (341)$$

Therefore equation (339) becomes

$$\frac{D}{Dt}\left(\frac{1}{2}v^2\right) = -\frac{D}{Dt}h_{\text{th}}. \quad (342)$$

Combining the two material derivatives,

$$\frac{D}{Dt}\left(\frac{1}{2}v^2 + h_{\text{th}}\right) = 0. \quad (343)$$

Hence, along a streamline,

$$\boxed{\frac{1}{2}v^2 + h_{\text{th}} = \text{const.}} \quad (344)$$

Physical significance: Along the path of a fluid element, kinetic energy per unit mass and enthalpy per unit mass trade off while their sum remains fixed.

12.8.3 Incompressible Form

For an incompressible fluid, $\rho = \rho_0$ and

$$dh_{\text{th}} = \frac{1}{\rho_0} dp. \quad (345)$$

Integrating,

$$h_{\text{th}} = \frac{p}{\rho_0} + \text{const.} \quad (346)$$

Substituting into equation (344),

$$\boxed{\frac{1}{2}v^2 + \frac{p}{\rho_0} = \text{const} \quad \text{along a streamline.}} \quad (347)$$

Physical significance: In a steady incompressible ideal flow, regions of larger speed have smaller pressure along the same streamline.

Multiplying by ρ_0 gives the common pressure form:

$$\boxed{\frac{1}{2}\rho_0 v^2 + p = \text{const} \quad \text{along a streamline.}} \quad (348)$$

Physical significance: Dynamic pressure $\frac{1}{2}\rho_0 v^2$ and static pressure p exchange like kinetic and potential energy densities.

If a conservative body force such as gravity is retained, a potential term is added. For $\mathbf{g} = -g\hat{\mathbf{z}}$,

$$\boxed{\frac{1}{2}v^2 + \frac{p}{\rho_0} + gz = \text{const} \quad \text{along a streamline.}} \quad (349)$$

Physical significance: Height, pressure, and speed are three reservoirs of mechanical energy per unit mass.

12.9 Summary

The ideal-fluid model developed in this lecture is

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \quad \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \mathbf{g}, \quad f(p, \rho) = 0.} \quad (350)$$

Physical significance: These are mass conservation, Newton's second law, and mechanical closure.

The central new mathematical object is the material derivative:

$$\boxed{\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla.} \quad (351)$$

Physical significance: It is the derivative seen by a moving fluid element.

For an impermeable ideal boundary,

$$\boxed{\mathbf{v} \cdot \hat{\mathbf{n}} = 0, \quad \mathbf{v}_{\parallel} \text{ need not vanish.}} \quad (352)$$

Physical significance: Ideal fluids cannot cross walls, but without viscosity they may slide along them.

Under the additional assumptions of steady, ideal, isentropic flow with no nonconservative forcing,

$$\boxed{\frac{1}{2}v^2 + h_{\text{th}} = \text{const}} \quad (353)$$

Physical significance: Bernoulli's equation is the energy conservation law obtained from Euler's equation along a streamline.

Further Reading

- L. D. Landau and E. M. Lifshitz, *Fluid Mechanics*, Secs. 1–5: ideal fluids, Euler's equation, and Bernoulli's theorem.
- G. K. Batchelor, *An Introduction to Fluid Dynamics*, Ch. 1: kinematics, material derivative, and inviscid equations.
- D. J. Acheson, *Elementary Fluid Dynamics*, Chs. 1–2: continuity, Euler's equation, and elementary Bernoulli applications.
- Any analytical mechanics text, section on energy conservation: the derivation by dotting Newton's equation with velocity is the template for Bernoulli's theorem.

13 Lecture 13 – Bernoulli’s Equation and Applications

Date: May 20, 2026

Overview

This lecture has two goals. First, we re-derive Bernoulli’s equation carefully, including the gravitational-potential term that was omitted in the previous session. Second, we apply it to three canonical problems: a draining tank (Torricelli’s theorem), the Venturi effect, and the lift on an airfoil. The lecture closes with a preview of the next topic: systematic derivation of conservation laws from an action principle, which requires switching from the Eulerian to the Lagrangian description of fluid motion.

13.1 Complete Derivation of Bernoulli’s Equation

Physical Motivation

Bernoulli’s equation is the energy-conservation law of an ideal fluid along a streamline. Last lecture the derivation was done rapidly and the gravitational potential term was dropped. We now redo it from scratch. Euler’s equation is the starting point; dotting it with the velocity vector converts a force law into an energy law, exactly as in particle mechanics.

13.1.1 Setup: Steady Flow with a Conservative Body Force

We assume

- *Steady flow*: all partial time derivatives vanish, $\partial_t(\cdot) = 0$.
- *Ideal (inviscid) fluid*: stress is purely isotropic, $\sigma_{ij} = -p\delta_{ij}$.
- *Conservative body force*: the body-force acceleration is $\mathbf{g} = -\nabla\Phi$, where $\Phi(\mathbf{x})$ is the external potential (e.g. $\Phi = gz$ for uniform gravity).
- *Barotropic fluid*: pressure and density satisfy $p = p(\rho)$, so the specific enthalpy

$$h_{\text{th}} = \int \frac{dp}{\rho(p)}$$

is well-defined, and $\nabla h_{\text{th}} = \frac{1}{\rho} \nabla p$.

Euler’s equation is

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \mathbf{g}. \quad (354)$$

Under steady flow $\partial_t \mathbf{v} = 0$, and substituting the conservative body force,

$$(\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla h_{\text{th}} - \nabla \Phi. \quad (355)$$

13.1.2 Dot with $\mathbf{v}$

Take the scalar product of (355) with $\mathbf{v}$:

$$\mathbf{v} \cdot [(\mathbf{v} \cdot \nabla) \mathbf{v}] = -\mathbf{v} \cdot \nabla h_{\text{th}} - \mathbf{v} \cdot \nabla \Phi. \quad (356)$$

The left side simplifies because

$$\mathbf{v} \cdot [(\mathbf{v} \cdot \nabla) \mathbf{v}] = (\mathbf{v} \cdot \nabla) \frac{1}{2} v^2. \quad (357)$$

For steady flow, $\partial_t = 0$, so the material derivative satisfies $\frac{D}{Dt}(\cdot) = \mathbf{v} \cdot \nabla(\cdot)$. Thus the right side can also be written as material derivatives:

$$\mathbf{v} \cdot \nabla h_{\text{th}} = \frac{D}{Dt} h_{\text{th}}, \quad (358)$$

$$\mathbf{v} \cdot \nabla \Phi = \frac{D}{Dt} \Phi. \quad (359)$$

(These equalities rely on steady flow, i.e., $\partial_t h_{\text{th}} = \partial_t \Phi = 0$.)

Equation (356) therefore becomes

$$\frac{D}{Dt} \left(\frac{1}{2} v^2 \right) = - \frac{D}{Dt} h_{\text{th}} - \frac{D}{Dt} \Phi, \quad (360)$$

which rearranges to

$$\frac{D}{Dt} \left(\frac{1}{2} v^2 + h_{\text{th}} + \Phi \right) = 0. \quad (361)$$

The material derivative is zero along the path of any fluid element, so the quantity in the parenthesis is constant along every streamline:

Physical Law 13.1 Bernoulli's Equation (general barotropic form)

For steady, ideal, barotropic flow in a conservative external field Φ ,

$$\boxed{\frac{1}{2} v^2 + h_{\text{th}} + \Phi = \text{const along a streamline,}} \quad (362)$$

where $h_{\text{th}} = \int dp/\rho$ is the specific enthalpy and Φ is the external potential.

Significance: The three terms are kinetic energy per unit mass, pressure-work energy per unit mass, and gravitational potential energy per unit mass. Their sum is conserved along any streamline. The previously missing term is Φ .

13.1.3 Important Special Cases

Incompressible fluid with uniform gravity. For $\rho = \rho_0 = \text{const}$, $h_{\text{th}} = p/\rho_0$, and $\Phi = gz$:

$$\boxed{\frac{1}{2} v^2 + \frac{p}{\rho_0} + gz = \text{const along a streamline.}} \quad (363)$$

Multiplied by ρ_0 ,

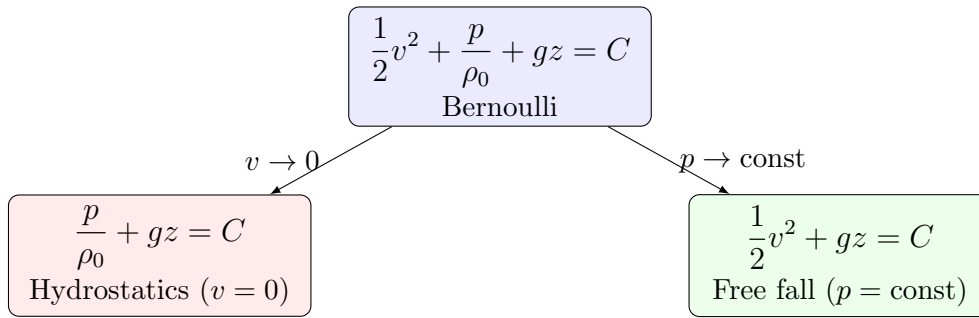
$$\boxed{\frac{1}{2} \rho_0 v^2 + p + \rho_0 gz = \text{const.}} \quad (364)$$

Static limit ($\mathbf{v} = 0$). The kinetic term vanishes and Bernoulli reduces to the hydrostatic equation:

$$p + \rho_0 gz = \text{const} \quad \implies \quad \nabla p = -\rho_0 g \hat{\mathbf{z}} = \rho_0 \mathbf{g}. \quad (365)$$

Significance: Bernoulli unifies hydrostatics and free-fall energy conservation into one equation.

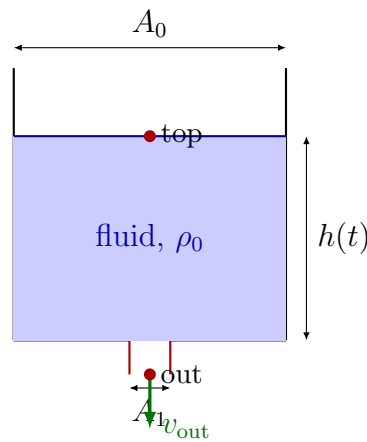
Physical interpretation. Bernoulli = hydrostatics + free-fall in two limits:



13.2 Application 1: Draining Tank (Torricelli's Theorem)

Problem Statement

A tank with large cross-sectional area A_0 is filled with an incompressible fluid to height h . A small tap of cross-section $A_1 \ll A_0$ is opened at the bottom. Find (a) the exit speed v_{out} , (b) the total draining time, and (c) the profile of the falling stream below the tap.



13.2.1 Part (a): Exit Speed — Torricelli's Theorem

Mathematical Setup

Apply Bernoulli along a streamline connecting the top surface to the exit. Choose $z = 0$ at the tap:

Quantity	Top surface	Exit (just outside tap)
p	p_0 (atmosphere)	p_0 (boundary condition)
z	h	0
v	$v_{\text{top}} \approx 0$	v_{out}

The boundary condition $p_{\text{out}} = p_0$ follows from the no-stress condition at a free surface: just outside the tap the fluid is in contact with the atmosphere.

The approximation $v_{\text{top}} \approx 0$ is justified by the continuity equation:

$$\rho_0 v_{\text{top}} A_0 = \rho_0 v_{\text{out}} A_1 \quad \implies \quad v_{\text{top}} = v_{\text{out}} \frac{A_1}{A_0} \ll v_{\text{out}}. \quad (366)$$

Substituting into (363):

$$\frac{1}{2} v_{\text{top}}^2 + \frac{p_0}{\rho_0} + gh = \frac{1}{2} v_{\text{out}}^2 + \frac{p_0}{\rho_0} + 0. \quad (367)$$

The pressure and (to leading order) kinetic-energy terms at the top cancel, giving:

Physical Law 13.2 Torricelli's Theorem

$$v_{\text{out}} = \sqrt{2gh}, \quad (368)$$

where h is the height of the fluid surface above the tap.

Significance: The exit speed equals the speed a body would acquire falling freely through the same height h . Bernoulli's equation converts gravitational potential energy into kinetic energy, just as in free fall — the pressure at entry and exit is the same, so pressure does no net work.

13.2.2 Part (b): Draining Time**Physical Motivation**

As the fluid drains, the surface drops: $h = h(t)$. To find how long draining takes we use continuity to relate $\dot{h}$ to v_{out} .

The rate at which the fluid surface drops times the tank area equals the volume flux out of the tap:

$$A_0 \left(-\frac{dh}{dt} \right) = A_1 v_{\text{out}}(h) = A_1 \sqrt{2gh}. \quad (369)$$

Rearranging gives a separable ODE:

$$\frac{dh}{\sqrt{h}} = -\frac{A_1}{A_0} \sqrt{2g} dt. \quad (370)$$

Integrating from h_0 (initial level) to 0 (empty):

$$\int_{h_0}^0 \frac{dh}{\sqrt{h}} = -\frac{A_1}{A_0} \sqrt{2g} \int_0^T dt, \quad (371)$$

$$\left[2\sqrt{h} \right]_0^{h_0} = \frac{A_1}{A_0} \sqrt{2g} T. \quad (372)$$

Note:

The integrand $h^{-1/2}$ is integrable at $h \rightarrow 0$ (its integral $2\sqrt{h}$ remains finite), so the tank drains in *finite* time despite the exit velocity going to zero as the tank empties.

Physical Law 13.3 Total Draining Time

$$T = \frac{A_0}{A_1} \sqrt{\frac{2h_0}{g}}. \quad (373)$$

Significance: The draining time scales with the area ratio A_0/A_1 (a large tank drains slowly through a small hole) and with $\sqrt{h_0/g}$ (the free-fall timescale for height h_0).

13.2.3 Part (c): Stream Profile Below the Tap**Physical Motivation**

Open a tap and the stream narrows as it falls. Why? Gravity accelerates the fluid, increasing its speed. Continuity then requires the cross-sectional area to shrink proportionally.

Let z be the distance *below* the tap (so the tap is at $z = 0$). The speed of the jet at depth z is found by applying Bernoulli again between the tap exit and a point at depth z along the free jet.

Both points are exposed to atmosphere ($p = p_0$):

$$\begin{aligned}\frac{1}{2}v_{\text{out}}^2 + 0 &= \frac{1}{2}v(z)^2 - gz \\ v(z) &= \sqrt{v_{\text{out}}^2 + 2gz}.\end{aligned}\tag{374}$$

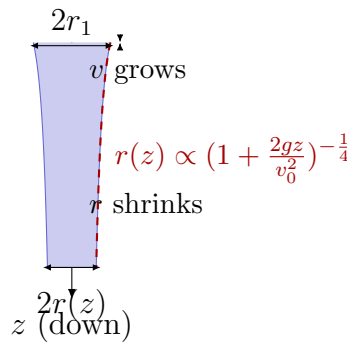
By the continuity equation (incompressible):

$$A(z)v(z) = A_1 v_{\text{out}} \quad \Longrightarrow \quad A(z) = A_1 \frac{v_{\text{out}}}{\sqrt{v_{\text{out}}^2 + 2gz}}.\tag{375}$$

Since $A = \pi r^2$, the radius profile is

$$r(z) = r_1 \left(1 + \frac{2gz}{v_{\text{out}}^2}\right)^{-1/4},\tag{376}$$

where $r_1 = \sqrt{A_1/\pi}$.

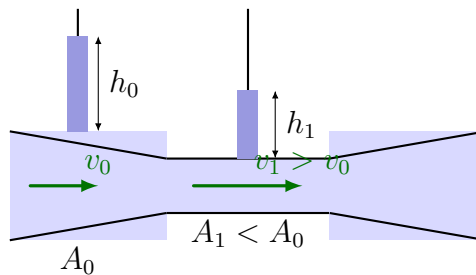


Significance: The stream narrows as $(1 + 2gz/v_0^2)^{-1/4}$ — slower than $1/v$ because $r \propto \sqrt{A}$. Surface tension maintains the cylindrical shape by minimizing surface area (a circle has the smallest perimeter for a given cross-sectional area), so the stream does not split until it is thin enough for surface tension to form droplets.

13.3 Application 2: The Venturi Effect

Physical Motivation

Giovanni Venturi (Italian physicist, 1801) placed a small tube perpendicular to a constricted pipe. He observed that the liquid in the tube rose to a *different* height compared to a tube placed in the wide section. This is a direct measurement of pressure via Bernoulli's equation.



13.3.1 Analysis

By continuity:

$$v_0 A_0 = v_1 A_1 \implies v_1 = v_0 \frac{A_0}{A_1} > v_0. \quad (377)$$

Apply Bernoulli along the central streamline (horizontal pipe, so $\Delta z \approx 0$):

$$\frac{1}{2} \rho_0 v_0^2 + p_0 = \frac{1}{2} \rho_0 v_1^2 + p_1. \quad (378)$$

Since $v_1 > v_0$, we must have $p_1 < p_0$.

The pressure in each section equals the atmospheric pressure plus the hydrostatic head of the rising column:

$$p_0 = p_{\text{atm}} + \rho_0 g h_0, \quad (379)$$

$$p_1 = p_{\text{atm}} + \rho_0 g h_1. \quad (380)$$

Therefore $p_1 < p_0$ implies $h_1 < h_0$.

Physical Concept 13.1: Venturi Tube

A device in which a constriction in a pipe causes an increase in fluid velocity and a corresponding drop in pressure, measurable by the height of fluid in a perpendicular manometer tube. The height difference satisfies

$$h_0 - h_1 = \frac{v_1^2 - v_0^2}{2g} = \frac{v_0^2}{2g} \left[\left(\frac{A_0}{A_1} \right)^2 - 1 \right]. \quad (381)$$

Significance: The Venturi tube converts a velocity measurement into a pressure (height) measurement. It is one of the earliest flow-rate meters.

Consequence 13.4 Venturi Effect

$$\text{speed} \uparrow \iff \text{pressure} \downarrow \quad (\text{same streamline, same height}). \quad (382)$$

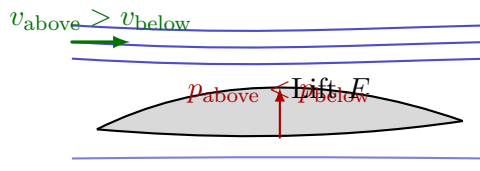
13.4 Application 3: Bernoulli and Airfoil Lift

Physical Motivation

Airplane wings have been flying for over a century, yet the precise mechanism of lift is still debated. Here we present the standard Bernoulli-based argument, noting its limitations.

13.4.1 Standard Argument

In the frame of the wing, the airflow approaches from the left. The wing is asymmetric: the upper surface is more curved (convex) than the lower surface.



The upper streamlines are compressed into a narrower region $\rightarrow$ by continuity, $v_{\text{above}} > v_{\text{below}}$. By Bernoulli (neglecting the small height difference and gravitational term),

$$p_{\text{above}} + \frac{1}{2}\rho v_{\text{above}}^2 = p_{\text{below}} + \frac{1}{2}\rho v_{\text{below}}^2 \implies p_{\text{above}} < p_{\text{below}}. \quad (383)$$

The pressure difference creates an upward net force:

$$F_{\text{lift}} = (p_{\text{below}} - p_{\text{above}}) A_{\text{wing}}. \quad (384)$$

Remark 8. This argument is **contested**. A flat plate tilted at an angle of attack also generates lift, which suggests that streamline curvature due to wing shape is not the only mechanism. Additionally, for air (a compressible gas), density variations, viscosity, and boundary-layer effects all contribute. The argument is presented here as a qualitative illustration of Bernoulli's equation, not as a complete theory of lift.

13.5 Preview: Conservation Laws and Lagrangian Description

Physical Motivation

Bernoulli's equation is one conservation law. In mechanics, Noether's theorem tells us that every continuous symmetry of the action yields a conservation law. Can we derive fluid conservation laws from symmetry in a systematic way?

13.5.1 The Two Descriptions of Fluid Motion

The current Eulerian description tracks fields at fixed points $\mathbf{x}$:

$$\rho(\mathbf{x}, t), \quad p(\mathbf{x}, t), \quad \mathbf{v}(\mathbf{x}, t). \quad (385)$$

An action principle, however, requires tracking individual fluid particles — this is the *Lagrangian description*:

$$\mathbf{x} = \mathbf{x}(\mathbf{X}, t), \quad (386)$$

where $\mathbf{X}$ labels a particle by its initial position and $\mathbf{x}$ is its current position.

13.5.2 Key Points for Next Lecture

- An action principle exists for ideal (inviscid) fluids; it does not exist for viscous fluids (friction is dissipative).
- Landau & Lifshitz derive conservation laws by guessing the energy density, energy flux, and momentum flux — then verifying they satisfy continuity-like equations using the equations of motion.
- A more systematic route uses the action and Noether's theorem.
- The first step is to switch to the Lagrangian description, which is the topic of Lecture 14.

Note:

The Eulerian description is convenient for measurement (velocities at fixed sensors) but is inconvenient for the action principle. The Lagrangian description follows material elements and is the natural setting for variational mechanics.

Summary

The complete Bernoulli equation for incompressible flow under gravity is

$$\boxed{\frac{1}{2}\rho_0 v^2 + p + \rho_0 g z = \text{const along a streamline.}} \quad (387)$$

Three applications:

1. **Draining tank:** exit speed $v_{\text{out}} = \sqrt{2gh}$ (Torricelli), drain time $T = \frac{A_0}{A_1} \sqrt{2h_0/g}$, stream narrows as $r \propto (1 + 2gz/v_0^2)^{-1/4}$.
2. **Venturi:** higher speed in narrow section $\Rightarrow$ lower pressure, measured by a manometer column.
3. **Lift:** standard Bernoulli argument qualitatively explains lift; full theory is more complex.

Further Reading

- L. D. Landau and E. M. Lifshitz, *Fluid Mechanics*, Secs. 5–7: Bernoulli’s theorem, energy flux, and its applications.
- D. J. Acheson, *Elementary Fluid Dynamics*, Ch. 1: Bernoulli’s streamline theorem and applications.
- G. K. Batchelor, *An Introduction to Fluid Dynamics*, Ch. 3: flow through orifices, jets, and tubes.

Lecture 14 – Circulation, Vorticity, and Two Descriptions of Fluid Motion

Date: May 25, 2026

Overview

This lecture opens a new chapter in the fluid-dynamics portion of the course. Two intertwined threads run through the session. The first is *Kelvin's circulation theorem* — a conservation law that constrains when and where vortical motion can appear in ideal flow. The second is the shift from the Eulerian (field) description to the Lagrangian (particle-tracking) description, which is the natural language for an action principle. Both threads converge in the action functional for an ideal fluid, presented at the end of the lecture.

Topics:

1. Brief review: Venturi clarification and second explanation of airfoil lift
2. Kelvin's circulation theorem (1869)
3. Helmholtz decomposition: longitudinal (irrotational) vs. transverse (solenoidal) parts
4. Vorticity and differential forms of Kelvin's theorem
5. Eulerian vs. Lagrangian descriptions of fluid motion
6. Streamlines and pathlines
7. Conservation of energy and momentum in ideal flow
8. Action principle for ideal fluids (preview)

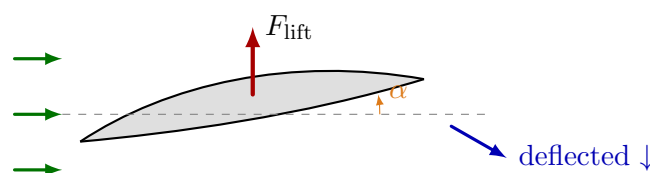
14.1 Brief Review: Venturi and Airfoil Lift

Venturi: Height Reference

A question left from Lecture 13: *relative to which point* do we measure height in the Venturi manometer columns? Answer: follow the *central streamline*. Apply Bernoulli between a point in the wide section (pressure p_0 , speed v_0) and a point in the narrow section (pressure p_1 , speed v_1). For a horizontal pipe the height terms cancel and the manometer-column height difference $h_0 - h_1$ directly encodes $\frac{1}{2}(v_1^2 - v_0^2)/g$.

Airfoil Lift: Angle of Attack

Last lecture gave the Bernoulli explanation for lift (upper surface more curved $\Rightarrow$ streamlines compressed $\Rightarrow v \uparrow, p \downarrow$). A complementary explanation uses the *angle of attack* α :



The tilted wing deflects air downward. By Newton's third law the air pushes the wing upward. This mechanism works even for a flat plate, and even when the wing flies inverted — as long as α is large enough. Both mechanisms (Bernoulli curvature + angle-of-attack deflection) contribute to real lift; the complete story requires solving the Navier-Stokes equations.

14.2 Kelvin's Circulation Theorem

Physical Motivation

Bernoulli's equation is a conservation law *along a streamline*. Can we find a global conservation law capturing the *rotational* character of flow? Kelvin's theorem answers yes: the circulation around any closed material curve is conserved in ideal barotropic flow.

14.2.1 Definition: Circulation

Physical Concept 14.1: Circulation

Given a closed oriented curve C , the *circulation* is

$$\Gamma(C) = \oint_C \mathbf{v} \cdot d\mathbf{l}, \quad (388)$$

where $d\mathbf{l}$ is the line element along C .

Significance: Circulation is the fluid analogue of the line integral of the magnetic field. By Stokes' theorem, $\Gamma = \iint_S \omega \cdot d\mathbf{S}$ where $\omega = \nabla \times \mathbf{v}$ is the vorticity (§14.4). A nonzero Γ signals enclosed vortical motion.

14.2.2 Material Curves

The crucial ingredient is that C is a *material curve*: a closed curve made of specific fluid particles that are carried along with the flow.



14.2.3 Key Kinematic Lemma

Mathematical Result 14.1 Rate of Change of a Material Line Element

If $d\mathbf{l}$ connects two adjacent fluid particles on a material curve, then

$$\frac{D(d\mathbf{l})}{Dt} = d\mathbf{v}, \quad (389)$$

where $d\mathbf{v}$ is the velocity difference between the two particles.

Proof: At time t , the two adjacent particles sit at $\mathbf{x}$ and $\mathbf{x} + d\mathbf{l}$ with velocities $\mathbf{v}(\mathbf{x})$ and $\mathbf{v}(\mathbf{x} + d\mathbf{l})$. After time dt they are at $\mathbf{x} + \mathbf{v} dt$ and $(\mathbf{x} + d\mathbf{l}) + \mathbf{v}(\mathbf{x} + d\mathbf{l}) dt$. The new separation is

$$d\mathbf{l}(t + dt) = d\mathbf{l}(t) + [\mathbf{v}(\mathbf{x} + d\mathbf{l}) - \mathbf{v}(\mathbf{x})] dt = d\mathbf{l}(t) + d\mathbf{v} dt.$$

Dividing by dt gives (389). ■

14.2.4 Proof of Kelvin's Theorem

Differentiate $\Gamma = \oint_C \mathbf{v} \cdot d\mathbf{l}$ along the material curve:

$$\frac{d\Gamma}{dt} = \oint_C \frac{D\mathbf{v}}{Dt} \cdot d\mathbf{l} + \oint_C \mathbf{v} \cdot \frac{D(d\mathbf{l})}{Dt}. \quad (390)$$

Term II. Using Lemma ??:

$$\oint_C \mathbf{v} \cdot d\mathbf{v} = \oint_C d\left(\frac{1}{2}v^2\right) = 0 \quad (\text{exact differential, closed loop}). \quad (391)$$

Term I. For an ideal barotropic fluid ($p = p(\rho)$) in a conservative field $\mathbf{g} = -\nabla\Phi$, Euler's equation gives

$$\frac{D\mathbf{v}}{Dt} = -\frac{1}{\rho}\nabla p - \nabla\Phi = -\nabla h_{\text{th}} - \nabla\Phi, \quad (392)$$

using $\nabla h_{\text{th}} = \frac{1}{\rho}\nabla p$. Therefore:

$$\oint_C \frac{D\mathbf{v}}{Dt} \cdot d\mathbf{l} = - \oint_C \nabla(h_{\text{th}} + \Phi) \cdot d\mathbf{l} = - \oint_C d(h_{\text{th}} + \Phi) = 0. \quad (393)$$

Both terms vanish:

Physical Law 14.2 Kelvin's Circulation Theorem (1869)

For an ideal, barotropic fluid in a conservative external field,

$$\boxed{\frac{d\Gamma}{dt} = 0 \quad \Longleftrightarrow \quad \Gamma(C(t)) = \text{const},} \quad (394)$$

where $C(t)$ is any closed material curve. The proof uses two ingredients: (1) the kinematic identity $D(d\mathbf{l})/Dt = d\mathbf{v}$, and (2) Euler's equation for ideal barotropic flow.

Significance: Ideal flow can neither create nor destroy circulation. If vortices are absent initially, they remain absent; vortices present at $t = 0$ cannot dissipate without viscosity. This is a topological conservation law: it holds for every material loop simultaneously.

14.2.5 Physical Discussion

No spontaneous vortex generation. Ideal flow starting irrotationally ($\omega = \mathbf{0}$ everywhere) stays irrotational forever. Real flows (waterfalls, pipe flow) generate vortices because viscosity creates vorticity at solid walls, violating the ideal-fluid assumption.

Historical note. William Thomson (1824–1907), later Lord Kelvin, proved the theorem in 1869. He is also known for the Kelvin temperature scale, the Second Law of Thermodynamics, and the first transatlantic telegraph cable. He is *not* J.J. Thomson (discoverer of the electron).

14.3 Helmholtz Decomposition

Physical Motivation

Kelvin's theorem conserves circulation, which is sensitive only to the *rotational* part of $\mathbf{v}$. The Helmholtz decomposition separates any vector field into a rotational part and an irrotational part, making this precise.

Physical Concept 14.2: Helmholtz Decomposition (Longitudinal-Transverse)

Any smooth vector field $\mathbf{V}$ (decaying at infinity) decomposes uniquely as

$$\mathbf{V} = \mathbf{V}_{\parallel} + \mathbf{V}_{\perp}, \quad (395)$$

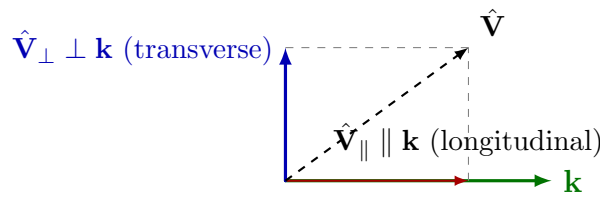
where

$$\mathbf{V}_{\parallel} \text{ (longitudinal) : } \nabla \times \mathbf{V}_{\parallel} = \mathbf{0} \Rightarrow \exists \phi : \mathbf{V}_{\parallel} = \nabla \phi, \quad (396)$$

$$\mathbf{V}_{\perp} \text{ (transverse) : } \nabla \cdot \mathbf{V}_{\perp} = 0. \quad (397)$$

14.3.1 Fourier-Space Picture

In Fourier space, decomposition by direction relative to $\mathbf{k}$ is immediate:



- $\hat{\mathbf{V}}_{\parallel} \parallel \mathbf{k}$: oscillation parallel to propagation — *compressional* (longitudinal) wave (e.g. sound, P-wave).
- $\hat{\mathbf{V}}_{\perp} \perp \mathbf{k}$: oscillation perpendicular to propagation — *shear* (transverse) wave (e.g. S-wave).

Note:

The Helmholtz decomposition appears in electromagnetism (Coulomb gauge, transverse photons), elasticity (Lec 7: P- and S-waves), and fluid mechanics. Students typically encounter it first in electromagnetism; the inventors of classical electrodynamics (Faraday, Helmholtz, Maxwell) knew it from fluid mechanics.

14.3.2 Connection to Circulation

Circulation probes only the transverse part of $\mathbf{v}$:

$$\Gamma = \oint_C \mathbf{v} \cdot d\mathbf{l} = \underbrace{\oint_C \mathbf{v}_{\parallel} \cdot d\mathbf{l}}_{=\oint_C d\phi = 0} + \oint_C \mathbf{v}_{\perp} \cdot d\mathbf{l}. \quad (398)$$

Kelvin's theorem is a conservation law for the transverse (rotational) component of $\mathbf{v}$ only.

14.4 Vorticity

Physical Concept 14.3: Vorticity

The *vorticity* is the curl of the velocity field:

$$\omega = \nabla \times \mathbf{v}. \quad (399)$$

A fluid element spinning at angular velocity $\boldsymbol{\Omega}$ has $\omega = 2\boldsymbol{\Omega}$.

Since $\nabla \times \mathbf{v}_{\parallel} = \mathbf{0}$, vorticity sees only the transverse part: $\omega = \nabla \times \mathbf{v}_{\perp}$.

14.4.1 Stokes' Theorem: Connecting Integral and Differential

Physical Law 14.3 Stokes' Theorem for Circulation

$$\Gamma(C) = \oint_C \mathbf{v} \cdot d\mathbf{l} = \iint_S \boldsymbol{\omega} \cdot d\mathbf{S}, \quad (400)$$

where S is any surface bounded by C .

Significance: This converts Kelvin's integral conservation law into a local statement. Kelvin: $d\Gamma/dt = 0 \iff$ the vorticity flux through any material surface is conserved.

14.4.2 Differential Forms of Kelvin's Theorem (stated without proof)

Two equivalent local formulations (see L&L §2):

Spatial (Eulerian) form:

$$\frac{\partial \boldsymbol{\omega}}{\partial t} + \nabla \times (\boldsymbol{\omega} \times \mathbf{v}) = \mathbf{0}. \quad (401)$$

Material (Lagrangian) form — vorticity equation:

$$\frac{D}{Dt} \boldsymbol{\omega} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{v}. \quad (402)$$

The right side is the *vortex-stretching term*: as a vortex tube stretches, its cross-section shrinks (incompressibility) and its vorticity intensifies.

Note:

For 2D incompressible flow ($\boldsymbol{\omega} = \omega \hat{\mathbf{z}}$, all quantities independent of z): the stretching term $(\boldsymbol{\omega} \cdot \nabla) \mathbf{v} = \omega \partial_z \mathbf{v} = 0$ vanishes. Therefore $D\omega/Dt = 0$ — vorticity is merely advected, never amplified, in 2D ideal flow.

14.5 Two Descriptions of Fluid Motion

Physical Motivation

All equations of motion so far have used fields $\rho(\mathbf{x}, t)$, $p(\mathbf{x}, t)$, $\mathbf{v}(\mathbf{x}, t)$ evaluated at a fixed point in space. This is the *Eulerian* description. An alternative: track each fluid particle individually. This is the *Lagrangian* description. Both are complete and equivalent, with complementary advantages.

14.5.1 Eulerian (Spatial) Description

Physical Concept 14.4: Eulerian Description

The fundamental variable is $\mathbf{v}(\mathbf{x}, t)$ — the velocity of whichever fluid particle happens to occupy position $\mathbf{x}$ at time t . The label $\mathbf{x}$ is a fixed location in space, *not* a specific particle.

14.5.2 Lagrangian (Material) Description

Physical Concept 14.5: Lagrangian Description

Each fluid particle is labeled by its initial position $\mathbf{X}$ (the Lagrangian label). The fundamental variable is the *trajectory map*

$$\mathbf{x} = \mathbf{x}(\mathbf{X}, t), \quad (403)$$

giving the position at time t of the particle that started at $\mathbf{X}$. The Eulerian velocity is recovered via

$$\mathbf{v}(\mathbf{x}, t) = \left. \frac{\partial \mathbf{x}(\mathbf{X}, t)}{\partial t} \right|_{\mathbf{x}=\mathbf{x}(\mathbf{X}, t)}. \quad (404)$$

Significance: The Lagrangian picture treats fluid mechanics as “mechanics of infinitely many particles,” one per label $\mathbf{X}$. It is the natural language for Noether’s theorem and the action principle (§14.8).

14.5.3 Equivalence and Conversion

Lagrangian $\rightarrow$ Eulerian. Given $\mathbf{x}(\mathbf{X}, t)$:

1. Compute $\mathbf{v}(\mathbf{x}, t) = \partial_t \mathbf{x}(\mathbf{X}, t)|_{\mathbf{x}}$.
2. Algebraically invert $\mathbf{x}(\mathbf{X}, t)$ to express $\mathbf{X}$ as a function of $(\mathbf{x}, t)$.

This is relatively easy (algebraic inversion).

Eulerian $\rightarrow$ Lagrangian. Given $\mathbf{v}(\mathbf{x}, t)$, find each particle’s trajectory by solving the initial-value problem

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}(\mathbf{x}(t), t), \quad \mathbf{x}(0) = \mathbf{X}. \quad (405)$$

This is a nonlinear ODE (the unknown $\mathbf{x}$ appears on both sides) — generally hard, especially in turbulent flows.

14.5.4 Advantages

	Eulerian	Lagrangian
Fundamental variable	$\mathbf{v}(\mathbf{x}, t)$	$\mathbf{x}(\mathbf{X}, t)$
Named after	Euler	Lagrange
Main advantage	Directly measurable	Enables action principle
Main difficulty	Nonlinear advection ($\mathbf{v} \cdot \nabla$) $\mathbf{v}$	ODE to convert to Eulerian

Note:

The material derivative $D/Dt = \partial_t + (\mathbf{v} \cdot \nabla)$ is exactly the time derivative *at fixed* $\mathbf{X}$ (i.e., following a particle). The definition of circulation on a material curve already uses the material derivative, so Kelvin’s theorem lives naturally in the Lagrangian description.

14.6 Streamlines and Pathlines

Physical Concept 14.6: Streamline

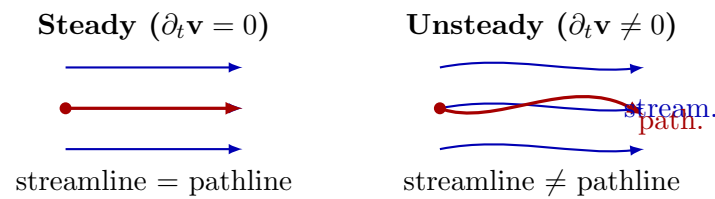
A *streamline* is a curve that is everywhere tangent to $\mathbf{v}(\mathbf{x}, t_0)$ at a *fixed* instant t_0 . It satisfies

$$\frac{dx_1}{v_1} = \frac{dx_2}{v_2} = \frac{dx_3}{v_3} \equiv d\lambda, \quad (406)$$

i.e., $d\mathbf{x}/d\lambda = \mathbf{v}(\mathbf{x}, t_0)$ with time *frozen*.

Physical Concept 14.7: Pathline (Trajectory)

A *pathline* is the actual trajectory of a fluid particle, i.e., the solution of (405). It follows the same particle through a time-varying velocity field.



Physical Law 14.4 Streamlines Equal Pathlines for Steady Flow

If $\partial_t \mathbf{v} = 0$, streamlines and pathlines coincide.

Proof: For steady flow $\mathbf{v}(\mathbf{x}, t) = \mathbf{v}(\mathbf{x})$. The pathline ODE $\dot{\mathbf{x}} = \mathbf{v}(\mathbf{x})$ has the same form as the streamline equation (406) (with $\lambda \rightarrow t$), so their solution curves are identical. ■

14.7 Conservation Laws: Energy and Momentum

Physical Motivation

We now have Bernoulli (Lec 13) and Kelvin (this lecture) as two conservation laws. Two more — energy and momentum — exist for ideal flow. They are stated here; derivations combine the continuity equation and Euler's equation (see L&L §6–7).

14.7.1 Energy Conservation

Define the *energy density per unit volume*:

$$\mathcal{E} = \rho\left(\frac{1}{2}v^2 + \epsilon\right) + \rho\Phi, \quad (407)$$

where ϵ is the internal (thermodynamic) energy per unit mass.

The *energy flux*:

$$\mathbf{j}_E = \left[\rho\left(\frac{1}{2}v^2 + \epsilon\right) + p + \rho\Phi\right]\mathbf{v} = (\mathcal{E} + p)\mathbf{v}. \quad (408)$$

Note:

The extra $p\mathbf{v}$ term (not present in $\mathcal{E}\mathbf{v}$) arises because pressure performs work as fluid moves: $p\mathbf{v}$ is the power per unit area transmitted to adjacent fluid. It converts internal energy ϵ into specific enthalpy $h_{\text{th}} = \epsilon + p/\rho$ in the flux.

Physical Law 14.5 Local Energy Conservation — Ideal Fluid

$$\boxed{\frac{\partial \mathcal{E}}{\partial t} + \nabla \cdot \mathbf{j}_E = 0.} \quad (409)$$

14.7.2 Momentum Conservation

The *momentum flux tensor* (stress tensor for an ideal fluid) is

$$\Pi_{ij} = \rho v_i v_j + p \delta_{ij}. \quad (410)$$

The two terms are: (1) $\rho v_i v_j$ — convective transport of the i -component of momentum in the j -direction; (2) $p \delta_{ij}$ — isotropic pressure contribution.

Physical Law 14.6 Local Momentum Conservation — Ideal Fluid

Without body forces,

$$\boxed{\frac{\partial(\rho v_i)}{\partial t} + \frac{\partial \Pi_{ij}}{\partial x_j} = 0, \quad \Pi_{ij} = \rho v_i v_j + p \delta_{ij}.} \quad (411)$$

Note:

With a conservative body force $f_i = -\rho \partial_i \Phi$ (gravity), a source term $-\rho \partial_i \Phi$ appears on the right. This is Euler's equation in conservation form.

14.8 Action Principle for Ideal Fluids (Preview)**Physical Motivation**

By Noether's theorem, every continuous symmetry of an action implies a conservation law. Can we derive the fluid conservation laws this way? Yes — but only for ideal (inviscid) fluids, since viscosity is dissipative and has no action. The action must be written in the Lagrangian description.

14.8.1 Compressible Ideal Fluid

The action for a compressible ideal fluid in Lagrangian variables is

$$S = \int dt \int d^3 \mathbf{X} \rho_0(\mathbf{X}) \left[\frac{1}{2} \left| \frac{\partial \mathbf{x}}{\partial t} \right|^2 - \epsilon(\rho) - \Phi(\mathbf{x}) \right], \quad (412)$$

where:

- $\mathbf{X}$ labels particles (Lagrangian variable); the integral is over all particles.
- $\rho_0(\mathbf{X})$ is the initial density (constant in time for each particle).
- $\frac{1}{2} |\partial_t \mathbf{x}|^2 = \frac{1}{2} v^2$ is kinetic energy per unit mass.
- $\epsilon(\rho)$ is internal energy per unit mass, where ρ is determined by mass conservation:

$$\rho(\mathbf{x}, t) = \frac{\rho_0(\mathbf{X})}{J}, \quad J = \det \left(\frac{\partial x_i}{\partial X_j} \right), \quad (413)$$

with J the Jacobian of the map $\mathbf{X} \rightarrow \mathbf{x}$. Mass conservation requires $\rho d^3 \mathbf{x} = \rho_0 d^3 \mathbf{X}$.

Varying (412) with respect to $\mathbf{x}(\mathbf{X}, t)$ (with generalized Euler-Lagrange equations for fields) yields Euler's equation. Energy and momentum conservation follow from time and space translation symmetry via Noether's theorem.

14.8.2 Incompressible Ideal Fluid

For incompressible flow, $\nabla \cdot \mathbf{v} = 0$ is equivalent to $J = 1$ at all times. This constraint is imposed by a Lagrange multiplier $p(\mathbf{x}, t)$:

$$S_{\text{inc}} = \int dt \int d^3\mathbf{X} \left[\frac{1}{2} \left| \frac{\partial \mathbf{x}}{\partial t} \right|^2 - \Phi(\mathbf{x}) \right] - \int dt \int d^3\mathbf{x} p(\mathbf{x}, t) (J - 1). \quad (414)$$

Physical Law 14.7 Pressure as Lagrange Multiplier

In (414), the pressure $p(\mathbf{x}, t)$ is precisely the Lagrange multiplier enforcing $J = 1$ (incompressibility $\nabla \cdot \mathbf{v} = 0$). Varying with respect to p gives $J = 1$; varying with respect to $\mathbf{x}$ gives Euler's equation.

Significance: In an incompressible ideal fluid, *pressure is not a thermodynamic variable* — it is a constraint force (like the normal force in constrained particle mechanics) that enforces $\nabla \cdot \mathbf{v} = 0$. Its value adjusts whatever is needed to keep the fluid incompressible.

Summary

- **Kelvin:** $d\Gamma/dt = 0$ for ideal barotropic flow on any material curve.
- **Vorticity:** $\omega = \nabla \times \mathbf{v}$; satisfies $D\omega/Dt = (\omega \cdot \nabla)\mathbf{v}$ (3D), $D\omega/Dt = 0$ (2D).
- **Helmholtz:** $\mathbf{v} = \mathbf{v}_{\parallel} + \mathbf{v}_{\perp}$; Kelvin involves only $\mathbf{v}_{\perp}$.
- **Eulerian:** $\mathbf{v}(\mathbf{x}, t)$ — measurable. **Lagrangian:** $\mathbf{x}(\mathbf{X}, t)$ — enables action principle.
- **Streamlines = pathlines** for steady flow.
- **Energy:** $\partial_t \mathcal{E} + \nabla \cdot [(\mathcal{E} + p)\mathbf{v}] = 0$.
- **Momentum:** $\partial_t(\rho v_i) + \partial_j(\rho v_i v_j + p\delta_{ij}) = 0$.
- **Pressure in incompressible flow** is the Lagrange multiplier for $\nabla \cdot \mathbf{v} = 0$.

Further Reading

- L. D. Landau and E. M. Lifshitz, *Fluid Mechanics* (Vol. 6): §2 (Kelvin's theorem, vorticity dynamics), §6–7 (energy and momentum conservation).
- V. I. Arnold, *Mathematical Methods of Classical Mechanics*: action principle and Noether's theorem for fluids.

15 Lecture 15 – Potential Flow

Date: May 27, 2026

Overview

With Euler's equation and the continuity equation in hand, we have a closed system for ideal flow, but the equations are nonlinear and we have not yet exhibited concrete flow fields. This lecture introduces a powerful simplifying assumption — *irrotational flow* (vorticity $\omega = \mathbf{0}$ everywhere) — that reduces the dynamics to a single scalar potential ϕ . Layering on incompressibility collapses the entire problem onto Laplace's equation $\nabla^2\phi = 0$, which we already know how to solve from electrostatics. The lecture closes with three worked examples: a point source, a source near a wall (image method), and a time-dependent flow from a syringe.

Topics:

1. Motivation from Kelvin: when is $\omega = \mathbf{0}$?
2. The velocity potential ϕ , its dimensions, and equipotential surfaces
3. Bernoulli-like equation from Euler's equation
4. Incompressible potential flow $\Rightarrow$ Laplace's equation
5. Boundary conditions: rigid wall and sources/sinks
6. Solution methods (numerical, symmetry, Fourier, Green functions, images)
7. Example 1: point source in 3D
8. Example 2: point source near a rigid wall (image method, stagnation point)
9. Example 3: time-dependent incompressible flow in a syringe

15.1 Irrotational Flow: Motivation and Definition

Conceptual Background

Kelvin's theorem (Lec 14) tells us that an ideal barotropic fluid cannot *generate* vorticity: $D\Gamma/Dt = 0$ along any material curve. In many natural setups (uniform flow at infinity, a body moving through fluid at rest) the vorticity is zero in the asymptotic region. Kelvin then guarantees it remains zero everywhere the fluid eventually visits. This makes *irrotational flow* a physically realistic regime, not merely a mathematical convenience.

Physical Concept 15.1: Irrotational (Potential) Flow

A flow is *irrotational* if

$$\omega \equiv \nabla \times \mathbf{v} = \mathbf{0} \quad \text{everywhere in the fluid.} \quad (415)$$

Significance: A uniform stream at infinity has $\nabla \times \mathbf{v} = \mathbf{0}$. Combined with Kelvin's theorem, this is automatically inherited throughout the flow for an ideal fluid – so flows past bodies in an otherwise undisturbed stream are irrotational.

15.1.1 Velocity Potential

Because $\nabla \times \mathbf{v} = \mathbf{0}$, the velocity field is a gradient:

Physical Concept 15.2: Velocity Potential

In an irrotational flow there exists a scalar field $\phi(\mathbf{x}, t)$, called the *velocity potential*, such that

$$\mathbf{v} = \nabla \phi. \quad (416)$$

(Note the + sign; in mechanics one writes $\mathbf{F} = -\nabla U$, but here the convention is opposite.)

Significance: A vector field with 3 components has been replaced by a single scalar ϕ – this is the entire payoff of the irrotational assumption. The Helmholtz decomposition (Lec 14) of $\mathbf{v} = \mathbf{v}_{\parallel} + \mathbf{v}_{\perp}$ has collapsed onto its longitudinal piece, with $\mathbf{v}_{\perp} = \mathbf{0}$.

15.1.2 Dimensions of ϕ

From $\mathbf{v} = \nabla \phi$:

$$[\phi] = [\mathbf{v}] \cdot [\text{length}] = \frac{L^2}{T}. \quad (417)$$

This is *not* the dimension of a mechanical potential energy. It is the dimension of **angular momentum per unit mass**, or equivalently, of **action per unit mass**. In Hamilton–Jacobi theory the action S satisfies $p = \partial S / \partial x$, exactly mirroring $v = \partial \phi / \partial x$. Thus ϕ is closer in spirit to the Hamilton–Jacobi action than to a force-potential.

Note:

The name “potential” is appropriate in the sense that ϕ contains the potential to recover the full velocity field. But conceptually ϕ behaves like an action density, not a potential energy. Kol attributes the formulation to Euler (1752).

15.1.3 Equipotential Surfaces

Since $\mathbf{v} = \nabla \phi$ and the gradient is perpendicular to level sets, *surfaces of constant ϕ are perpendicular to the flow*:

$$\mathbf{v} \perp \{\phi = \text{const}\}. \quad (418)$$

This generalises the familiar electrostatics statement that **E**-lines are perpendicular to equipotentials.

15.2 Bernoulli-Like Equation

Mathematical Setup

We now translate Euler’s equation into a single scalar equation for ϕ . The key tool is an algebraic identity that rewrites the nonlinear convective term $(\mathbf{v} \cdot \nabla) \mathbf{v}$ in a form well-suited to potential flow.

15.2.1 Vector Identity for the Convective Term

Mathematical Result 15.1 Convective-Term Identity

For any smooth vector field $\mathbf{v}$,

$$(\mathbf{v} \cdot \nabla) \mathbf{v} = \nabla \left(\frac{1}{2} v^2 \right) - \mathbf{v} \times \boldsymbol{\omega}, \quad (419)$$

where $\boldsymbol{\omega} = \nabla \times \mathbf{v}$.

Proof: In index notation,

$$v_j \partial_j v_i = v_j (\partial_j v_i - \partial_i v_j) + v_j \partial_i v_j = -\epsilon_{ijk} v_j \omega_k + \partial_i (\tfrac{1}{2} v^2),$$

using $\epsilon_{ijk} \omega_k = \partial_i v_j - \partial_j v_i$ (curl in components) and $v_j \partial_i v_j = \partial_i (\tfrac{1}{2} v_j v_j)$. ■

15.2.2 Euler's Equation in Potential Form

Recall Euler's equation for an ideal fluid in a conservative external field $\mathbf{g} = -\nabla \Phi_g$:

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p - \nabla \Phi_g. \quad (420)$$

For a barotropic fluid, $\frac{1}{\rho} \nabla p = \nabla h_{\text{th}}$, where $h_{\text{th}} = \int dp/\rho$ is the specific enthalpy (per unit mass). Using Lemma 15.1 and dropping the $\mathbf{v} \times \boldsymbol{\omega}$ term (irrotational $\Rightarrow \boldsymbol{\omega} = \mathbf{0}$):

$$\frac{\partial \mathbf{v}}{\partial t} + \nabla (\tfrac{1}{2} v^2) = -\nabla h_{\text{th}} - \nabla \Phi_g. \quad (421)$$

Substitute $\mathbf{v} = \nabla \phi$, so $\partial_t \mathbf{v} = \nabla (\partial_t \phi)$:

$$\nabla \left[\frac{\partial \phi}{\partial t} + \tfrac{1}{2} v^2 + h_{\text{th}} + \Phi_g \right] = \mathbf{0}. \quad (422)$$

A function whose gradient vanishes everywhere is independent of space. We absorb that arbitrary time-function into ϕ (which is defined only up to such a function, since adding $f(t)$ to ϕ does not change $\mathbf{v}$):

Physical Law 15.2 Bernoulli-Like Equation for Potential Flow

In an irrotational, barotropic, ideal flow in a conservative body force,

$$\boxed{\frac{\partial \phi}{\partial t} + \tfrac{1}{2} v^2 + h_{\text{th}} + \Phi_g = f(t)}, \quad (423)$$

where $f(t)$ is an arbitrary function of time alone.

Significance: Euler's vector equation has been integrated to a single scalar relation. The novel term, absent in ordinary Bernoulli, is the unsteady contribution $\partial_t \phi$. In steady flow ($\partial_t \phi = 0$) this is *stronger* than Bernoulli: the constant holds throughout the entire fluid, not merely along each streamline.

Note:

Pay attention: Φ_g is the gravitational potential (capital), distinct from the velocity potential ϕ (small). In this chapter ϕ is reserved for the velocity potential, so we will keep using capital Φ_g for gravity.

15.2.3 Summary of Equations for Potential Flow

The complete system of three scalar equations for three unknowns (ϕ, ρ, p) :

$$\frac{\partial \phi}{\partial t} + \tfrac{1}{2} |\nabla \phi|^2 + h_{\text{th}}(\rho) + \Phi_g = f(t), \quad (424)$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \nabla \phi) = 0, \quad (425)$$

$$p = p(\rho) \quad (\text{equation of state}). \quad (426)$$

This is already a major reduction: the vector unknown $\mathbf{v}$ has been traded for a scalar ϕ . One further assumption simplifies dramatically.

15.3 Incompressible Potential Flow

Mathematical Setup

Take $\rho = \text{const}$ in addition to irrotational. The equation of state becomes trivial, the continuity equation collapses to $\nabla \cdot \mathbf{v} = 0$, and Bernoulli-like decouples from the field equation.

15.3.1 Laplace's Equation

With $\rho = \text{const}$, the continuity equation reduces to

$$\nabla \cdot \mathbf{v} = 0 \implies \nabla \cdot (\nabla \phi) = 0, \quad (427)$$

which is

Physical Law 15.3 Laplace's Equation for Incompressible Potential Flow

$$\boxed{\nabla^2 \phi = 0.} \quad (428)$$

Significance: The full nonlinear problem of an ideal incompressible irrotational flow has collapsed onto Laplace's equation – the very same equation governing electrostatic potentials in source-free regions. Existence, uniqueness, and a vast library of solutions are inherited from electrostatics. Crucially, the equation is *linear*, so **superposition holds**: any linear combination of solutions is itself a solution.

15.3.2 Decoupling and the Role of Pressure

Notice the structural change. In (424)–(425) the unknowns ϕ and ρ were coupled. Now ρ is a constant and (428) involves only ϕ . The pressure has been demoted to an output: once ϕ is known, p is determined from the Bernoulli-like equation

$$\frac{\partial \phi}{\partial t} + \frac{1}{2} |\nabla \phi|^2 + \frac{p}{\rho} + \Phi_g = f(t), \quad (429)$$

(in the incompressible case $h_{\text{th}} = p/\rho$ since ρ is constant). **Solve Laplace for ϕ ; then read off p .**

Note:

This is the same role pressure plays in the Lagrangian for incompressible flow (Lec 14): it is a constraint force enforcing $\nabla \cdot \mathbf{v} = 0$, not a thermodynamic variable.

15.3.3 Boundary Conditions

To solve $\nabla^2 \phi = 0$ we need boundary data.

Rigid impermeable wall. The fluid cannot penetrate a solid surface, so the normal component of $\mathbf{v}$ vanishes there: $v_n = 0$. In terms of ϕ ,

$$\boxed{\partial_n \phi \equiv \hat{\mathbf{n}} \cdot \nabla \phi = 0 \quad \text{on the wall.}} \quad (430)$$

This is a **Neumann** boundary condition – in contrast with electrostatics, where rigid conductors typically impose Dirichlet ($\phi_{\text{elec}} = \text{const}$).

Source (or sink). A localized source at $\mathbf{x}_0$ injects volume into the flow at a rate F (volume per unit time); a sink has $F < 0$. As an equation,

$$\nabla^2 \phi = F \delta^{(3)}(\mathbf{x} - \mathbf{x}_0). \quad (431)$$

Equivalently, one may treat the source as a boundary at a small sphere around $\mathbf{x}_0$ with prescribed outward flux F . The sign $+F$ on the right (versus the $-\rho$ in electrostatics) follows from $\mathbf{v} = +\nabla \phi$.

15.3.4 Solution Methods

How does one actually solve $\nabla^2\phi = 0$ with given boundary data? The truthful answer is that there is no general analytic method; the universally applicable technique is *numerical*.

1. **Numerical (relaxation, finite differences).** Discretise space and iterate $\phi_{ijk} \rightarrow \frac{1}{6}(\phi_{i\pm 1,j,k} + \phi_{i,j\pm 1,k} + \phi_{i,j,k\pm 1})$. *Always works*. Physically, relaxation minimises the electrostatic-like energy density $\frac{1}{2}|\nabla\phi|^2$.
2. **Separation of variables.** Works when the geometry has symmetry (Cartesian, spherical, cylindrical).
3. **Fourier methods.** For translationally-invariant geometries; a special case of separation.
4. **Method of images.** Replace a wall by a fictitious source whose superposed field automatically satisfies the wall boundary condition.
5. **Green's functions.** For arbitrary source distributions: $\phi = \int G(\mathbf{x}, \mathbf{x}') \rho_{\text{src}}(\mathbf{x}') d^3x'$; the image method is a special case.
6. **Superposition.** Linearity allows building complicated solutions from elementary ones (point sources, dipoles, uniform flow, vortex lines).

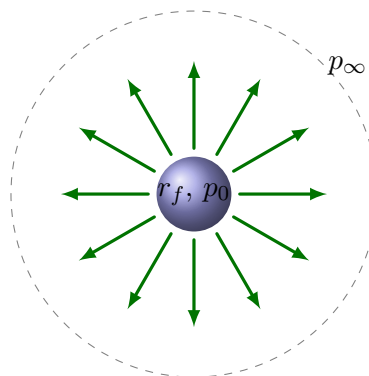
Note:

Physics is the art of finding the simple limits where analytic understanding is possible. The general case is the domain of numerics; analytic techniques work only when geometry/symmetry cooperate. The set of analytically tractable problems is small but instructive.

15.4 Example 1: Point Source in 3D

Physical Motivation

A small spherical reservoir at radius r_f ejects fluid radially with volumetric flow rate F into an unbounded incompressible irrotational fluid. We compute the velocity, the velocity potential, and the pressure field, and find a physically required lower bound on r_f .



15.4.1 Symmetry and the Form of $\mathbf{v}$

Spherical symmetry forces $\mathbf{v} = v(r)\hat{\mathbf{r}}$ and $\phi = \phi(r)$. Mass conservation (or, equivalently, Gauss' theorem applied to a sphere of radius r) requires that the same volume per unit time crosses every concentric sphere:

$$F = v(r) \cdot 4\pi r^2 \implies v(r) = \frac{F}{4\pi r^2}. \quad (432)$$

15.4.2 Velocity Potential

Integrating $v = d\phi/dr$:

$$\boxed{\phi(r) = -\frac{F}{4\pi r}, \quad \mathbf{v}(\mathbf{x}) = \frac{F}{4\pi r^2} \hat{\mathbf{r}}.} \quad (433)$$

Significance: This is structurally identical to the Coulomb potential of a point charge, but with reversed sign convention ($\mathbf{v} = +\nabla\phi$). For a sink, $F < 0$ and the formulas hold unchanged.

15.4.3 Pressure Field

Apply the steady incompressible Bernoulli-like equation (no time dependence, $\Phi_g = 0$):

$$\frac{p(r)}{\rho} + \frac{1}{2}v(r)^2 = \text{const.} \quad (434)$$

Match at the reservoir surface where $r = r_f$, $p = p_0$, $v = F/(4\pi r_f^2)$:

$$p(r) = p_0 + \frac{\rho}{2}[v(r_f)^2 - v(r)^2] = p_0 + \frac{\rho F^2}{32\pi^2} \left[\frac{1}{r_f^4} - \frac{1}{r^4} \right]. \quad (435)$$

At infinity $v \rightarrow 0$ and

$$p_\infty = p_0 + \frac{\rho F^2}{32\pi^2 r_f^4}. \quad (436)$$

15.4.4 Minimum Source Size

Pressure is bounded below by zero (no negative absolute pressures). Demanding $p_0 \geq 0$ in (436):

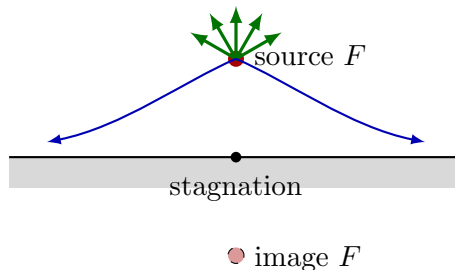
$$\boxed{r_f^4 \geq \frac{\rho F^2}{32\pi^2 p_\infty}.} \quad (437)$$

Significance: The source cannot be arbitrarily small. Pushing more volume through a smaller hole eventually demands negative pressures at the reservoir surface, which is unphysical (cavitation in practice). Intuitively: as $r \rightarrow r_f$, the speed $v \propto 1/r_f^2$ blows up; Bernoulli then demands the pressure p_0 to drop arbitrarily low.

15.5 Example 2: Point Source Near a Rigid Wall

Physical Motivation

Place a point source at distance d from an infinite rigid wall. The wall is impermeable, so streamlines must bend tangentially as they approach it. We solve by the method of images.



15.5.1 Image Method

The wall condition $\partial_n \phi = 0$ is satisfied if the flow is symmetric across the wall. Place an image source of *the same sign* $F' = F$ at the mirror position:

$$\phi(\mathbf{x}) = -\frac{F}{4\pi} \left[\frac{1}{|\mathbf{x} - d\hat{\mathbf{z}}|} + \frac{1}{|\mathbf{x} + d\hat{\mathbf{z}}|} \right]. \quad (438)$$

Significance: Note the sign contrast with electrostatics. For a grounded conductor in electrostatics, $\phi_{\text{elec}} = 0$ on the plane and one uses an image of *opposite* sign. Here we want vanishing *normal derivative*, which is achieved by an image of the *same* sign.

15.5.2 Stagnation Point

By symmetry, fluid leaving the source straight toward the wall must, upon reaching the wall, have no tangential preference. Yet $v_n = 0$ on the wall by construction, so all components of $\mathbf{v}$ vanish on the symmetry axis exactly at the wall:

Physical Concept 15.3: Stagnation Point

A point in the flow where $\mathbf{v} = \mathbf{0}$. Here it occurs at the foot of the perpendicular from the source to the wall.

Significance: Stagnation points are where Bernoulli sends pressure to a maximum. In the source-with-wall configuration the maximum pressure is attained at the stagnation point on the wall.

15.5.3 Equipotential Surfaces – Qualitative Picture

- Close to the source: nearly spherical surfaces centred at the source.
- Very far away: the two sources merge into a single effective source of strength $2F$, and equipotentials are again nearly spherical, centred at the wall.
- In between: a one-parameter family that smoothly interpolates – closed spheres around the source, transitioning through a critical surface that touches the wall, and finally half-spheres terminating on the wall.

15.5.4 Pressure Maximum from Bernoulli

In steady incompressible flow with no gravity,

$$p(\mathbf{x}) = \text{const} - \frac{1}{2}\rho v(\mathbf{x})^2. \quad (439)$$

Maximum pressure occurs at $v = 0$, i.e. at the stagnation point:

$$p_{\text{max}} = p_{\text{stagnation}}. \quad (440)$$

15.6 Local Flow Near a Stagnation Point

Mathematical Setup

Independently of any specific source distribution, we can characterise *universal* features of potential flow near any stagnation point $\mathbf{x}_0$, where $\mathbf{v}(\mathbf{x}_0) = \mathbf{0}$. Place the origin at $\mathbf{x}_0$ and Taylor-expand ϕ .

15.6.1 Taylor Expansion of ϕ

Write

$$\phi(\mathbf{x}) = \phi(0) + x_i \partial_i \phi|_0 + \frac{1}{2} x_i x_j \partial_i \partial_j \phi|_0 + \mathcal{O}(x^3). \quad (441)$$

The constant $\phi(0)$ is irrelevant (gauge). The linear term has coefficient $\nabla \phi|_0 = \mathbf{v}(\mathbf{x}_0) = \mathbf{0}$ at a stagnation point. Hence near $\mathbf{x}_0$,

$$\phi(\mathbf{x}) = \frac{1}{2} B_{ij} x_i x_j + \mathcal{O}(x^3), \quad B_{ij} = \partial_i \partial_j \phi|_0. \quad (442)$$

15.6.2 Constraints on the Quadratic Form

Laplace's equation: $\nabla^2 \phi = 0 \Rightarrow \text{tr } B = B_{11} + B_{22} + B_{33} = 0$.

Rigid wall (if present): If the stagnation point sits on a rigid wall $z = 0$ with normal $\hat{\mathbf{z}}$, then $v_z = \partial_z \phi = 0$ on the wall (not just at the origin). This forces $B_{13} = B_{23} = 0$ – no off-diagonal mixing of z with the wall-tangential directions.

After diagonalising the wall-tangential block, the canonical form near a wall-stagnation point in 2D is

$$\boxed{\phi(x, z) = \frac{1}{2} A (x^2 - z^2),} \quad (443)$$

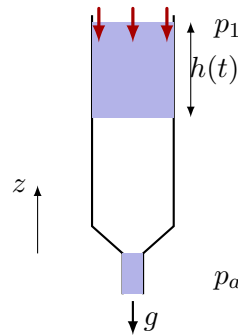
yielding $\mathbf{v} = (Ax, -Az)$ – a hyperbolic flow with the wall as a separatrix.

Significance: The local structure of potential flow near any stagnation point on a wall is universal – a quadratic potential, equivalently a hyperbolic velocity field. Streamlines are hyperbolae $xz = \text{const}$.

15.7 Example 3: Time-Dependent Syringe

Physical Motivation

A vertical syringe of cross-section A contains fluid up to height $h(t)$. A piston applies constant overpressure p_1 on top; the bottom opens to atmospheric pressure p_a and gravity g pulls the fluid down. We want $h(t)$. This example uses incompressible time-dependent potential flow, so the $\partial_t \phi$ term in the Bernoulli-like equation contributes.



15.7.1 Continuity: v Depends Only on Time

In a uniform vertical column, $\mathbf{v} = -v(z, t)\hat{\mathbf{z}}$ (downwards). Incompressibility gives

$$\nabla \cdot \mathbf{v} = -\frac{\partial v}{\partial z} = 0 \implies v = v(t) \text{ only.} \quad (444)$$

Physically: by mass conservation the column moves as a rigid slab; what enters at the top must leave at the bottom at the same rate.

15.7.2 Velocity Potential

From $\mathbf{v} = -v(t)\hat{\mathbf{z}}$ and $\mathbf{v} = \nabla \phi$:

$$\phi(z, t) = -v(t) z. \quad (445)$$

15.7.3 Bernoulli-Like at Top and Bottom

The Bernoulli-like equation reads, in incompressible form,

$$\frac{\partial \phi}{\partial t} + \frac{1}{2}v^2 + \frac{p}{\rho} + g z = f(t). \quad (446)$$

Evaluate at:

Top of the column ($z = h(t)$, $p = p_1$):

$$-\dot{v}(t) h + \frac{1}{2}v^2 + \frac{p_1}{\rho} + g h = f(t).$$

Bottom of the syringe ($z = 0$, $p = p_a$):

$$0 + \frac{1}{2}v^2 + \frac{p_a}{\rho} = f(t).$$

Subtracting:

$$-\dot{v}(t) h + g h + \frac{p_1 - p_a}{\rho} = 0 \implies \dot{v}(t) = g + \frac{p_1 - p_a}{\rho h(t)}. \quad (447)$$

15.7.4 Equation of Motion for $h(t)$

Now $v(t) = -\dot{h}(t)$ (the column descends), so $\dot{v} = -\ddot{h}$. Substituting:

$$\boxed{\ddot{h}(t) = -g - \frac{p_1 - p_a}{\rho h(t)}}. \quad (448)$$

This is a second-order ODE for $h(t)$, with initial conditions $h(0) = h_0$, $\dot{h}(0) = 0$.

15.7.5 Effective-Potential Interpretation

Equation (448) has the form $\ddot{h} = -\partial_h V_{\text{eff}}(h)$ with

$$V_{\text{eff}}(h) = g h + \frac{p_1 - p_a}{\rho} \ln h. \quad (449)$$

The motion of the fluid level $h(t)$ is identical to that of a unit-mass “particle” in V_{eff} . Both terms pull the particle toward $h = 0$: gravity contributes the linear part, the overpressure contributes the logarithmic part. The fluid level accelerates as it drops, and the syringe empties in finite time.

Significance: A time-dependent fluid problem with internal volume conservation reduces to one-dimensional Lagrangian mechanics. This is the cleanest example of how the $\partial_t \phi$ term in the Bernoulli-like equation produces inertial-like dynamics for a geometric variable (here, the fluid level).

Summary

- **Potential flow:** $\omega = \mathbf{0} \Rightarrow \mathbf{v} = \nabla \phi$; $[\phi] = L^2/T$ (action per unit mass).
- **Bernoulli-like:** $\partial_t \phi + \frac{1}{2}v^2 + h_{\text{th}} + \Phi_g = f(t)$; equals to a constant everywhere (not just on streamlines).
- **Incompressible + irrotational** $\Rightarrow \nabla^2 \phi = 0$ – linear, Laplace, superposition.
- **BCs:** rigid wall $\partial_n \phi = 0$ (Neumann); source $\nabla^2 \phi = F \delta^{(3)}$.

- **Solution methods:** numerical relaxation, separation, Fourier, images, Green functions, superposition.
- **Point source:** $\phi = -F/(4\pi r)$, $v = F/(4\pi r^2)$; minimum reservoir size from $p_0 \geq 0$.
- **Image method (wall):** image of *same sign* satisfies Neumann.
- **Stagnation point:** $\mathbf{v} = \mathbf{0}$, p maximal; local potential is quadratic with $\text{tr } B = 0$.
- **Syringe:** $\phi = -v(t)z$, EOM $\ddot{h} = -g - (p_1 - p_a)/(\rho h)$, reducible to a particle in an effective potential.

Further Reading

- L. D. Landau and E. M. Lifshitz, *Fluid Mechanics* (Vol. 6): §10–11 (potential flow, irrotational flow).
- D. J. Acheson, *Elementary Fluid Dynamics*: Ch. 4 (potential flow, image method, time-dependent Bernoulli).
- J. D. Jackson, *Classical Electrodynamics*: Ch. 2 (Laplace's equation, boundary-value methods, images) – mathematically identical machinery.

16 Lecture 16 – Potential Flow Around a Sphere and D’Alembert’s Paradox

Date: June 1, 2026

Overview

Lecture 15 reduced ideal, incompressible, irrotational flow to a single linear field equation, $\nabla^2\phi = 0$ (§15.3). This lecture exhibits the single most important closed-form solution of that program: the flow generated by a *sphere moving through a fluid otherwise at rest*. The payoff is threefold. First, the boundary-value problem turns out to be mathematically identical to the electrostatics of a conducting sphere in a uniform field, so the solution is a *pure dipole*. Second, integrating the (unsteady) Bernoulli pressure over the sphere yields a force that is *exactly zero whenever the sphere moves at constant velocity* – this is **D’Alembert’s paradox**. Third, when the sphere accelerates, the surviving term reveals that the fluid endows the sphere with an **added (effective) mass** equal to half the displaced fluid mass. We close by promoting the effective mass to a symmetric rank-2 tensor for general bodies, and by previewing how viscosity resolves the paradox.

Topics:

1. Recap of the potential-flow program (Laplace + Bernoulli-like pressure)
2. Physical setup: a sphere moving through a fluid at rest; two reference frames
3. Qualitative expectations: streamlines, stagnation points, the drag question
4. Boundary-value problem in the sphere frame: Laplace + Neumann at $r = R$ + dipole at infinity
5. Multipole expansion; why the velocity BC selects the pure $\ell = 1$ dipole
6. Matching the Neumann condition at $r = R$; solving for the dipole strength
7. Galilean transformation to the fluid frame
8. Streamline / equipotential analysis and the stagnation points
9. Generalization to arbitrary motion $\mathbf{r}_s(t)$
10. The force from the unsteady Bernoulli pressure; parity and the vanishing of drag
11. Added mass: factor $\frac{1}{2}$ for the sphere; the effective-mass tensor M_{ij}
12. D’Alembert’s paradox and its resolution by viscosity

16.1 Recap: The Potential-Flow Program

Conceptual Background

Before tackling the sphere, recall precisely *why* the flow problem became tractable. We made a chain of assumptions, each linearising or decoupling the dynamics one step further.

Step 1 – Irrotational $\Rightarrow$ scalar potential. We defined the *vorticity* $\omega \equiv \nabla \times \mathbf{v}$. For an ideal (inviscid) fluid, Kelvin’s circulation theorem (Lec 14) guarantees that if the flow is irrotational far from any obstacle – as it is for a body moving through a fluid at rest – it remains irrotational everywhere the fluid visits. A curl-free field is a gradient, so

$$\omega = \nabla \times \mathbf{v} = \mathbf{0} \implies \mathbf{v} = \nabla\phi, \quad (450)$$

the identity $\nabla \times \nabla\phi \equiv \mathbf{0}$ following from the commutativity of partial derivatives $\epsilon_{ijk}\partial_j\partial_k\phi = 0$. A three-component vector field is replaced by one scalar.

Step 2 – Incompressible $\Rightarrow$ Laplace. For a constant-density fluid the continuity equation collapses to $\nabla \cdot \mathbf{v} = 0$. Substituting (450),

$$\nabla \cdot \mathbf{v} = \nabla \cdot (\nabla \phi) = \nabla^2 \phi = 0. \quad (451)$$

Significance: The full *nonlinear* Euler problem has collapsed onto the *linear* Laplace equation. Linearity is the crucial gain: it licenses superposition, which is exactly what makes the multipole method below work.

Step 3 – Pressure from the unsteady Bernoulli relation. Pressure is no longer a dynamical unknown; it is read off after ϕ is known. The Bernoulli-like equation for incompressible irrotational flow (Lec 15) is

$$\frac{\partial \phi}{\partial t} + \frac{1}{2}v^2 + \frac{p}{\rho} + \Phi_g = f(t), \quad (452)$$

where Φ_g is the gravitational potential and $f(t)$ an arbitrary function of time. We will drop gravity ($\Phi_g = 0$) throughout, since for aerodynamic/hydrodynamic drag it is irrelevant (or, for ships, treated separately as buoyancy). Solving for the pressure:

$$p = \rho f(t) - \rho \frac{\partial \phi}{\partial t} - \frac{1}{2}\rho v^2. \quad (453)$$

Significance: Each term has a clean meaning. $\rho f(t)$ is a spatially uniform reference pressure set by the far field; $-\rho \partial_t \phi$ is the genuinely *unsteady* contribution, present only when the flow changes in time; $-\frac{1}{2}\rho v^2$ is the familiar Bernoulli suction – fast fluid is low-pressure fluid. The whole force calculation later hinges on which of these terms survive integration over the sphere.

Note:

The program is asymmetric: (451) is *one equation in one unknown ϕ* and contains no time derivative – time enters *only* through the boundary conditions. Once $\phi(\mathbf{x}, t)$ is found, (453) delivers the pressure everywhere. This “solve-then-read-off” structure is what makes the moving-sphere problem soluble in closed form.

16.2 Physical Setup and Two Reference Frames

Physical Motivation

Consider a rigid sphere of radius R moving with velocity $\mathbf{v}_0$ through an unbounded incompressible fluid that is at rest at infinity. This is the prototype of every engineering problem of motion through a fluid: a submarine or ship hull through water, an aircraft or projectile through air. We choose a sphere not because real vehicles are spherical – they are streamlined precisely to *minimise* drag and avoid turbulent wakes – but because the sphere is the simplest geometry admitting a full *analytic* solution. In general fluid dynamics one is forced to numerics; here we get an exact answer that builds intuition.

There are two natural frames, and keeping them straight is essential.

Physical Concept 16.1: Fluid Frame (lab frame)

The frame in which the fluid is at rest at infinity. The boundary condition at infinity is

$$\mathbf{v} \xrightarrow{r \rightarrow \infty} \mathbf{0}. \quad (454)$$

The sphere translates through this frame with velocity $\mathbf{v}_0$.

Significance: This is the “true” frame – far from the sphere the water is genuinely still. There is *no* incoming stream; the only disturbance is what the sphere stirs up locally.

Physical Concept 16.2: Sphere Frame

The frame instantaneously co-moving with the sphere, in which the sphere is at rest and the fluid streams past. The boundary condition at infinity becomes

$$\mathbf{v} \xrightarrow{r \rightarrow \infty} -\mathbf{v}_0, \quad (455)$$

i.e. a uniform stream of velocity $-\mathbf{v}_0$.

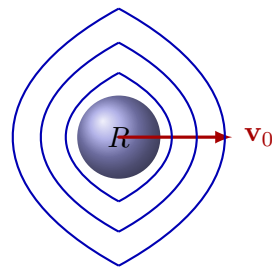
Significance: This is the “wind-tunnel” frame: the aircraft sits still and giant fans blow air past it. It is the more convenient frame to solve in, because the geometry (a fixed sphere in a uniform stream) is static. The asterisk we attach to the name signals a caveat: if the sphere accelerates, this frame is *non-inertial*; it is genuinely inertial only at the instant we freeze the velocity.

The two frames are related by a Galilean boost. If ϕ and $\phi^{(\text{fluid})}$ denote the potentials in the sphere and fluid frames respectively, then since velocities add as $\mathbf{v}^{(\text{fluid})} = \mathbf{v} + \mathbf{v}_0$,

$$\phi^{(\text{fluid})}(\mathbf{x}, t) = \phi(\mathbf{x}, t) + \mathbf{v}_0 \cdot \mathbf{x}, \quad (456)$$

because $\nabla(\mathbf{v}_0 \cdot \mathbf{x}) = \mathbf{v}_0$ reproduces the extra uniform velocity. We will solve in the sphere frame and then add the $\mathbf{v}_0 \cdot \mathbf{x}$ piece to descend to the fluid frame.

Panel A: Fluid Frame
(fluid at rest at ∞)



closed dipole loops

Panel B: Sphere Frame
(uniform stream $-\mathbf{v}_0$)

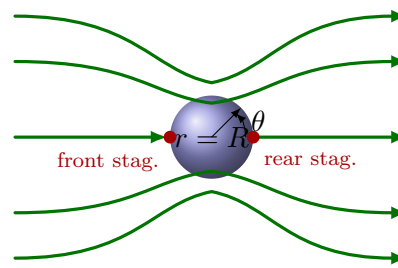


Figure 9: Two frames for the moving sphere. **Panel A** (fluid frame): the fluid is still at infinity; the sphere pushes fluid ahead and pulls it in behind, producing closed dipole loops centred on the sphere. **Panel B** (sphere frame): a uniform stream $-\mathbf{v}_0$ diverts around the fixed sphere; the central streamline terminates at the *front* ($\theta = \pi$) and *rear* ($\theta = 0$) stagnation points where $\mathbf{v} = \mathbf{0}$.

Qualitative Expectations

Before solving, the lecture pauses to anticipate the answer – a healthy physicist’s habit.

- **Streamlines.** The sphere must push fluid out of its way at the front and let fluid fill in behind it. In the fluid frame this gives the closed loops of Panel A; in the sphere frame, fluid divides and flows around the obstacle (Panel B).
- **Stagnation points.** In the sphere frame, the streamline aiming dead-centre at the sphere cannot pass through it; by symmetry it must terminate on the surface. The velocity there is zero – a *stagnation point*. There are two, fore and aft.
- **Drag?** Everyday experience says yes: a ball dropped in glycerin reaches a terminal velocity, so there must be a retarding *drag force*. We write a question mark next to this expectation – and we will find, paradoxically, that the ideal-fluid calculation gives *zero drag*.

16.3 The Boundary-Value Problem in the Sphere Frame

Mathematical Setup

We solve in the sphere frame. The unknown is the velocity potential ϕ , governed by Laplace's equation (451). A unique solution requires boundary conditions at the two boundaries of the domain: infinity and the sphere surface $r = R$.

Far-field boundary condition. In the sphere frame the fluid streams uniformly with $\mathbf{v} \rightarrow -\mathbf{v}_0$. Integrating $\mathbf{v} = \nabla\phi$ with $\mathbf{v} = -\mathbf{v}_0$ constant:

$$\phi \xrightarrow{r \rightarrow \infty} -\mathbf{v}_0 \cdot \mathbf{x} + \text{const.} \quad (457)$$

Choosing the z -axis along $\mathbf{v}_0$, with θ the polar angle from $+z$, we have $\mathbf{v}_0 \cdot \mathbf{x} = v_0 r \cos \theta$, so

$$\phi \xrightarrow{r \rightarrow \infty} -v_0 r \cos \theta. \quad (458)$$

(The additive constant does not change $\mathbf{v} = \nabla\phi$, so we discard it – a “gauge” choice.)

Surface boundary condition (impermeability). The sphere is a rigid, impermeable solid: fluid cannot cross its surface, so the *normal* component of velocity must vanish there. The outward normal on a sphere is radial, $\hat{\mathbf{n}} = \hat{\mathbf{r}}$, so the normal velocity is just the radial derivative of ϕ :

$$\boxed{v_n = \hat{\mathbf{n}} \cdot \nabla\phi = \left. \frac{\partial\phi}{\partial r} \right|_{r=R} = 0.} \quad (459)$$

Significance: This is a **Neumann** boundary condition (a condition on the normal derivative, not the value, of ϕ). It encodes the single physical fact that the wall is solid. This contrasts with the analogous electrostatics problem, where a conductor imposes a *Dirichlet* condition ($\phi_{\text{elec}} = \text{const}$).

16.3.1 Electrostatic Analogy

Conceptual Background

The problem “ $\nabla^2\phi = 0$, with $\nabla\phi \rightarrow \text{const}$ at infinity and a sphere as inner boundary” is one you have already solved in electromagnetism: a *conducting sphere in a uniform external field* $\mathbf{E}_0$. There, the sphere distorts the field and develops an *induced dipole*. The two problems share the same equation and the same far-field structure ($\nabla\phi \rightarrow \text{const}$); they differ *only* in the surface condition:

	Sphere in flow	Conducting sphere in $\mathbf{E}_0$
Equation	$\nabla^2\phi = 0$	$\nabla^2\phi_{\text{el}} = 0$
At ∞	$\nabla\phi \rightarrow -\mathbf{v}_0$	$\nabla\phi_{\text{el}} \rightarrow -\mathbf{E}_0$
At $r = R$	$\partial_r\phi = 0$ (Neumann)	$\phi_{\text{el}} = \text{const}$ (Dirichlet)

D'Alembert solved this flow problem in 1750, long before electrostatics existed; the modern reader gets the solution almost for free by analogy. In both cases the response of the sphere is a pure *dipole*.

16.3.2 Multipole Expansion and Selection of the Dipole

Mathematical Setup

The general axisymmetric solution of Laplace's equation, written as a sum over spherical harmonics $Y_{\ell m}$, is

$$\phi(r, \theta, \varphi) = \sum_{\ell, m} \left(a_{\ell m} r^\ell + \frac{b_{\ell m}}{r^{\ell+1}} \right) Y_{\ell m}(\theta, \varphi). \quad (460)$$

For each multipole ℓ the radial dependence is a pure power law: a *growing* solution r^ℓ (sourced from infinity) and a *decaying* solution $r^{-(\ell+1)}$ (sourced from the origin). The angular dependence is carried entirely by the spherical harmonics.

Why a pure dipole? The far-field condition (458) has the angular dependence $\cos \theta$. But $\cos \theta$ is (proportional to) the $\ell = 1$, $m = 0$ spherical harmonic – the **dipole**. Listing the lowest harmonics:

$$Y_0 \propto 1 \text{ (monopole)}, \quad Y_1 \propto \cos \theta \text{ (dipole)}, \quad Y_2 \propto (3 \cos^2 \theta - 1) \text{ (quadrupole)}. \quad (461)$$

The far field is purely $\ell = 1$. Because the surface condition is imposed at a *spherical* boundary $r = R$ – which respects full rotational symmetry – it cannot mix different ℓ values: a $\cos \theta$ forcing can only excite a $\cos \theta$ response. Hence the *entire* solution lives in the $\ell = 1$ sector. We therefore take the trial form (keeping the growing r^1 piece fixed by the far field, plus the decaying r^{-2} dipole tail):

$$\phi(r, \theta) = \cos \theta \left(a r + \frac{b}{r^2} \right) = -v_0 \cos \theta \left(r + \frac{b}{r^2} \right), \quad (462)$$

where matching (458) fixes the growing coefficient to $a = -v_0$ (so that as $r \rightarrow \infty$ only the $-v_0 r \cos \theta$ survives). The single remaining unknown is the dipole strength b .

Significance: A *vector* boundary datum at infinity (the uniform velocity $\mathbf{v}_0$, a directional object) maps onto exactly the dipole harmonic, because a dipole is the unique $\ell = 1$ object and $\ell = 1$ is the unique multipole carrying a single preferred direction. This is why “uniform stream past a sphere” is synonymous with “pure dipole.”

16.3.3 Matching the Neumann Condition

Mathematical Setup

Impose (459) on the trial (462). The normal derivative is the radial derivative:

$$\frac{\partial \phi}{\partial r} = -v_0 \cos \theta \left(1 - \frac{2b}{r^3} \right). \quad (463)$$

Evaluate at $r = R$ and demand it vanish for *all* θ :

$$\left. \frac{\partial \phi}{\partial r} \right|_{r=R} = -v_0 \cos \theta \left(1 - \frac{2b}{R^3} \right) = 0. \quad (464)$$

Since $\cos \theta$ is not identically zero, the bracket must vanish:

$$1 - \frac{2b}{R^3} = 0 \implies b = \frac{R^3}{2}. \quad (465)$$

Significance: The condition collapses to a single algebraic equation for b precisely because the $\cos \theta$ factors out of every term – a direct dividend of the multipole selection. There is exactly one dipole strength that makes the sphere impermeable.

16.3.4 The Solution in the Sphere Frame

Substituting $b = R^3/2$ into (462):

$$\boxed{\phi(r, \theta) = -v_0 \cos \theta \left(r + \frac{R^3}{2r^2} \right) \quad (\text{sphere frame}).} \quad (466)$$

Significance: This is the exact velocity potential for uniform flow past a sphere. The first term is the undisturbed stream; the second is the dipole correction the sphere adds to enforce impermeability. One checks the two boundary conditions directly: as $r \rightarrow \infty$ the dipole term dies and $\phi \rightarrow -v_0 r \cos \theta$; at $r = R$, $\partial_r \phi \propto (1 - 2b/R^3) = 0$, so no fluid penetrates the surface.

16.4 Transforming to the Fluid Frame

Mathematical Setup

To get the potential in the (lab) fluid frame we apply the Galilean rule (456), adding $\mathbf{v}_0 \cdot \mathbf{x} = v_0 r \cos \theta$:

$$\phi^{(\text{fluid})} = \phi + v_0 r \cos \theta = -v_0 \cos \theta \left(r + \frac{R^3}{2r^2} \right) + v_0 r \cos \theta. \quad (467)$$

The two uniform-stream pieces $\mp v_0 r \cos \theta$ cancel exactly, leaving only the dipole:

$$\boxed{\phi^{(\text{fluid})}(r, \theta) = -\frac{v_0 R^3 \cos \theta}{2r^2} \quad (\text{fluid frame}).} \quad (468)$$

Significance: In the fluid frame the potential is a *pure dipole* – nothing else. The growing r^{-1} “incoming-from-infinity” multipole has been removed, consistent with the physical fact that there is no flow at infinity in this frame. Of the two kinds of multipole in (460), the fluid frame retains only the $1/r^2$ dipole *sourced at the origin* (decaying outward); the growing dipole “arriving from infinity,” which was present in the sphere frame as the uniform stream, has cancelled.

Note:

Multipoles come in two species: those *sourced at the origin*, decaying as $r^{-(\ell+1)}$, and those *arriving from infinity*, growing as r^ℓ . A monopole field goes as $1/r^2$ in velocity, a dipole as $1/r^3$. The fluid-frame solution is the cleanest statement of the physics: a moving sphere is, to the surrounding still fluid, exactly a hydrodynamic dipole.

16.5 Streamlines, Equipotentials, and Stagnation Points

Physical Motivation

With ϕ in hand we can verify the qualitative picture of fig. 9 and locate the stagnation points quantitatively. The velocity components in spherical coordinates follow from $\mathbf{v} = \nabla \phi$:

$$v_r = \frac{\partial \phi}{\partial r}, \quad v_\theta = \frac{1}{r} \frac{\partial \phi}{\partial \theta}, \quad v_\varphi = \frac{1}{r \sin \theta} \frac{\partial \phi}{\partial \varphi} = 0, \quad (469)$$

the azimuthal component vanishing because the solution is axisymmetric ($\partial_\varphi \phi = 0$).

16.5.1 Stagnation Points

Mathematical Setup

Work in the sphere frame, (466). The tangential component is

$$v_\theta = \frac{1}{r} \frac{\partial \phi}{\partial \theta} = \frac{1}{r} v_0 \sin \theta \left(r + \frac{R^3}{2r^2} \right) \propto \sin \theta, \quad (470)$$

which vanishes at $\theta = 0$ and $\theta = \pi$ – the poles. The radial component is

$$v_r = \frac{\partial \phi}{\partial r} = -v_0 \cos \theta \left(1 - \frac{R^3}{r^3} \right), \quad (471)$$

which we *constructed* to vanish on the surface $r = R$ (the bracket is $1 - R^3/r^3$). Therefore at the two surface poles ($r = R$, $\theta = 0$) and ($r = R$, $\theta = \pi$) *both* components vanish (and $v_\varphi = 0$ always), so

$$\boxed{\mathbf{v} = \mathbf{0} \quad \text{at the front and rear poles of the sphere.}} \quad (472)$$

Significance: These are the two *stagnation points* anticipated qualitatively. The rear pole (in the direction of motion) sits at $\theta = 0$, the front pole at $\theta = \pi$. By Bernoulli (453), $v = 0$ means *maximum* pressure: the fluid presses hardest exactly fore and aft. The perfect fore–aft symmetry of these high-pressure zones is the seed of D’Alembert’s paradox.

16.5.2 Equipotential Surfaces

In the fluid frame, the dipole equipotentials $\phi^{(\text{fluid})} = \text{const}$ satisfy, from (468),

$$\frac{\cos \theta}{r^2} = \text{const} \implies r^2 = r_{\max}^2 \cos \theta, \quad (473)$$

where $r_{\max}$ is the radius reached at $\theta = 0$. Each such surface passes through the origin (at $\theta = \pi/2$, $r \rightarrow 0$) and is a closed lobe; different constants give nested, scaled copies of the same lobe, with a mirror family for $\cos \theta < 0$. The full surface is the body of revolution obtained by rotating the lobe about the z -axis (the motion axis), giving a closed dimpled surface that pinches to a singular point at the origin. The streamlines (blue in fig. 9) are everywhere perpendicular to these equipotentials, since $\mathbf{v} = \nabla \phi$.

Note:

In lecture the professor first mis-drew the equipotential surface as a torus; the correct surface is the figure of revolution about the *motion axis*, a closed lobe pinched at the origin, not a doughnut. The streamline picture and all formulae were correct.

16.6 Generalization to Arbitrary Motion

Conceptual Background

So far the sphere moved at *constant* velocity. We now allow the sphere centre to follow an arbitrary trajectory $\mathbf{r}_s(t)$ with instantaneous velocity $\dot{\mathbf{r}}_s(t)$. The key structural observation is that *time does not appear in Laplace's equation* (451) – it enters only through the boundary conditions. So at each instant the spatial problem is the *same* dipole problem, with two updates: (i) re-centre the coordinates on the moving sphere, $\mathbf{x} \rightarrow \mathbf{x} - \mathbf{r}_s(t)$; (ii) replace the constant $\mathbf{v}_0$ by the instantaneous velocity $\dot{\mathbf{r}}_s(t)$.

16.6.1 Vectorial Rewriting

Mathematical Setup

First rewrite the fluid-frame dipole (468) covariantly. Note $v_0 \cos \theta = \mathbf{v}_0 \cdot \hat{\mathbf{r}}$, hence

$$\frac{v_0 \cos \theta}{r^2} = \frac{\mathbf{v}_0 \cdot \hat{\mathbf{r}}}{r^2} = \frac{\mathbf{v}_0 \cdot \mathbf{x}}{r^3}, \quad (474)$$

using $\hat{\mathbf{r}} = \mathbf{x}/r$. Thus the constant-velocity fluid-frame potential is

$$\phi^{(\text{fluid})} = -\frac{R^3}{2} \frac{\mathbf{v}_0 \cdot \mathbf{x}}{r^3}. \quad (475)$$

16.6.2 The Accelerating Sphere

Now promote to arbitrary motion. Measuring position relative to the instantaneous sphere centre, $\mathbf{x} \rightarrow \mathbf{x} - \mathbf{r}_s(t)$, and the velocity to the instantaneous value $\dot{\mathbf{r}}_s(t)$:

$$\phi(\mathbf{x}, t) = -\frac{R^3}{2} \frac{\dot{\mathbf{r}}_s(t) \cdot (\mathbf{x} - \mathbf{r}_s(t))}{|\mathbf{x} - \mathbf{r}_s(t)|^3}. \quad (476)$$

Significance: This is the exact potential for a sphere in *arbitrary* translational motion through ideal fluid. The crucial point: the surface boundary condition $v_n = 0$ depends only on the instantaneous *velocity* of the sphere, never on its acceleration – a solid wall is impermeable regardless of how fast it accelerates. Acceleration enters the physics only later, through the *time derivative* $\partial_t \phi$ in the pressure (453), never through the geometry of ϕ itself. This is exactly why the static-looking dipole suffices even for accelerated motion.

16.7 The Force on the Sphere

Conceptual Background

How does the fluid exert a force on the sphere? Exactly as in elasticity: neighbouring fluid elements exert short-range *contact* stresses on one another, described by the *stress tensor* $\boldsymbol{\sigma}$. In an ideal fluid the stress has no deviatoric (shear) part – there is no viscosity – so it is purely isotropic:

$$\sigma_{ij} = -p \delta_{ij}. \quad (477)$$

The coefficient of the isotropic stress, with the conventional minus sign, is the *pressure*. The total force the fluid exerts on the sphere is the surface integral of this stress over the sphere's surface. With the area element $d\mathbf{S} = \hat{\mathbf{n}} dS$ pointing *outward* (out of the fluid, into the sphere is $-\hat{\mathbf{n}}$), the force on the sphere is

$$\mathbf{F} = - \oint_{r=R} p \hat{\mathbf{n}} dS = - \oint_{r=R} p(\theta, \varphi) \hat{\mathbf{r}} R^2 d\Omega. \quad (478)$$

Significance: Drag and lift are nothing more than the imbalance of pressure pushing on different parts of the surface. If the pressure pattern is fore–aft symmetric, the net push is zero. The whole question of drag thus reduces to the *symmetry* of the pressure field.

16.7.1 Assembling the Pressure

Mathematical Setup

From the unsteady Bernoulli relation (453) (incompressible, gravity dropped),

$$p = \rho f(t) - \rho \frac{\partial \phi}{\partial t} - \frac{1}{2} \rho v^2. \quad (479)$$

There are three contributions. We examine which survive the surface integral (478). The decisive tool is the **parity symmetry** $\mathbf{x} \rightarrow -\mathbf{x}$ about the sphere centre: it maps each surface point to its antipode. A term in p contributes *zero net force* if it produces *equal* pressure at antipodal points, because the outward normals $\hat{\mathbf{r}}$ and $-\hat{\mathbf{r}}$ then cancel in the integral.

(1) The $\rho f(t)$ term. Spatially uniform: equal at all points, hence equal at antipodes. It pushes inward equally from all directions and integrates to zero force,

$$- \oint \rho f(t) \hat{\mathbf{r}} R^2 d\Omega = -\rho f(t) R^2 \oint \hat{\mathbf{r}} d\Omega = \mathbf{0}, \quad (480)$$

since $\oint \hat{\mathbf{r}} d\Omega = \mathbf{0}$ (the average of the unit normal over a sphere vanishes).

(2) The $-\frac{1}{2}\rho v^2$ term. The velocity field is dipolar, like the field of an electric dipole, so it has the property $\mathbf{v}(-\mathbf{x}) = -\mathbf{v}(\mathbf{x})$ (odd under parity). Therefore its square is *even*:

$$v^2(-\mathbf{x}) = v^2(\mathbf{x}). \quad (481)$$

Equal pressure at antipodal points $\Rightarrow$ zero net force. *The Bernoulli suction does not produce drag.*

(3) The unsteady term $-\rho \partial_t \phi$. This is the only term that can survive. Differentiate (476). Two pieces depend on time: the explicit velocity $\dot{\mathbf{r}}_s$ and the position $\mathbf{r}_s$. Writing $\mathbf{r}' = \mathbf{x} - \mathbf{r}_s$,

$$\frac{\partial \phi}{\partial t} = -\frac{R^3}{2} \left[\frac{\ddot{\mathbf{r}}_s \cdot \mathbf{r}'}{r'^3} + \dot{\mathbf{r}}_s \cdot \frac{\partial}{\partial t} \left(\frac{\mathbf{r}'}{r'^3} \right) \right]. \quad (482)$$

The first piece is proportional to the *acceleration* $\ddot{\mathbf{r}}_s$; under parity $\mathbf{r}' \rightarrow -\mathbf{r}'$ it is *odd*, so it gives equal-and-opposite – actually *unequal* – pressure at antipodes and can yield a nonzero force. The second

piece, $\dot{\mathbf{r}}_s \cdot \partial_t(\mathbf{r}'/r'^3)$, is built from velocities only; since $\partial_t(\mathbf{r}'/r'^3) = -(\dot{\mathbf{r}}_s \cdot \nabla)(\mathbf{r}'/r'^3)$ it is quadratic in velocity and even under parity, contributing zero force by the same argument as term (2).

Significance: Of the three pressure contributions, only the piece proportional to the sphere's *acceleration* can exert a net force. In particular, **at constant velocity** ($\ddot{\mathbf{r}}_s = 0$) **every term integrates to zero – the net force vanishes.**

Physical Law 16.1 D'Alembert's Paradox (No Steady Drag)

An ideal (inviscid, incompressible, irrotational) fluid exerts *zero drag* on a sphere moving at constant velocity:

$$\mathbf{F}_{\text{drag}} = \mathbf{0} \quad \text{for} \quad \dot{\mathbf{r}}_s = \text{const.} \quad (483)$$

Significance: Drag, by definition, is a force opposing steady motion. The ideal-fluid model predicts *none* – in flat contradiction with the everyday observation that objects slow down in fluids. This is the famous paradox D'Alembert reached in 1750. Its root is the perfect fore-aft symmetry of the pressure field (Panel A): the high pressure at the front stagnation point is exactly matched by equal high pressure at the rear, so the pushes cancel.

16.8 Added Mass (Effective Mass)

Physical Motivation

The surviving acceleration term does produce a force. Let us compute it. Physically, an accelerating sphere must *drag the surrounding fluid along with it*; setting that fluid into motion costs an extra force beyond what is needed to accelerate the bare sphere. The sphere therefore behaves as if it were heavier – it has an *added mass*.

16.8.1 A Tensor Integral over the Sphere

Mathematical Setup

The force (478) from the acceleration term requires the angular integral of $\hat{r}_i \hat{r}_j$ over the sphere. We compute it once and for all.

Mathematical Result 16.2 Angular Average of $\hat{r}_i \hat{r}_j$

Over a sphere of radius R ,

$$\oint \hat{r}_i \hat{r}_j d\Omega = \frac{4\pi}{3} \delta_{ij}. \quad (484)$$

Proof: By rotational invariance the integral can only depend on the one available isotropic tensor, δ_{ij} : write $\oint \hat{r}_i \hat{r}_j d\Omega = C \delta_{ij}$. Take the trace ($i = j$, sum): the left side becomes $\oint \hat{r}_i \hat{r}_i d\Omega = \oint 1 d\Omega = 4\pi$ (since $\hat{r}_i \hat{r}_i = |\hat{\mathbf{r}}|^2 = 1$), while the right side is $C \delta_{ii} = 3C$. Hence $3C = 4\pi$, giving $C = 4\pi/3$. ■

Significance: This is the workhorse identity: the angular average of two unit-normal components is isotropic, with coefficient $4\pi/3$. It is exactly what converts the dipolar pressure pattern into a clean force along the acceleration direction.

16.8.2 Evaluating the Force

Mathematical Setup

Keep only the surviving acceleration piece of the pressure. From (479) and the first term of (482), on the surface $r' = R$,

$$p|_{\text{accel}} = -\rho \left. \frac{\partial \phi}{\partial t} \right|_{\text{accel}} = +\frac{\rho R^3}{2} \frac{\ddot{\mathbf{r}}_s \cdot \hat{\mathbf{r}}}{R^2} = \frac{\rho R}{2} \ddot{r}_{s,j} \hat{r}_j. \quad (485)$$

Insert into the force (478), with $dS = R^2 d\Omega$ and $\hat{\mathbf{n}} = \hat{\mathbf{r}}$:

$$F_i = - \oint p \hat{r}_i R^2 d\Omega = -\frac{\rho R}{2} \ddot{r}_{s,j} R^2 \oint \hat{r}_i \hat{r}_j d\Omega. \quad (486)$$

Apply (484):

$$F_i = -\frac{\rho R^3}{2} \ddot{r}_{s,j} \cdot \frac{4\pi}{3} \delta_{ij} = -\frac{4\pi}{3} \cdot \frac{\rho R^3}{2} \ddot{r}_{s,i}. \quad (487)$$

Recognise $V_{\text{sph}} = \frac{4}{3}\pi R^3$ as the volume of the sphere:

$$\boxed{\mathbf{F} = -\frac{1}{2} \rho V_{\text{sph}} \ddot{\mathbf{r}}_s.} \quad (488)$$

Significance: The fluid pushes *against* the acceleration of the sphere (note the minus sign), with magnitude $\frac{1}{2}\rho V_{\text{sph}}$ times the acceleration. It behaves exactly like an extra inertia. There is *no* velocity-dependent (drag) term – consistent with ?? 16.1.

16.8.3 The Effective Mass

Conceptual Background

Put this force into Newton's second law for the sphere of true mass m under an external force $\mathbf{F}_{\text{ext}}$:

$$m \ddot{\mathbf{r}}_s = \mathbf{F}_{\text{fluid}} + \mathbf{F}_{\text{ext}} = -\frac{1}{2}\rho V_{\text{sph}} \ddot{\mathbf{r}}_s + \mathbf{F}_{\text{ext}}. \quad (489)$$

Move the fluid force to the left:

$$(m + \frac{1}{2}\rho V_{\text{sph}}) \ddot{\mathbf{r}}_s = \mathbf{F}_{\text{ext}}. \quad (490)$$

This is Newton's law with a modified inertia:

Physical Concept 16.3: Effective (Added) Mass of a Sphere

A sphere accelerating through ideal fluid responds to external force as if it had mass

$$\boxed{m_{\text{eff}} = m + \frac{1}{2}\rho V_{\text{sph}} = m + \frac{1}{2}m_{\text{displaced}}.} \quad (491)$$

Significance: The sphere “feels heavier” by exactly *half* the mass of the fluid it displaces. The factor $\frac{1}{2}$ is specific to the sphere; it comes from the dipole field the sphere generates. The physical origin: to accelerate, the sphere must also accelerate the cloud of fluid moving with it, and that co-moving fluid stores extra kinetic energy.

Note:

Compare with buoyancy (Lec 11): there, the displaced fluid mass appears in a *static* force (Archimedes). Here the displaced fluid mass appears in a *dynamic*, inertial role – and is multiplied by $\frac{1}{2}$. Same ρV , completely different physics: buoyancy is gravitational, added mass is inertial.

16.8.4 Generalization: The Effective-Mass Tensor

Conceptual Background

For a non-spherical body the co-moving fluid – and hence the added inertia – depends on the *direction* of acceleration: a flat plate accelerated broadside drags far more fluid than the same plate accelerated edge-on. The added mass is therefore a **symmetric rank-2 tensor** M_{ij} :

$$F_i = -M_{ij} \ddot{r}_{s,j}, \quad M_{ij} = M_{ji}. \quad (492)$$

Its symmetry is most transparent from the kinetic energy of the entrained fluid, which is a quadratic form in the body velocity:

$$T_{\text{fluid}} = \frac{1}{2} M_{ij} v_i v_j. \quad (493)$$

Significance: Instead of a scalar mass multiplying v^2 , a general body carries a *matrix* of masses contracting $v_i v_j$. For the sphere, isotropy forces $M_{ij} = \frac{1}{2} \rho V_{\text{sph}} \delta_{ij}$, recovering the scalar result (491). The persistence of $\mathbf{F}_{\text{drag}} = \mathbf{0}$, however, is general: any body with reflection symmetry exhibits zero ideal-fluid drag.

Note:

The general added mass is of order $\rho \times (\text{body volume})$ but the precise coefficient depends on the flow field the body's shape generates – for the sphere it happens to be exactly $\frac{1}{2}$.

16.9 Resolution of D'Alembert's Paradox

Conceptual Background

A paradox in physics is a productive crisis: two strong arguments collide, and resolving the collision teaches something deep. Here, Euler's ideal-flow equations *rigorously* predict zero drag (?? 16.1), while everyday experience – a ball thrown into water decelerating to terminal velocity – *demands* drag. One of the two must be giving way, and we know which.

The culprit is the **ideal-fluid approximation itself**. We assumed the stress was purely isotropic, $\sigma_{ij} = -p \delta_{ij}$, with *no* shear (viscous) part. Real fluids have *viscosity*: tangential stresses that act at the body surface, enforce a no-slip boundary condition, and – crucially – break the fore-aft symmetry of the flow by generating a wake behind the body. Once the viscous shear stress is restored to the stress tensor (the Navier–Stokes equations), the pressure symmetry is destroyed and a genuine drag force appears.

Claim 16.1 Viscosity Resolves the Paradox

Drag is absent in ideal flow precisely because the deviatoric (shear) stress was discarded. Restoring viscosity – moving from Euler to Navier–Stokes – breaks the fore-aft symmetry, produces a wake, and yields a nonzero drag force consistent with experiment.

Significance: D'Alembert (1750) could not have spotted this: the concept of viscosity as a momentum-diffusing shear stress was not formulated until Navier and Stokes nearly a century later. The paradox is thus a precise diagnostic – it isolates *exactly* the term (viscous shear) whose omission renders the model unphysical, and motivates the viscous-flow theory we turn to next.

Summary

- **Setup.** A sphere moving through still fluid; solve $\nabla^2 \phi = 0$ with $\partial_r \phi|_R = 0$ (Neumann, impermeable wall) and $\phi \rightarrow -v_0 r \cos \theta$ at infinity (uniform stream in sphere frame).
- **Multipole selection.** The $\cos \theta$ far field is the $\ell = 1$ harmonic; the spherical boundary cannot mix multipoles, so the solution is a *pure dipole*. Dipole strength fixed by Neumann: $b = R^3/2$.
- **Solutions.** Sphere frame $\phi = -v_0 \cos \theta (r + R^3/2r^2)$; fluid frame $\phi = -v_0 R^3 \cos \theta / 2r^2$ (pure dipole). Related by Galilean boost $\phi^{\text{fluid}} = \phi + \mathbf{v}_0 \cdot \mathbf{x}$.
- **Stagnation points.** $\mathbf{v} = \mathbf{0}$ at front and rear poles; pressure maximal there.
- **Arbitrary motion.** $\phi = -\frac{R^3}{2} \dot{\mathbf{r}}_s \cdot (\mathbf{x} - \mathbf{r}_s) / |\mathbf{x} - \mathbf{r}_s|^3$; boundary condition needs only the instantaneous velocity, never acceleration.

- **Force.** From $p = \rho f(t) - \rho \partial_t \phi - \frac{1}{2} \rho v^2$: the uniform and v^2 terms cancel by parity; only the acceleration term survives. **Zero drag** at constant velocity (D'Alembert's paradox).
- **Added mass.** $\mathbf{F} = -\frac{1}{2} \rho V_{\text{sph}} \ddot{\mathbf{r}}_s$, so $m_{\text{eff}} = m + \frac{1}{2} \rho V_{\text{sph}}$ (half the displaced fluid mass). Generalizes to a symmetric tensor M_{ij} with $T_{\text{fluid}} = \frac{1}{2} M_{ij} v_i v_j$.
- **Resolution.** The paradox is cured by viscosity: restoring shear stress (Navier–Stokes) breaks fore–aft symmetry and produces real drag.

Further Reading

- L. D. Landau and E. M. Lifshitz, *Fluid Mechanics* (Vol. 6): §10–11 (flow past a sphere, D'Alembert's paradox, induced/added mass).
- D. J. Acheson, *Elementary Fluid Dynamics*: Ch. 4–5 (potential flow past a sphere, added mass, D'Alembert).
- J. D. Jackson, *Classical Electrodynamics*: §2.5 (conducting sphere in a uniform field – the mathematically identical problem).

Lecture 17 – Planar Potential Flow and the Complex Potential

Date: June 3, 2026

Overview

The previous lectures built the theory of ideal, incompressible, irrotational flow and reduced it to a single linear field equation $\nabla^2\phi = 0$ (§16.1). Lecture 16 then exhibited its flagship three-dimensional solution – flow past a sphere – and discovered D’Alembert’s paradox. This lecture restricts attention to *planar* (two-dimensional) flow, and the payoff is spectacular: in two dimensions the equations of ideal potential flow are *exactly* the Cauchy–Riemann equations, so the entire machinery of complex analysis becomes available. We will find that every analytic function of a complex variable $z = x + iy$ is a planar potential flow, and that the real and imaginary parts of that function are two physically meaningful potentials: the velocity potential ϕ and a brand-new object, the *stream function* ψ . We develop two small pieces of planar vector algebra (the Hodge star and the wedge product), use them to introduce ψ , prove the equations of motion collapse to analyticity, give ψ its physical interpretation (streamlines and flux), and then harvest a sequence of exact solutions almost for free: uniform flow, source/sink, vortex, flow around a corner, and flow around a disk.

Topics:

1. Definition and motivation of planar flow (rivers, islands, airfoils)
2. Two planar vector operations: the Hodge star $*$ (90° rotation) and the wedge product $\wedge$ (signed area)
3. The stream function ψ from incompressibility; the velocity potential ϕ from irrotationality
4. Equations of motion = Cauchy–Riemann equations; both potentials harmonic
5. Boundary conditions: Neumann on ϕ becomes Dirichlet ($\psi = \text{const}$) on the wall
6. The complex velocity potential $w = \phi + i\psi$ and analyticity
7. Recovering velocity ($dw/dz = \bar{v}$) and pressure (Bernoulli)
8. Physical meaning of ψ : streamlines and the flux identity $\Delta\psi = Q$
9. Worked solutions: uniform flow, source/sink, vortex, corner flow, flow past a disk

17.1 What is Planar Flow, and Why Study It Separately?

Physical Motivation

We remain inside the fluid-mechanics half of the course, in the ideal-flow chapter (no viscosity yet). A flow is called *planar* (two-dimensional) when the velocity field has no component out of a fixed plane and depends only on the two in-plane coordinates.

Physical Concept 17.1: Planar (Two-Dimensional) Flow

A flow is *planar* if, choosing the flow plane to be the xy -plane,

$$\mathbf{v} = (v_x(x, y), v_y(x, y), 0), \quad (494)$$

i.e. the velocity points only along x and y and is independent of z .

Significance: A genuinely three-dimensional vector field is reduced to a two-component field on a two-dimensional domain. This is not merely a convenience: in exactly two dimensions a special analytic structure appears (the link to complex functions) that has no analogue in three dimensions.

Physical examples. Many engineering flows are planar to excellent approximation because one direction is long and uniform:

- **A bend in a river:** at different heights the flow pattern is essentially identical, so we may study only the in-plane dependence.
- **An island in a river:** the obstacle (which may be elliptical, circular, ...) is uniform in depth – the planar analogue of last lecture’s flow around a sphere.
- **An aircraft wing / aerofoil:** along most of the span the cross-sectional profile and the flow around it are constant, so the flow about each section is planar. (Adding a small *angle of attack* makes the model more realistic.)

We already *know* what an ideal fluid is, and we may again assume the flow is irrotational (a potential flow). The reason this planar case deserves its own dedicated lecture is the hidden *analytic structure* tied to complex functions, which will let us solve several flows exactly.

17.2 Mathematical Interlude: Two Operations on Planar Vectors

Mathematical Setup

Beyond the familiar operations on vectors (addition, scalar multiplication, the dot product), the plane admits two further operations that exist *only* because in two dimensions every vector defines a *unique* perpendicular line. In three dimensions a vector’s perpendicular is a whole plane, so these operations have no clean two-dimensional flavour there. We build the theory on these two.

Physical Concept 17.2: The Hodge Star $*$ (Rotation by 90°)

For a planar vector $\mathbf{u}$, define $*\mathbf{u} \equiv \hat{\mathbf{z}} \times \mathbf{u}$. In components, with $\mathbf{u} = (u_x, u_y)$,

$$*\mathbf{u} = (-u_y, u_x). \quad (495)$$

Geometrically, $*\mathbf{u}$ is $\mathbf{u}$ rotated by 90° in the positive (counterclockwise) sense.

Significance: The star turns a vector into the vector perpendicular to it (same length, rotated $+90^\circ$). It is the single tool that lets us trade "tangent to a curve" for "normal to a curve," which is exactly the trade needed to relate streamlines to flux. The definition via $\hat{\mathbf{z}} \times \mathbf{u}$ momentarily leaves the plane (a blemish); an equivalent purely-planar definition is the component rule (495).

Physical Concept 17.3: The Wedge (Exterior) Product $\wedge$

For two planar vectors $\mathbf{u}, \mathbf{v}$, define the *scalar*

$$\mathbf{u} \wedge \mathbf{v} \equiv \hat{\mathbf{z}} \cdot (\mathbf{u} \times \mathbf{v}) = u_x v_y - u_y v_x. \quad (496)$$

Significance: The wedge of two planar vectors returns the *signed area* of the parallelogram they span. The sign records orientation: $\mathbf{u} \wedge \mathbf{v} > 0$ when $\mathbf{v}$ lies counterclockwise from $\mathbf{u}$. This is the two-dimensional shadow of the three-dimensional cross product, collapsed to its single out-of-plane component.

Algebraic properties. Both operations are linear in each argument (they distribute over linear combinations). Beyond linearity we record three identities used repeatedly below.

$$*(*\mathbf{u}) = -\mathbf{u}, \quad (497)$$

because rotating twice by 90° reverses the vector; the wedge is antisymmetric,

$$\mathbf{v} \wedge \mathbf{u} = -\mathbf{u} \wedge \mathbf{v}, \quad (498)$$

visible immediately from (496); and the two operations are linked by

$$\boxed{\mathbf{u} \wedge (*\mathbf{v}) = \mathbf{u} \cdot \mathbf{v}.} \quad (499)$$

Significance: Identity (499) says: wedging with the rotated vector is the same as dotting with the original. It is the algebraic engine that turns the wedge-form equations of motion into the familiar dot-product form, and it is what underlies the flux interpretation of ψ .

Proof: [Proof of (499)] With $*\mathbf{v} = (-v_y, v_x)$, apply (496) using $\mathbf{w} = *\mathbf{v}$, so $w_x = -v_y$, $w_y = v_x$:

$$\mathbf{u} \wedge (*\mathbf{v}) = u_x w_y - u_y w_x = u_x v_x - u_y (-v_y) = u_x v_x + u_y v_y = \mathbf{u} \cdot \mathbf{v}.$$

■

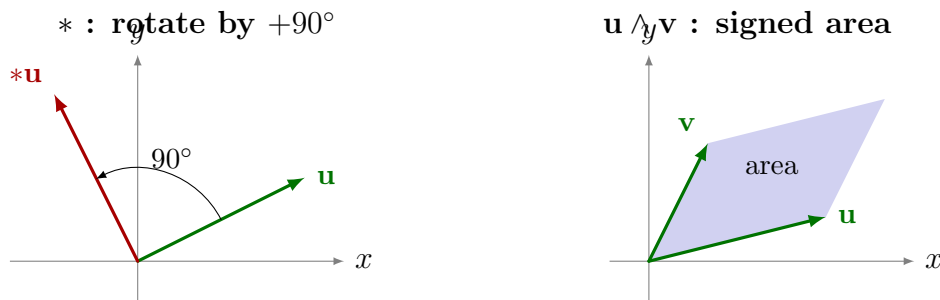


Figure 10: The two planar operations. **Left:** the Hodge star rotates $\mathbf{u}$ counterclockwise by 90° , $*\mathbf{u} = (-u_y, u_x)$. **Right:** the wedge $\mathbf{u} \wedge \mathbf{v} = u_x v_y - u_y v_x$ is the signed area of the parallelogram spanned by $\mathbf{u}$ and $\mathbf{v}$, positive when $\mathbf{v}$ is counterclockwise from $\mathbf{u}$.

17.3 The Two Potentials: ϕ and the Stream Function ψ

Conceptual Background

Planar potential flow obeys two scalar constraints, and each will buy us one scalar potential. The flow is *irrotational* (a potential flow) and *incompressible*.

Irrotationality $\Rightarrow$ velocity potential ϕ . The assumption $\omega = \nabla \times \mathbf{v} = \mathbf{0}$ is, in the plane, a single scalar equation, since the curl of a planar field has only a z -component:

$$\omega_z = \partial_x v_y - \partial_y v_x = \nabla \wedge \mathbf{v} = 0. \quad (500)$$

As in three dimensions, a curl-free field is a gradient, so there exists a *velocity potential* ϕ with

$$\mathbf{v} = \nabla \phi, \quad v_x = \partial_x \phi, \quad v_y = \partial_y \phi. \quad (501)$$

Incompressibility $\Rightarrow$ stream function ψ . The assumption $\nabla \cdot \mathbf{v} = 0$ is the other scalar equation,

$$\nabla \cdot \mathbf{v} = \partial_x v_x + \partial_y v_y = 0. \quad (502)$$

In three dimensions this single relation is comparatively weak, but in two dimensions it has a powerful consequence. It states that $\partial_x v_x = -\partial_y v_y$, which is exactly the integrability condition guaranteeing the existence of a function ψ whose derivatives are $(-v_y, v_x)$. We therefore *define* the *stream function* ψ by

Physical Concept 17.4: Stream Function ψ

For an incompressible planar flow there exists a scalar $\psi(x, y)$ with

$$\partial_y \psi = v_x, \quad \partial_x \psi = -v_y. \quad (503)$$

Significance: Where ϕ exists because the flow is *irrotational*, ψ exists because the flow is *incompressible*. They are different functions built from different physical assumptions. ψ is defined precisely so that $\nabla \cdot \mathbf{v} = 0$ becomes the trivial equality of mixed partials, $\partial_x \partial_y \psi - \partial_y \partial_x \psi = 0$, which holds automatically.

Vector form. Comparing $\nabla \psi = (\partial_x \psi, \partial_y \psi) = (-v_y, v_x)$ with the star (495), the definition of ψ is just

$$\boxed{\nabla \psi = * \mathbf{v}.} \quad (504)$$

Significance: $\nabla \psi$ is *not* the velocity (that is $\nabla \phi$); it is the velocity rotated by 90° . Hence $\nabla \psi \perp \mathbf{v}$, which already foreshadows that the level curves of ψ run *along* the flow.

17.4 Equations of Motion Are the Cauchy–Riemann Equations

Mathematical Setup

We now have two unknown fields, ϕ and ψ . They are not independent: both are built from the same velocity. Eliminating $\mathbf{v}$ between (501) and (504),

$$\nabla \psi = * \mathbf{v} = * \nabla \phi. \quad (505)$$

Writing this out component by component, with $* \nabla \phi = (-\partial_y \phi, \partial_x \phi)$:

$$\boxed{\partial_x \phi = \partial_y \psi, \quad \partial_y \phi = -\partial_x \psi.} \quad (506)$$

Significance: These are precisely the **Cauchy–Riemann equations** of complex analysis – the conditions for a complex function with real part ϕ and imaginary part ψ to be *analytic* (holomorphic). The full nonlinear fluid problem has, in two dimensions, collapsed into the statement “a certain complex function is analytic.” This is the central result of the lecture.

Both potentials are harmonic. The system (506) already *contains* the equation of motion. Take the curl-form: in vector language, applying the star to (505) and using $** = -1$ gives $* \nabla \psi = -\nabla \phi$, and the consistency of these two gradients (equality of mixed partials) forces

$$\nabla^2 \phi = 0, \quad \nabla^2 \psi = 0. \quad (507)$$

Concretely, differentiate the first of (506) by x and the second by y and add: $\partial_x^2 \phi + \partial_y^2 \phi = \partial_x \partial_y \psi - \partial_y \partial_x \psi = 0$, and symmetrically for ψ . *Significance:* We recover the Laplace equation $\nabla^2 \phi = 0$ from Lecture 15 – but now it comes packaged with a *second* harmonic field ψ . In complex analysis this is the familiar statement that the real and imaginary parts of an analytic function are automatically harmonic (*conjugate harmonics*). Solving the flow has become: find an analytic function meeting the boundary conditions.

Note:

Historical note: these equations were first written down by d'Alembert (1752) in the context of planar fluid flow, and by Euler in the context of complex functions; Cauchy and Riemann later gave the modern formulation that turned complex analysis into a complete theory. The deep link between planar ideal flow and complex functions is thus original to fluid mechanics, not a later borrowing.

17.5 Boundary Conditions

Physical Motivation

At a solid, impermeable wall the fluid cannot penetrate, so the *normal* velocity vanishes: $v_n = \partial_n \phi = 0$. This is a Neumann condition on ϕ – awkward to impose. The stream function converts it into something far simpler.

If the normal derivative of ϕ vanishes along the wall, then by the 90° relation (504) the *tangential* derivative of ψ along the wall vanishes: ψ does not change as we move along the boundary. Hence

$$\boxed{\psi = \text{const on a solid boundary.}} \quad (508)$$

Significance: The clumsy Neumann condition on ϕ becomes a clean *Dirichlet* condition on ψ : the wall is a streamline ($\psi = \text{const}$), which makes perfect sense because no fluid crosses it. For a single, simply-connected boundary we are free to choose the constant to be $\psi = 0$; only differences of ψ carry physical meaning.

17.6 The Complex Velocity Potential

Mathematical Setup

The Cauchy–Riemann equations (506) invite us to package ϕ and ψ into a single complex function. We identify the physical plane with the complex plane,

$$z = x + iy, \quad (509)$$

(x the real axis, y the imaginary axis – a *complexification* of the geometric plane), and define

Physical Concept 17.5: Complex Velocity Potential

$$w(z) \equiv \phi(x, y) + i\psi(x, y). \quad (510)$$

By the Cauchy–Riemann equations (506), w is an *analytic* function of z .

Significance: The entire flow is encoded in one analytic function. Conversely *any* analytic $w(z)$ furnishes a legitimate planar potential flow: take $\phi = \text{Re } w$, $\psi = \text{Im } w$. The boundary condition (508) becomes $\text{Im } w = \text{const}$ on solid walls – in the cases below, simply $\text{Im } w = 0$, i.e. w is *real* on the boundary. The recipe for solving a flow is now: *find an analytic function that is real on the given boundary*.

17.6.1 Recovering the Velocity Field

Mathematical Setup

Given $w(z)$, the velocity is obtained elegantly through the complex derivative. Because w is analytic, dw/dz may be computed by differentiating in any direction; choosing the x -direction,

$$\frac{dw}{dz} = \partial_x w = \partial_x \phi + i \partial_x \psi = v_x + i(-v_y) = v_x - iv_y. \quad (511)$$

Defining the *complex velocity* $\underline{\zeta} \equiv v_x + iv_y$, this is its complex conjugate:

$$\boxed{\frac{dw}{dz} = v_x - iv_y = \bar{\underline{\zeta}}.} \quad (512)$$

Significance: One complex differentiation delivers *both* velocity components at once (as the conjugate $\bar{\underline{\zeta}}$). The appearance of the conjugate is the small price paid for the convention $w = \phi + i\psi$; one simply remembers that the velocity is read off as the conjugate of dw/dz . The pressure then follows, exactly as before, by substituting the velocity into the (Bernoulli-like) pressure relation (453).

The solution recipe. The complete procedure is now:

1. Find an analytic $w(z)$ that is real (or constant imaginary part) on the solid boundary.
2. Differentiate: $dw/dz = \bar{\underline{\zeta}}$ gives the velocity field.
3. Insert $\underline{\zeta}$ into the Bernoulli-like equation to obtain the pressure.

17.7 Example: Uniform Flow

Worked Example

The simplest flow: uniform velocity $\underline{\zeta}_0 = v_{0x} + iv_{0y}$ everywhere, no boundary. We seek w with $dw/dz = \bar{\underline{\zeta}}_0$, a constant. Integrating,

$$\boxed{w(z) = \bar{\underline{\zeta}}_0 z} \quad (513)$$

(plus an irrelevant additive constant, since w is defined only up to a constant – it does not affect the flow). *Significance:* The most elementary analytic function, a linear one, is uniform flow. For real v_0 directed along x (so $\bar{\underline{\zeta}}_0 = v_0$), $w = v_0 z = v_0 x + iv_0 y$, giving

$$\phi = v_0 x, \quad \psi = v_0 y. \quad (514)$$

The equipotentials $\phi = \text{const}$ are vertical lines; the streamlines $\psi = \text{const}$ are horizontal lines parallel to the flow. The two families are orthogonal, as they must be since $\nabla\psi = *\nabla\phi$.

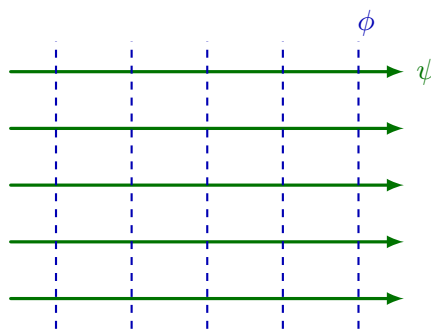


Figure 11: Uniform flow $w = v_0 z$ (with v_0 real). Solid green: streamlines $\psi = v_0 y = \text{const}$ (the flow direction). Dashed blue: equipotentials $\phi = v_0 x = \text{const}$, orthogonal to the streamlines.

17.8 Example: Source and Sink

Physical Motivation

A point from which fluid emerges and spreads radially (a *source*); the planar analogue of last lecture's three-dimensional source. Let Q be the *area flux* (volume per unit time, per unit depth) issuing from the origin: $Q > 0$ a source, $Q < 0$ a sink.

Mathematical Setup

By symmetry the velocity is purely radial, $\mathbf{v} = v_r(r) \hat{\mathbf{r}}$. A Gaussian "pillbox" – here a circle of radius r – carries the entire flux, so $v_r \cdot (2\pi r) = Q$, giving

$$v_r = \frac{Q}{2\pi r}. \quad (515)$$

Note the planar $1/r$ falloff (contrast the $1/r^2$ of a 3D point source): this is exactly the field of a charged *wire* in electrostatics, not a point charge. The velocity potential satisfies $\partial_r \phi = v_r = Q/2\pi r$, so $\phi = \frac{Q}{2\pi} \ln r$. Recognising $\ln r = \operatorname{Re} \ln z$, the analytic completion is

$$\boxed{w(z) = \frac{Q}{2\pi} \ln z.} \quad (516)$$

Significance: The 2D source is the complex logarithm. Splitting into real and imaginary parts with $z = re^{i\theta}$ (so $\ln z = \ln r + i\theta$):

$$\phi = \frac{Q}{2\pi} \ln r, \quad \psi = \frac{Q}{2\pi} \theta. \quad (517)$$

The stream function $\psi \propto \theta$ is *multivalued* – defined only modulo Q (it jumps by Q on circling the origin once), the hallmark of a source enclosed by the contour. The equipotentials $\phi = \text{const}$ are circles $r = \text{const}$; the streamlines $\psi = \text{const}$ are radial rays $\theta = \text{const}$, along which fluid flows outward. The velocity follows from $dw/dz = Q/2\pi z$, a simple pole at the origin whose residue $Q/2\pi$ encodes the source strength.

17.9 Example: The Point Vortex

Conceptual Background

A new flow is obtained *for free* by multiplying the source potential by i (equivalently $1/i$): multiplying an analytic function by a constant keeps it analytic, hence it is still a legitimate flow. Multiplication by i swaps the roles of ϕ and ψ – so circles become streamlines and rays become equipotentials. Take

$$\boxed{w(z) = \frac{\Gamma}{2\pi i} \ln z = -\frac{i\Gamma}{2\pi} \ln z.} \quad (518)$$

Significance: Splitting with $\ln z = \ln r + i\theta$,

$$\phi = \frac{\Gamma}{2\pi} \theta, \quad \psi = -\frac{\Gamma}{2\pi} \ln r. \quad (519)$$

Now the *streamlines* $\psi = \text{const}$ are circles $r = \text{const}$ – fluid circulates around the origin – and the equipotentials are radial rays. This is a **point vortex**. The velocity is purely azimuthal, $v_\theta = \frac{1}{r} \partial_\theta \phi = \Gamma/2\pi r$.

Identifying Γ as the circulation. Compute the line integral of $\mathbf{v}$ around any closed loop encircling the origin once:

$$\oint \mathbf{v} \cdot d\mathbf{l} = \int_0^{2\pi} v_\theta(r) d\theta = \int_0^{2\pi} \frac{\Gamma}{2\pi r} r d\theta = \Gamma. \quad (520)$$

Significance: The constant Γ is exactly the *circulation* of Lecture 14 (Kelvin's theorem). Although the flow is irrotational everywhere away from the origin ($\omega = \mathbf{0}$ for $r > 0$), the circulation around any loop enclosing the singular point is the nonzero Γ – all the vorticity is concentrated at the origin. The vortex is thus the flow-field counterpart of a magnetic field around a current-carrying wire.

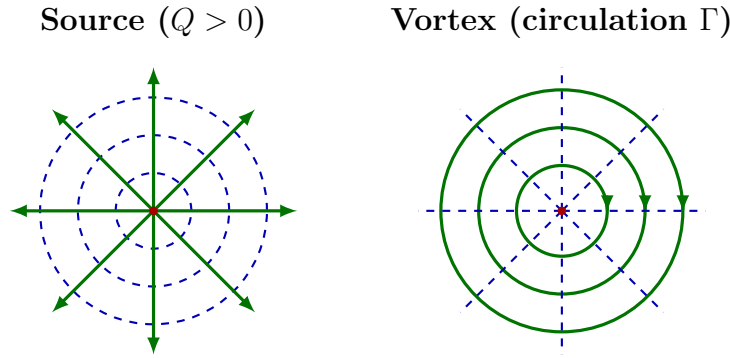


Figure 12: Source vs. vortex – related by multiplication of w by i . **Left:** source $w = \frac{Q}{2\pi} \ln z$: radial streamlines (green), circular equipotentials (dashed). **Right:** vortex $w = \frac{\Gamma}{2\pi i} \ln z$: circular streamlines (green), radial equipotentials (dashed). The roles of ϕ and ψ are exchanged.

17.10 Physical Meaning of the Stream Function

Conceptual Background

The examples reveal a universal pattern: the level curves $\psi = \text{const}$ are the *streamlines* of the flow, and differences of ψ measure flux. We now prove both statements in general.

Streamlines. On a curve $\psi = \text{const}$, $\nabla\psi$ is normal to the curve. But $\nabla\psi = *\mathbf{v}$ is $\mathbf{v}$ rotated by 90° , so $\mathbf{v}$ is *tangent* to the curve. Hence the fluid flows along level curves of ψ – they are streamlines. This is the origin of the name *stream function*.

Flux identity. Consider any path from A to B and ask how much fluid crosses it per unit time (per unit depth). For a line element $d\mathbf{l}$ the (rightward) normal element is $d\mathbf{S} = -*d\mathbf{l}$, and the area flux is

$$Q_{A \rightarrow B} = \int_A^B \mathbf{v} \cdot d\mathbf{S}. \quad (521)$$

Using $\nabla\psi = *\mathbf{v}$ together with the link (499), the integrand becomes a perfect differential:

$$\mathbf{v} \cdot d\mathbf{S} = \mathbf{v} \cdot (-*d\mathbf{l}) = -\mathbf{v} \wedge (*d\mathbf{l}) = \mathbf{v} \wedge d\mathbf{l} = \nabla\psi \cdot d\mathbf{l} = d\psi, \quad (522)$$

where we used $** = -1$ and $\mathbf{v} \wedge d\mathbf{l} = (*\mathbf{v}) \cdot d\mathbf{l} = \nabla\psi \cdot d\mathbf{l}$. Integrating,

$$\boxed{Q_{A \rightarrow B} = \psi_B - \psi_A = \Delta\psi.} \quad (523)$$

Significance: The difference in ψ between two points equals the volume flux (per unit depth) of fluid flowing between the two streamlines through them. Two immediate corollaries: (i) on a streamline $\Delta\psi = 0$, so no fluid crosses it – consistent with it being a wall; (ii) the flux between two streamlines is independent of *where* we measure it (by continuity / incompressibility), so naming the two bounding streamlines already fixes the flux. This is the quantitative meaning of the level *value* of ψ , complementing the qualitative fact that its level *curves* are streamlines.

17.11 Example: Flow Around a Corner

Physical Motivation

Consider flow inside (or around) a rigid corner of opening angle α – the angle may be acute, obtuse, or reflex ($> \pi$). The walls are solid and impermeable, and we want the flow that turns the corner. We need an analytic function that is real on the bent boundary.

Mathematical Setup

The trick is conformal mapping: find a map that straightens the corner into a straight wall (the real axis) in the w -plane. Powers of z multiply angles, so a power map does the job. The boundary of a wedge of opening α is sent to the real line by

$$\boxed{w(z) = A z^{\pi/\alpha}} \quad (A \text{ real constant}), \quad (524)$$

because raising to the power π/α stretches the wedge angle α up to π (a straight wall). *Significance:* A single elementary analytic function solves the entire family of corner flows – one formula for every opening angle.

Right-angle corner. Take $\alpha = \pi/2$, so $\pi/\alpha = 2$ and (absorbing A)

$$w = Az^2 = A[(x^2 - y^2) + i 2xy]. \quad (525)$$

Then $\phi = A(x^2 - y^2)$ and the streamlines are

$$\psi = 2Axy = \text{const} \implies xy = \text{const}, \quad (526)$$

a family of rectangular *hyperbolae*. The boundary streamline $\psi = 0$ is $xy = 0$, i.e. the two half-axes – exactly the right-angle walls. The velocity is $\underline{\zeta} = dw/dz = 2Az$, so near the corner the flow speeds up linearly in distance from the vertex. *Significance:* The simplest nonlinear analytic function, z^2 , already describes flow turning a square corner; the hyperbolic streamlines hug the walls and the corner itself is a stagnation point ($\underline{\zeta} = 0$ at $z = 0$).

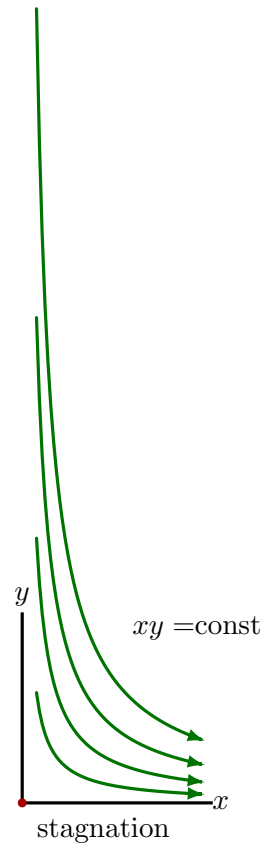


Figure 13: Flow around a right-angle corner, $w = Az^2$. The solid walls lie along the positive x - and y -axes (the streamline $\psi = 0$, i.e. $xy = 0$). Other streamlines are the hyperbolae $xy = \text{const}$ (green), which the fluid follows as it turns the corner. The vertex is a stagnation point.

17.12 Example: Flow Around a Disk

Physical Motivation

Finally, the planar analogue of last lecture's flow around a sphere: a uniform stream v_0 (real, along x) past a solid impermeable disk of radius R . We need an analytic $w(z)$ that (i) tends to uniform flow at infinity and (ii) is real on the circle $|z| = R$ (so the circle is the streamline $\psi = 0$).

Mathematical Setup

Start from uniform flow $v_0 z$ and add a correction that vanishes at infinity yet makes w real on the circle. On the unit circle $|z| = 1$, the conjugate is $\bar{z} = 1/z$, so $z + 1/z = z + \bar{z} = 2 \operatorname{Re} z$ is real there. Scaling from the unit circle to radius R ($z \rightarrow z/R$, and choosing the overall factor to recover $v_0 z$ at infinity) gives

$$w(z) = v_0 \left(z + \frac{R^2}{z} \right). \quad (527)$$

Significance: On $|z| = R$, writing $z = Re^{i\theta}$ gives $w = v_0(Re^{i\theta} + Re^{-i\theta}) = 2v_0 R \cos \theta$, which is real, so $\psi = 0$ there – the disk is a streamline, as required. At large $|z|$ the R^2/z term dies and $w \rightarrow v_0 z$, recovering the uniform stream. The correction term $v_0 R^2/z$ is precisely a **2D dipole**: in the plane the logarithm $\ln z$ is the monopole (source) and $1/z$ is the dipole, just as the sphere problem produced a $1/r^2$ dipole in three dimensions. The velocity is

$$\bar{\zeta} = \frac{dw}{dz} = v_0 \left(1 - \frac{R^2}{z^2} \right), \quad (528)$$

which vanishes at $z = \pm R$: the two *stagnation points* on the disk, fore and aft of the stream – the exact two-dimensional echo of the sphere's stagnation points.

Note:

The professor left the full streamline analysis of the disk as a guided exercise. The picture: a dividing streamline arrives along the axis, splits at the front stagnation point $z = -R$, runs along the upper and lower surfaces, and rejoins at the rear stagnation point $z = +R$; neighbouring streamlines bulge around the disk. Because the flow is fore-aft symmetric and inviscid, D'Alembert's paradox again applies – zero drag – with the same resolution (viscosity) to come.

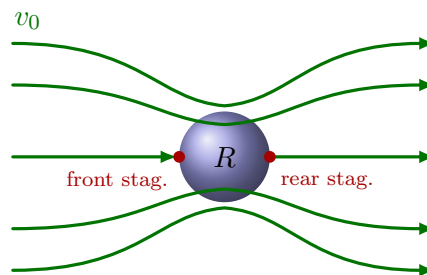


Figure 14: Uniform flow past a solid disk, $w = v_0(z + R^2/z)$. The circle $|z| = R$ is the streamline $\psi = 0$; the stream divides at the front stagnation point $z = -R$ and rejoins at the rear $z = +R$ (where $dw/dz = 0$). The R^2/z term is the 2D dipole, the planar analogue of the sphere's dipole.

Summary

- **Planar flow:** $\mathbf{v} = (v_x(x, y), v_y(x, y), 0)$. Examples: river bends, islands, aerofoil sections.
- **Two planar operations:** Hodge star $*\mathbf{u} = (-u_y, u_x)$ (rotate 90°); wedge $\mathbf{u} \wedge \mathbf{v} = u_x v_y - u_y v_x$ (signed area); linked by $\mathbf{u} \wedge *\mathbf{v} = \mathbf{u} \cdot \mathbf{v}$.

- **Two potentials:** irrotational $\Rightarrow \mathbf{v} = \nabla\phi$; incompressible $\Rightarrow$ stream function ψ with $\partial_y\psi = v_x$, $\partial_x\psi = -v_y$, i.e. $\nabla\psi = *\mathbf{v}$.
- **Equations of motion = Cauchy–Riemann:** $\partial_x\phi = \partial_y\psi$, $\partial_y\phi = -\partial_x\psi$, forcing $\nabla^2\phi = \nabla^2\psi = 0$.
- **Complex potential:** $w = \phi + i\psi$ is analytic in $z = x + iy$; velocity $dw/dz = v_x - iv_y = \bar{\underline{v}}$. Solid wall $\Rightarrow \psi = \text{const}$ ($\text{Im } w = \text{const}$).
- **Meaning of ψ :** level curves are streamlines; $\Delta\psi = \text{volume flux (per depth) between them}$.
- **Catalogue:** uniform $w = \bar{\underline{v}}_0 z$; source $w = \frac{Q}{2\pi} \ln z$; vortex $w = \frac{\Gamma}{2\pi i} \ln z$ (circulation Γ); corner $w = Az^{\pi/\alpha}$ (right angle $\rightarrow z^2$, hyperbolic streamlines); disk $w = v_0(z + R^2/z)$ (2D dipole, two stagnation points).

Further Reading

- L. D. Landau and E. M. Lifshitz, *Fluid Mechanics* (Vol. 6): §10 (two-dimensional flow, the complex potential, sources and vortices).
- D. J. Acheson, *Elementary Fluid Dynamics*: Ch. 4–5 (stream function, complex potential, flow past a cylinder).
- G. K. Batchelor, *An Introduction to Fluid Dynamics*: §6.5–6.6 (two-dimensional irrotational flow and conformal mapping).

Part III

Homework Solutions

Landau & Lifshitz Exercises

Eran Rehani

18 Chapter 1 – Fundamental Equations §7

Throughout these solutions we use the standard small-deformation theory of an isotropic linear elastic material. The displacement is denoted by $\mathbf{u}$, the infinitesimal strain by

$$u_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right),$$

and Hooke's law is written as

$$\sigma_{ij} = 2\mu u_{ij} + \lambda u_{kk} \delta_{ij}, \quad \mu = \frac{E}{2(1+\nu)}, \quad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}.$$

Here E is Young's modulus, ν is Poisson's ratio, and ρ is the mass density. Positive normal stress is tensile; pressure therefore gives a negative normal traction.

Problem 18.1. Determine the deformation of a long rod (with length L) standing vertically in a gravitational field.

Proof: [Solution] Let z be measured upward from the fixed lower end of the rod, so that $0 \leq z \leq L$. Gravity acts in the negative z -direction. Because the rod is long and slender, the leading deformation is a one-dimensional axial compression.

The governing equilibrium equation is force balance in the vertical direction:

$$\frac{d\sigma_{zz}}{dz} + \rho f_z = 0, \quad f_z = -g.$$

Therefore

$$\frac{d\sigma_{zz}}{dz} - \rho g = 0.$$

Before integrating, we state the boundary condition: the upper end is free, so

$$\sigma_{zz}(L) = 0.$$

Now integrate the equilibrium equation:

$$\frac{d\sigma_{zz}}{dz} = \rho g \quad \implies \quad \sigma_{zz}(z) = \rho g z + C.$$

Using $\sigma_{zz}(L) = 0$ gives

$$0 = \rho g L + C \quad \implies \quad C = -\rho g L.$$

Hence

$$\sigma_{zz}(z) = -\rho g (L - z).$$

This is negative, as expected, because each cross-section supports the weight of the material above it.

The constitutive equation for a uniaxial rod with free lateral surface is

$$\varepsilon_{zz} = \frac{\sigma_{zz}}{E}.$$

Since $\varepsilon_{zz} = \frac{du_z}{dz}$, we obtain

$$\frac{du_z}{dz} = -\frac{\rho g}{E} (L - z).$$

Before integrating the displacement, state the support condition:

$$u_z(0) = 0.$$

Now integrate:

$$u_z(z) = -\frac{\rho g}{E} \int_0^z (L - s) ds = -\frac{\rho g}{E} \left(Lz - \frac{z^2}{2} \right).$$

Thus the top of the rod moves by

$$u_z(L) = -\frac{\rho g L^2}{2E}.$$

The shortening of the rod is therefore

$$\Delta L = -u_z(L) = \frac{\rho g L^2}{2E}.$$

If the lateral surface is stress-free, then the transverse strains follow from Poisson's effect:

$$\varepsilon_{xx} = \varepsilon_{yy} = -\nu \varepsilon_{zz} = \nu \frac{\rho g}{E} (L - z).$$

So the rod is compressed vertically and expands laterally, with the largest strain at the bottom. ■

Problem 18.2. Determine the deformation of a hollow sphere (with external and internal radii R_1 and R_2) with a pressure p_1 inside and p_2 outside.

Proof: [Solution] To avoid ambiguity in the radius notation, write

$$a = R_{\text{in}}, \quad b = R_{\text{out}}, \quad a < b.$$

If the statement uses R_1 for the outer radius and R_2 for the inner radius, then $a = R_2$ and $b = R_1$.

The deformation is spherically symmetric, so

$$\mathbf{u} = u(r) \hat{\mathbf{r}}.$$

The governing equilibrium equation without body force is

$$\frac{d\sigma_{rr}}{dr} + \frac{2}{r} (\sigma_{rr} - \sigma_{\theta\theta}) = 0.$$

For spherical symmetry the non-zero strains are

$$u_{rr} = \frac{du}{dr}, \quad u_{\theta\theta} = u_{\phi\phi} = \frac{u}{r}.$$

Substituting these strains into Hooke's law gives

$$\sigma_{rr} = 2\mu \frac{du}{dr} + \lambda \left(\frac{du}{dr} + 2\frac{u}{r} \right), \quad \sigma_{\theta\theta} = 2\mu \frac{u}{r} + \lambda \left(\frac{du}{dr} + 2\frac{u}{r} \right).$$

The displacement equation obtained from equilibrium is

$$\frac{d}{dr} \left[\frac{1}{r^2} \frac{d}{dr} (r^2 u) \right] = 0.$$

Before solving it, write its general spherical solution:

$$u(r) = Ar + \frac{B}{r^2}.$$

Now calculate the strains:

$$u_{rr} = A - \frac{2B}{r^3}, \quad u_{\theta\theta} = u_{\phi\phi} = A + \frac{B}{r^3},$$

and therefore

$$u_{kk} = u_{rr} + u_{\theta\theta} + u_{\phi\phi} = 3A.$$

The radial stress becomes

$$\sigma_{rr} = 2\mu \left(A - \frac{2B}{r^3} \right) + 3\lambda A = (3\lambda + 2\mu)A - \frac{4\mu B}{r^3}.$$

It is useful to note the identity

$$3\lambda + 2\mu = \frac{E}{1 - 2\nu}.$$

Now state the pressure boundary conditions. Pressure acts inward on each surface, so

$$\sigma_{rr}(a) = -p_1, \quad \sigma_{rr}(b) = -p_2.$$

Substitute the stress formula:

$$(3\lambda + 2\mu)A - \frac{4\mu B}{a^3} = -p_1, \quad (3\lambda + 2\mu)A - \frac{4\mu B}{b^3} = -p_2.$$

Subtract the two equations:

$$4\mu B \left(\frac{1}{a^3} - \frac{1}{b^3} \right) = p_1 - p_2.$$

Since

$$\frac{1}{a^3} - \frac{1}{b^3} = \frac{b^3 - a^3}{a^3 b^3},$$

we find

$$B = \frac{(p_1 - p_2)a^3 b^3}{4\mu(b^3 - a^3)} = \frac{(1 + \nu)(p_1 - p_2)a^3 b^3}{2E(b^3 - a^3)}.$$

Next solve for A from the inner boundary condition:

$$(3\lambda + 2\mu)A = -p_1 + \frac{4\mu B}{a^3} = -p_1 + \frac{(p_1 - p_2)b^3}{b^3 - a^3}.$$

Putting this over a common denominator gives

$$(3\lambda + 2\mu)A = \frac{p_1 a^3 - p_2 b^3}{b^3 - a^3}.$$

Hence

$$A = \frac{1 - 2\nu}{E} \frac{p_1 a^3 - p_2 b^3}{b^3 - a^3}.$$

The radial displacement of the hollow sphere is therefore

$$u(r) = \frac{1 - 2\nu}{E} \frac{p_1 a^3 - p_2 b^3}{b^3 - a^3} r + \frac{1 + \nu}{2E} \frac{(p_1 - p_2)a^3 b^3}{b^3 - a^3} \frac{1}{r^2}.$$

The deformation of the inner and outer surfaces is obtained by substituting $r = a$ and $r = b$. ■

Problem 18.3. Determine the deformation of a solid sphere (with radius R) in its own gravitational field.

Proof: [Solution] Assume the sphere has uniform density ρ . Let $u(r)$ be the radial displacement, positive outward. The gravitational field inside a uniform sphere is inward and has magnitude

$$g(r) = g \frac{r}{R},$$

where g is the magnitude at the surface. Thus the body force per unit volume is

$$\rho \mathbf{f} = -\rho g \frac{r}{R} \hat{\mathbf{r}}.$$

The governing Navier equation for static isotropic elasticity is

$$\mu \nabla^2 \mathbf{u} + (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) + \rho \mathbf{f} = 0.$$

For a purely radial spherical displacement, $\nabla \times \mathbf{u} = 0$, so this is equivalently

$$(\lambda + 2\mu) \nabla (\nabla \cdot \mathbf{u}) - \rho g \frac{r}{R} \hat{\mathbf{r}} = 0.$$

Using

$$\nabla \cdot \mathbf{u} = \frac{1}{r^2} \frac{d}{dr} (r^2 u),$$

the radial equation is

$$(\lambda + 2\mu) \frac{d}{dr} \left[\frac{1}{r^2} \frac{d}{dr} (r^2 u) \right] = \rho g \frac{r}{R}.$$

Before integrating, define

$$C = \frac{\rho g}{(\lambda + 2\mu)R}.$$

Then the equation becomes

$$\frac{d}{dr} \left[\frac{1}{r^2} \frac{d}{dr} (r^2 u) \right] = Cr.$$

Integrate once:

$$\frac{1}{r^2} \frac{d}{dr} (r^2 u) = \frac{Cr^2}{2} + C_1.$$

Multiply by r^2 and integrate again:

$$\frac{d}{dr} (r^2 u) = \frac{Cr^4}{2} + C_1 r^2,$$

$$r^2 u = \frac{Cr^5}{10} + \frac{C_1 r^3}{3} + C_2.$$

Therefore

$$u(r) = \frac{C}{10} r^3 + \frac{C_1}{3} r + \frac{C_2}{r^2}.$$

Regularity at the center requires $C_2 = 0$.

Now impose the free-surface boundary condition. Since no pressure is applied on the surface,

$$\sigma_{rr}(R) = 0.$$

The strain components are

$$u_{rr} = \frac{du}{dr}, \quad u_{\theta\theta} = u_{\phi\phi} = \frac{u}{r},$$

and

$$\sigma_{rr} = 2\mu \frac{du}{dr} + \lambda \left(\frac{du}{dr} + 2\frac{u}{r} \right).$$

Writing the regular solution as

$$u(r) = \alpha r + \beta r^3, \quad \beta = \frac{C}{10},$$

we calculate

$$\frac{du}{dr} = \alpha + 3\beta r^2, \quad \frac{u}{r} = \alpha + \beta r^2.$$

Thus

$$\sigma_{rr} = 2\mu(\alpha + 3\beta r^2) + \lambda(3\alpha + 5\beta r^2) = (3\lambda + 2\mu)\alpha + (5\lambda + 6\mu)\beta r^2.$$

The condition $\sigma_{rr}(R) = 0$ gives

$$(3\lambda + 2\mu)\alpha + (5\lambda + 6\mu)\beta R^2 = 0,$$

so

$$\alpha = -\frac{5\lambda + 6\mu}{3\lambda + 2\mu} \beta R^2.$$

Using the Lamé constants in terms of E, ν , this simplifies to

$$\alpha = -\frac{\rho g R}{10(\lambda + 2\mu)} \frac{5\lambda + 6\mu}{3\lambda + 2\mu} = -\frac{(1 + \nu)(1 - 2\nu)}{10E(1 - \nu)} \rho g R \frac{3 - \nu}{1 + \nu}.$$

Also

$$\beta = \frac{\rho g}{10(\lambda + 2\mu)R} = \frac{(1 + \nu)(1 - 2\nu)}{10E(1 - \nu)} \frac{\rho g}{R}.$$

Therefore

$$u(r) = -\frac{(1 + \nu)(1 - 2\nu)\rho g R}{10E(1 - \nu)} r \left(\frac{3 - \nu}{1 + \nu} - \frac{r^2}{R^2} \right).$$

The displacement is negative for ordinary elastic materials, so the sphere contracts under its own gravity. At the surface,

$$u(R) = -\frac{(1 - 2\nu)\rho g R^2}{5E}.$$

Thus the new radius is

$$R_{\text{new}} = R + u(R) = R \left(1 - \frac{(1 - 2\nu)\rho g R}{5E} \right).$$

■

Problem 18.4. Determine the deformation of a cylindrical pipe (with external and internal radii R_2 and R_1), with a pressure p inside and no pressure outside.

Proof: [Solution] Let

$$a = R_1, \quad b = R_2, \quad a < b,$$

where a is the inner radius and b is the outer radius. We treat the pipe as long, so the state is axisymmetric and independent of z . For a long pipe the natural approximation is plane strain:

$$u_z = 0, \quad u_{zz} = 0.$$

The displacement is radial:

$$\mathbf{u} = u(r) \hat{\mathbf{r}}.$$

The governing radial equilibrium equation is

$$\frac{d\sigma_{rr}}{dr} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} = 0.$$

The non-zero in-plane strains are

$$u_{rr} = \frac{du}{dr}, \quad u_{\theta\theta} = \frac{u}{r}.$$

Solving the corresponding displacement equation gives

$$u(r) = Ar + \frac{B}{r}.$$

Now calculate the strains:

$$u_{rr} = A - \frac{B}{r^2}, \quad u_{\theta\theta} = A + \frac{B}{r^2}, \quad u_{zz} = 0.$$

Their trace is

$$u_{kk} = u_{rr} + u_{\theta\theta} + u_{zz} = 2A.$$

Hooke's law therefore gives

$$\begin{aligned} \sigma_{rr} &= 2\mu \left(A - \frac{B}{r^2} \right) + 2\lambda A = 2(\lambda + \mu)A - \frac{2\mu B}{r^2}, \\ \sigma_{\theta\theta} &= 2\mu \left(A + \frac{B}{r^2} \right) + 2\lambda A = 2(\lambda + \mu)A + \frac{2\mu B}{r^2}. \end{aligned}$$

Now state the pressure boundary conditions:

$$\sigma_{rr}(a) = -p, \quad \sigma_{rr}(b) = 0.$$

Substitute the stress formula:

$$2(\lambda + \mu)A - \frac{2\mu B}{a^2} = -p, \quad 2(\lambda + \mu)A - \frac{2\mu B}{b^2} = 0.$$

Subtract the second equation from the first:

$$-2\mu B \left(\frac{1}{a^2} - \frac{1}{b^2} \right) = -p.$$

Hence

$$B = \frac{pa^2b^2}{2\mu(b^2 - a^2)} = \frac{(1 + \nu)pa^2b^2}{E(b^2 - a^2)}.$$

From the outer boundary condition,

$$2(\lambda + \mu)A = \frac{2\mu B}{b^2} = \frac{pa^2}{b^2 - a^2}.$$

Since

$$2(\lambda + \mu) = \frac{E}{(1 + \nu)(1 - 2\nu)},$$

we get

$$A = \frac{(1 + \nu)(1 - 2\nu)pa^2}{E(b^2 - a^2)}.$$

Therefore the radial displacement is

$$u(r) = \frac{(1 + \nu)pa^2}{E(b^2 - a^2)} \left[(1 - 2\nu)r + \frac{b^2}{r} \right].$$

This expression is positive: the pipe expands outward. The stresses are

$$\sigma_{rr} = \frac{pa^2}{b^2 - a^2} \left(1 - \frac{b^2}{r^2} \right), \quad \sigma_{\theta\theta} = \frac{pa^2}{b^2 - a^2} \left(1 + \frac{b^2}{r^2} \right).$$

One can check the boundary values directly:

$$\sigma_{rr}(a) = \frac{pa^2}{b^2 - a^2} \left(1 - \frac{b^2}{a^2} \right) = -p, \quad \sigma_{rr}(b) = 0.$$

■

Problem 18.5. Determine the deformation of a cylinder rotating uniformly about its axis.

Proof: [Solution] Let the solid cylinder have radius R , density ρ , and angular velocity Ω . We again use the long-cylinder plane-strain approximation. The displacement is radial:

$$\mathbf{u} = u(r) \hat{\mathbf{r}}.$$

In the rotating steady state the centrifugal body force per unit volume is

$$\rho \mathbf{f} = \rho \Omega^2 r \hat{\mathbf{r}}.$$

The governing radial equilibrium equation with body force is

$$\frac{d\sigma_{rr}}{dr} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} + \rho \Omega^2 r = 0.$$

The displacement equation is obtained from the Navier equation:

$$\mu \left(u'' + \frac{u'}{r} - \frac{u}{r^2} \right) + (\lambda + \mu) \frac{d}{dr} \left(u' + \frac{u}{r} \right) + \rho \Omega^2 r = 0.$$

Because the right-hand side is proportional to r , try

$$u(r) = Ar + Cr^3.$$

Before substituting, compute the two differential combinations:

$$\begin{aligned} u' &= A + 3Cr^2, & u'' &= 6Cr, \\ u'' + \frac{u'}{r} - \frac{u}{r^2} &= 6Cr + \frac{A + 3Cr^2}{r} - \frac{Ar + Cr^3}{r^2} = 8Cr, \end{aligned}$$

and

$$\frac{d}{dr} \left(u' + \frac{u}{r} \right) = \frac{d}{dr} (A + 3Cr^2 + A + Cr^2) = 8Cr.$$

Substitute these into the displacement equation:

$$8\mu Cr + 8(\lambda + \mu)Cr + \rho \Omega^2 r = 0.$$

Cancel r :

$$8(\lambda + 2\mu)C + \rho\Omega^2 = 0,$$

so

$$C = -\frac{\rho\Omega^2}{8(\lambda + 2\mu)}.$$

The constant A is determined by the free-surface condition:

$$\sigma_{rr}(R) = 0.$$

For $u = Ar + Cr^3$,

$$u_{rr} = u' = A + 3Cr^2, \quad u_{\theta\theta} = \frac{u}{r} = A + Cr^2, \quad u_{zz} = 0.$$

Thus

$$\sigma_{rr} = 2\mu(A + 3Cr^2) + \lambda(2A + 4Cr^2) = 2(\lambda + \mu)A + 2(2\lambda + 3\mu)Cr^2.$$

Apply $\sigma_{rr}(R) = 0$:

$$2(\lambda + \mu)A + 2(2\lambda + 3\mu)CR^2 = 0,$$

$$A = -\frac{2\lambda + 3\mu}{\lambda + \mu}CR^2.$$

Using

$$\frac{2\lambda + 3\mu}{\lambda + \mu} = 3 - 2\nu, \quad \lambda + 2\mu = \frac{E(1 - \nu)}{(1 + \nu)(1 - 2\nu)},$$

we obtain

$$C = -\frac{(1 + \nu)(1 - 2\nu)}{8E(1 - \nu)}\rho\Omega^2,$$

and therefore

$$u(r) = \frac{(1 + \nu)(1 - 2\nu)\rho\Omega^2}{8E(1 - \nu)} [(3 - 2\nu)R^2r - r^3].$$

The expression is positive for $0 < r < R$, so the cylinder expands outward. At the surface,

$$u(R) = \frac{(1 + \nu)(1 - 2\nu)\rho\Omega^2 R^3}{4E}.$$

■

Problem 18.6. Derive the equations of equilibrium for an isotropic body (in the absence of body forces) in terms of the components of the stress tensor.

Proof: [Solution] The most general local balance of linear momentum for a static body is

$$\frac{\partial \sigma_{ij}}{\partial x_j} + \rho f_i = 0.$$

In the absence of body forces, $\mathbf{f} = 0$, so the governing equations are simply

$$\frac{\partial \sigma_{ij}}{\partial x_j} = 0.$$

Writing out the repeated index $j = 1, 2, 3$ gives the three component equations

$$\begin{aligned}\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} &= 0, \\ \frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} &= 0, \\ \frac{\partial \sigma_{zx}}{\partial x} + \frac{\partial \sigma_{zy}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} &= 0.\end{aligned}$$

For ordinary elastic continua without internal torques, the stress tensor is symmetric:

$$\sigma_{ij} = \sigma_{ji}.$$

Therefore these may also be written as

$$\boxed{\begin{aligned}\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} &= 0, \\ \frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} &= 0, \\ \frac{\partial \sigma_{xz}}{\partial x} + \frac{\partial \sigma_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} &= 0.\end{aligned}}$$

If one wants the same equilibrium equations in terms of displacement components for an isotropic material, insert Hooke's law

$$\sigma_{ij} = \lambda(\partial_k u_k)\delta_{ij} + \mu(\partial_i u_j + \partial_j u_i)$$

into $\partial_j \sigma_{ij} = 0$. The calculation is

$$\begin{aligned}\partial_j \sigma_{ij} &= \partial_j [\lambda(\partial_k u_k)\delta_{ij} + \mu(\partial_i u_j + \partial_j u_i)], \\ \partial_j \sigma_{ij} &= \lambda \partial_i (\partial_k u_k) + \mu \partial_i (\partial_j u_j) + \mu \partial_j \partial_j u_i.\end{aligned}$$

Since $\partial_j u_j = \partial_k u_k$, this becomes

$$\mu \nabla^2 u_i + (\lambda + \mu) \partial_i (\nabla \cdot \mathbf{u}) = 0.$$

Thus the displacement form is

$$\boxed{\mu \nabla^2 \mathbf{u} + (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) = 0}.$$

Using E and ν , divide by μ and use

$$\frac{\lambda + \mu}{\mu} = \frac{1}{1 - 2\nu}.$$

Then

$$\boxed{\nabla^2 \mathbf{u} + \frac{1}{1 - 2\nu} \nabla (\nabla \cdot \mathbf{u}) = 0}.$$

■

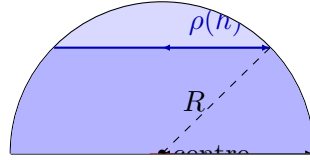
19 Fluid Mechanics — Draining Vessels (from Lecture 13)

Problem 19.1. A hemispherical bowl of radius R (curved side down, open flat face up) is filled completely with an incompressible fluid of density ρ_0 . A small circular hole of cross-sectional area $A_1 \ll A_0(h)$ is opened at the lowest point. Find the total time for the bowl to drain.

Extension: Repeat for a conical vessel with half-angle α at the apex (apex at the bottom), filled to height h_0 .

Proof: [Solution]

Setup and geometry. Let h denote the height of the fluid surface above the bottom of the bowl at time t . The bowl extends from $h = 0$ (apex) to $h = R$ (rim).



Step 1: Cross-sectional area $A_0(h)$. The sphere has its centre at height R above the bottom (i.e. at $h = R$ from the lowest point). At height h , the radial distance from the sphere's centre to the surface is R . The cross-sectional radius $\rho(h)$ satisfies

$$\rho(h)^2 + (R - h)^2 = R^2 \implies \rho(h)^2 = 2Rh - h^2.$$

Hence

$$A_0(h) = \pi \rho(h)^2 = \pi(2Rh - h^2). \quad (529)$$

Note $A_0(0) = 0$ (the bowl tapers to a point at the bottom) and $A_0(R) = \pi R^2$ (the rim has radius R).

Step 2: Exit velocity. By Torricelli's theorem (Lecture 13, Bernoulli between the free surface and the exit, with $A_1 \ll A_0(h)$),

$$v_{\text{out}}(h) = \sqrt{2gh}. \quad (530)$$

Step 3: ODE for $h(t)$. Conservation of mass (continuity) for the control volume of fluid inside the bowl:

$$\frac{d}{dt} \int_0^{h(t)} A_0(h') dh' = A_0(h) \dot{h} = -A_1 v_{\text{out}}(h).$$

Substituting (529) and (530),

$$\pi(2Rh - h^2) \frac{dh}{dt} = -A_1 \sqrt{2gh}. \quad (531)$$

Divide both sides by $\sqrt{h}$ (valid for $h > 0$) and rearrange:

$$(2R\sqrt{h} - h^{3/2}) dh = -\frac{A_1 \sqrt{2g}}{\pi} dt. \quad (532)$$

Step 4: Integrate. Integrate (532) with initial condition $h(0) = R$ and final condition $h(T) = 0$:

$$\int_R^0 (2R\sqrt{h} - h^{3/2}) dh = -\frac{A_1 \sqrt{2g}}{\pi} T.$$

The left side:

$$\int_0^R (2R\sqrt{h} - h^{3/2}) dh = \left[\frac{4R}{3} h^{3/2} - \frac{2}{5} h^{5/2} \right]_0^R = \frac{4R}{3} \cdot R^{3/2} - \frac{2}{5} \cdot R^{5/2} = R^{5/2} \left(\frac{4}{3} - \frac{2}{5} \right) = \frac{14}{15} R^{5/2}.$$

Therefore

$$T_{\text{hemisphere}} = \frac{14\pi}{15 A_1 \sqrt{2g}} R^{5/2}. \quad (533)$$

Is the drain time finite? As $h \rightarrow 0$, the cross-sectional area vanishes as $A_0 \sim 2\pi R h \rightarrow 0$, so the exit speed also goes to zero ($v_{\text{out}} \sim \sqrt{2gh} \rightarrow 0$). One might worry the drain time diverges, but the integrand near $h = 0$ behaves as

$$(2R\sqrt{h} - h^{3/2}) \approx 2R\sqrt{h},$$

and $\int_0 \sqrt{h} dh = \frac{2}{3}h^{3/2} \Big|_0 = 0$, so the integral converges. The bowl always drains in finite time.

Extension: Conical vessel. A cone with half-angle α (apex at the bottom), filled to height h_0 . At height h the cross-sectional radius is $h \tan \alpha$, so

$$A_0(h) = \pi h^2 \tan^2 \alpha.$$

The draining ODE (531) becomes

$$\pi h^2 \tan^2 \alpha \frac{dh}{dt} = -A_1 \sqrt{2gh} \quad \implies \quad h^{3/2} dh = -\frac{A_1 \sqrt{2g}}{\pi \tan^2 \alpha} dt.$$

Integrating from h_0 to 0:

$$\boxed{T_{\text{cone}} = \frac{2\pi \tan^2 \alpha}{5 A_1 \sqrt{2g}} h_0^{5/2}.} \quad (534)$$

Again finite: the integrand $h^{3/2}$ near $h = 0$ is integrable.

Comparison with the cylindrical tank. For a cylinder of cross-section A_0 (constant), filled to h_0 :

$$T_{\text{cylinder}} = \frac{A_0}{A_1} \sqrt{\frac{2h_0}{g}} \propto h_0^{1/2}.$$

For the cone, $T \propto h_0^{5/2}$; for the hemisphere (with $h_0 = R$ fixed), $T \propto R^{5/2}$. Vessels that taper toward the bottom drain more slowly at the end than a cylinder, but still empty in finite time. ■